

ARM740T

Datasheet



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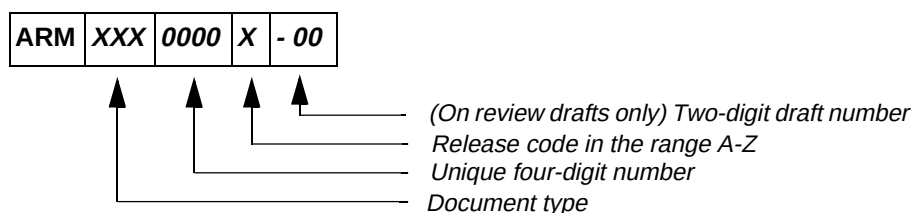
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1

Introduction

This chapter provides an introduction to the ARM740T.

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1.2	Block Diagram	1-3
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Introduction

1.1 Overview

The ARM740T is a general-purpose 32-bit microprocessor with:

- 8KB cache or 4KB variants
- write buffer
- Protection Unit

combined in a single macrocell.

The ARM740T is software-compatible with the ARM processor family and can be used with AMBA peripheral blocks, and has been optimised for use in embedded applications. The CPU within ARM740T is the ARM7.

ARM740T is a fully static part and has been designed to minimise power requirements. This makes it ideal for portable applications where both these features are essential.

The on-chip mixed data and instruction cache, and the write buffer, substantially raise the average execution speed and reduce the average amount of memory bandwidth required by the processor. This allows the external memory to support additional processors or *Direct Memory Access (DMA)* channels with minimal performance loss.

RISC architecture

The ARM740T architecture is based on *Reduced Instruction Set Computer (RISC)* principles, and the instruction set and related decode mechanism are greatly simplified compared with microprogrammed *Complex Instruction Set Computers (CISC)*.

1.2 Block Diagram

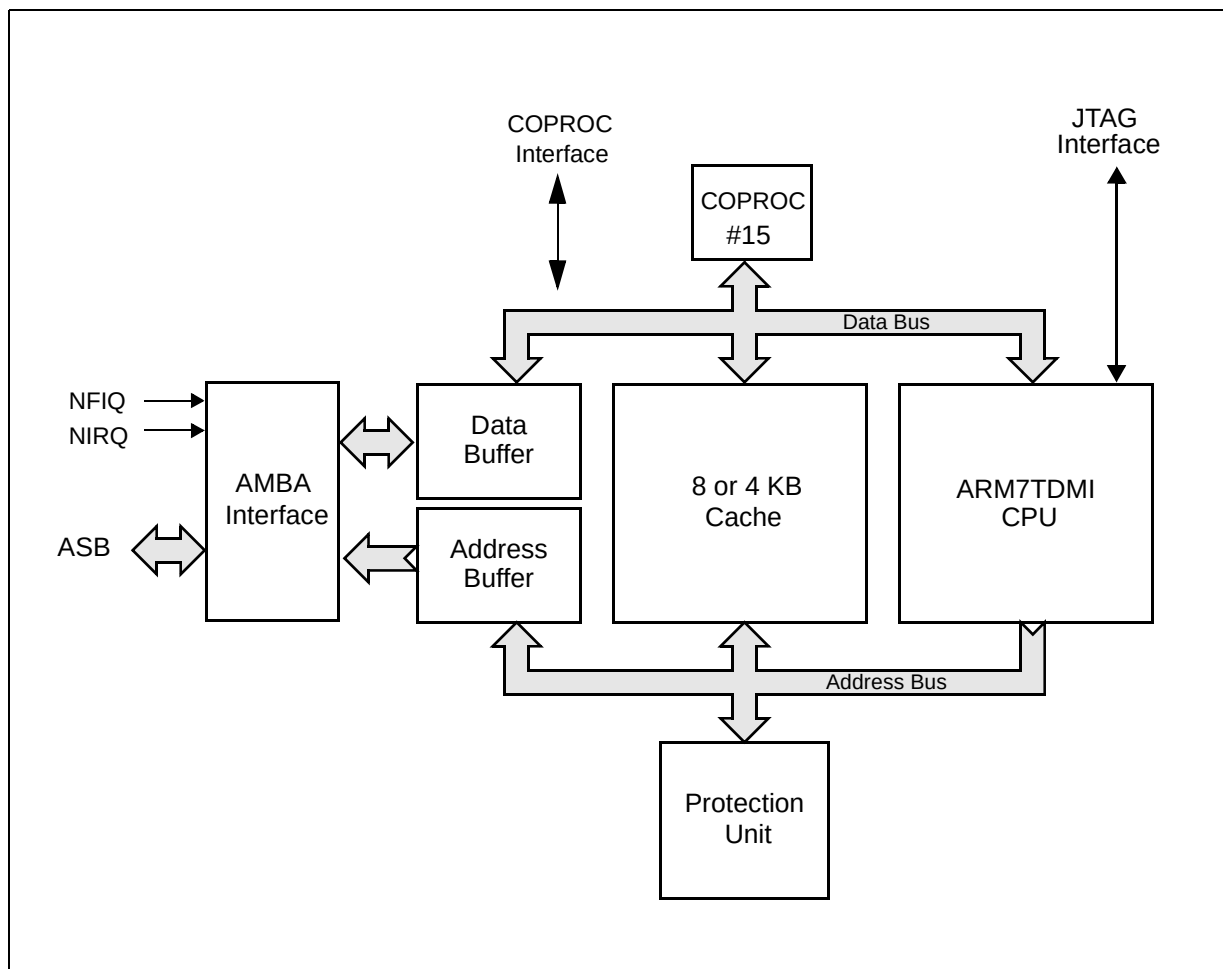


Figure 1-1: ARM740T block diagram

Introduction

1.3 Instruction Set Overview

The instruction set comprises ten basic instruction types:

- Two make use of the on-chip arithmetic logic unit, barrel shifter and multiplier to perform high-speed operations on the data in a bank of 31 registers, each 32 bits wide.
- Three classes of instruction control the data transfer between memory and the registers:
 - one optimised for flexibility of addressing
 - one for rapid context switching
 - one for swapping data
- Two control the flow and privilege level of execution.
- Three control external coprocessors which allow the functionality of the instruction set to be extended off-chip in an open and uniform way.

The ARM instruction set is a good target for compilers of many different high-level languages. Where required for critical code segments, assembly code programming is also straightforward, unlike some RISC processors which depend on sophisticated compiler technology to manage complicated instruction interdependencies.

1.3.1 ARM instruction set

This section gives an overview of the ARM instructions available. For full details of these instructions, please refer to the *ARM Architecture Reference Manual* (ARM DDI 0100).

Format summary

The ARM instruction set formats are shown below.

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
Data processing / PSRTransfer	Cond		0	0	I	Opcode				S	Rn				Rd				Operand 2															
Multiply	Cond		0	0	0	0	0	0	A	S	Rd				Rn				Rs				1	0	0	1	Rm							
Multiply Long	Cond		0	0	0	0	1	U	A	S	RdHi				RdLo				Rn				1	0	0	1	Rm							
Single Data Swap	Cond		0	0	0	1	0	B	0	0	Rn				Rd				0	0	0	0	1	0	0	1	Rm							
Branch and Exchange	Cond		0	0	0	1	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	1	Rn					
Halfword Data Transfer: register offset	Cond		0	0	0	P	U	0	W	L	Rn				Rd				0	0	0	0	1	S	H	1	Rm							
Halfword Data Transfer: immediate offset	Cond		0	0	0	P	U	1	W	L	Rn				Rd				Offset				1	S	H	1	Offset							
Single Data Transfer	Cond		0	1	I	P	U	B	W	L	Rn				Rd				Offset															
Undefined	Cond		0	1	1																									1				
Block Data Transfer	Cond		1	0	0	P	U	S	W	L	Rn				Register List																			
Branch	Cond		1	0	1	L	Offset																											
Coprocessor Data Transfer	Cond		1	1	0	P	U	N	W	L	Rn				CRd				CP#				Offset											
Coprocessor Data Operation	Cond		1	1	1	0	CP Opc				CRn				CRd				CP#				CP		0	CRm								
Coprocessor Register Transfer	Cond		1	1	1	0	CP Opc				L	CRn				Rd				CP#				CP		1	CRm							
Software Interrupt	Cond		1	1	1	1	Ignored by processor																											
	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		

Figure 1-2: ARM instruction set formats

Note Some instruction codes are not defined but do not cause the Undefined instruction trap to be taken; for example, a Multiply instruction with bit 6 changed to 1. These instructions should not be used, as their action may change in future ARM implementations.

Introduction

ARM instruction summary

The following table summarizes the ARM instruction set.

Mnemonic	Instruction	Action
ADC	Add with carry	$Rd := Rn + Op2 + \text{Carry}$
ADD	Add	$Rd := Rn + Op2$
AND	AND	$Rd := Rn \text{ AND } Op2$
B	Branch	$R15 := \text{address}$
BIC	Bit Clear	$Rd := Rn \text{ AND NOT } Op2$
BL	Branch with Link	$R14 := R15, R15 := \text{address}$
BX	Branch and Exchange	$R15 := Rn,$ $T \text{ bit} := Rn[0]$
CDP	Coprocessor Data Processing	(Coprocessor-specific)
CMN	Compare Negative	$CPSR \text{ flags} := Rn + Op2$
CMP	Compare	$CPSR \text{ flags} := Rn - Op2$
EOR	Exclusive OR	$Rd := (Rn \text{ AND NOT } Op2)$ $\text{OR } (Op2 \text{ AND NOT } Rn)$
LDC	Load coprocessor from memory	Coprocessor load
LDM	Load multiple registers	Stack manipulation (Pop)
LDR	Load register from memory	$Rd := (\text{address})$
MCR	Move CPU register to coprocessor register	$cRn := rRn \{<op>cRm\}$
MLA	Multiply Accumulate	$Rd := (Rm * Rs) + Rn$
MOV	Move register or constant	$Rd := Op2$
MRC	Move from coprocessor register to CPU register	$Rn := cRn \{<op>cRm\}$
MRS	Move PSR status/flags to register	$Rn := PSR$
MSR	Move register to PSR status/flags	$PSR := Rm$
MUL	Multiply	$Rd := Rm * Rs$
MVN	Move negative register	$Rd := 0xFFFFFFFF \text{ EOR } Op2$
ORR	OR	$Rd := Rn \text{ OR } Op2$
RSB	Reverse Subtract	$Rd := Op2 - Rn$
RSC	Reverse Subtract with Carry	$Rd := Op2 - Rn - 1 + \text{Carry}$

Table 1-1: ARM instruction summary

Mnemonic	Instruction	Action
SBC	Subtract with Carry	$Rd := Rn - Op2 - 1 + \text{Carry}$
STC	Store coprocessor register to memory	address := CRn
STM	Store Multiple	Stack manipulation (Push)
STR	Store register to memory	<address> := Rd
SUB	Subtract	$Rd := Rn - Op2$
SWI	Software Interrupt	OS call
SWP	Swap register with memory	$Rd := [Rn], [Rn] := Rm$
TEQ	Test bitwise equality	CPSR flags := Rn EOR Op2
TST	Test bits	CPSR flags := Rn AND Op2

Table 1-1: ARM instruction summary (Continued)

Introduction

1.3.2 THUMB Instruction Set

This section gives an overview of the THUMB instructions available. For full details of these instructions, please refer to the *ARM Architecture Reference Manual* (ARM DDI 0100).

Format summary

The THUMB instruction set formats are shown below.

		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Move shifted register	1	0	0	0	Op		Offset5					Rs		Rd			
Add/subtract	2	0	0	0	1	1	I	Op	Rn/offset3			Rs		Rd			
Move/compare/add/subtract immediate	3	0	0	1	Op		Rd			Offset8							
ALU operations	4	0	1	0	0	0	0	Op			Rs		Rd				
Hi register operations/branch exchange	5	0	1	0	0	0	1	Op		H1	H2	Rs/Hs		Rd/Hd			
PC-relative load	6	0	1	0	0	1	Rd			Word8							
Load/store with register offset	7	0	1	0	1	L	B	0	Ro			Rb		Rd			
Load/store sign-extended byte/halfword	8	0	1	0	1	H	S	1	Ro			Rb		Rd			
Load/store with immediate offset	9	0	1	1	B	L	Offset5					Rb		Rd			
Load/store halfword	10	1	0	0	0	L	Offset5					Rb		Rd			
SP-relative load/store	11	1	0	0	1	L	Rd			Word8							
Load address	12	1	0	1	0	SP	Rd			Word8							
Add offset to stack pointer	13	1	0	1	1	0	0	0	0	S	SWord7						
Push/pop registers	14	1	0	1	1	L	1	0	R	Rlist							
Multiple load/store	15	1	1	0	0	L	Rb			Rlist							
Conditional branch	16	1	1	0	1	Cond				Soffset8							
Software Interrupt	17	1	1	0	1	1	1	1	1	Value8							
Unconditional branch	18	1	1	1	0	0	Offset11										
Long branch with link	19	1	1	1	1	H	Offset										
		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Figure 1-3: THUMB instruction set formats

THUMB instruction summary

The following table summarizes the THUMB instruction set.

Mnemonic	Instruction	Lo register operand	Hi register operand	Condition codes set
ADC	Add with Carry	✓		✓
ADD	Add	✓	✓	✓(1)
AND	AND	✓		✓
ASR	Arithmetic Shift Right	✓		✓
B	Unconditional branch	✓		
Bxx	Conditional branch	✓		
BIC	Bit Clear	✓		✓
BL	Branch and Link			
BX	Branch and Exchange	✓	✓	
CMN	Compare Negative	✓		✓
CMP	Compare	✓	✓	✓
EOR	EOR	✓		✓
LDMIA	Load multiple	✓		
LDR	Load word	✓		
LDRB	Load byte	✓		
LDRH	Load halfword	✓		
LSL	Logical Shift Left	✓		✓
LDSB	Load sign-extended byte	✓		
LDSH	Load sign-extended halfword	✓		
LSR	Logical Shift Right	✓		✓
MOV	Move register	✓	✓	✓(2)
MUL	Multiply	✓		✓
MVN	Move Negative register	✓		✓
NEG	Negate	✓		✓
ORR	OR	✓		✓
POP	Pop registers	✓		
PUSH	Push registers	✓		

Table 1-2: THUMB instruction summary

Introduction

Mnemonic	Instruction	Lo register operand	Hi register operand	Condition codes set
ROR	Rotate Right	✓		✓
SBC	Subtract with Carry	✓		✓
STMIA	Store Multiple	✓		
STR	Store word	✓		
STRB	Store byte	✓		
STRH	Store halfword	✓		
SWI	Software Interrupt			
SUB	Subtract	✓		✓
TST	Test bits	✓		✓

Table 1-2: THUMB instruction summary (Continued)

- 1 The condition codes are unaffected by the format 5, 12 and 13 versions of this instruction.
- 2 The condition codes are unaffected by the format 5 version of this instruction.

2

Signal Description

This chapter describes the signals.

2.1	AMBA Interface Signals	2-2
2.2	Coprocessor Interface Signals	2-4
2.3	JTAG Signals	2-6
2.4	Debugger Signals	2-8
2.5	Miscellaneous Signals	2-9

Signal Description

2.1 AMBA Interface Signals

Name	Type	Drive	Source/ Destination	Description
AGNT	In	–	Arbiter	Access Grant This signal from the bus arbiter indicates that the bus master is currently the highest priority master requesting the bus. If AGNT is asserted at the end of a transfer (BWAIT LOW), the master will be granted the bus. AGNT changes during the LOW phase of BCLK , and remains valid through the HIGH phase.
AREQ	Out	D4		Access Request This signal indicates that the master requires the bus. This signal changes during the HIGH phase of BCLK . This signal is intended for use where the ARM740T is not the lowest priority or default bus master.
BA[31:0]	Out	D6	Current bus master	Bus Address This is the system address bus.
BCLK	In	–		Bus Clock This clock times all bus transfers.
BD[31:0]	InOut	D6	Bus master	Bus Data This is the bidirectional system data bus. The data bus is driven by the current bus master during write transfers, and by the appropriate bus slave during read cycles.
BERROR	InOut	D4	System decoder and current bus master	Bus Error This signal indicates a transfer error by the selected bus slave. When BERROR is HIGH, a transfer error has occurred. When BERROR is LOW, the transfer is successful. This signal is also used in combination with the BLAST signal to indicate a bus retract operation.
BLAST	InOut	D4	System decoder and current bus master	Bus Last This signal is driven by the selected bus slave to indicate if the current transfer should be the last of a burst sequence. When BLAST is HIGH, the next bus transfer must allow for sufficient time for address decoding. When BLAST is LOW, the next transfer may continue a burst sequence. This signal is also used in combination with the BERROR signal to indicate a bus retract operation.
BLOK	Out	D4	Arbiter	Bus Lock When HIGH, this signal indicates that the following transfer is to be indivisible and no other bus master should be given access to the bus.
BnRES	In	–	Reset state machine	Bus Reset This signal indicates the reset status of the bus.

Table 2-1: AMBA interface signal descriptions

Signal Description

Name	Type	Drive	Source/ Destination	Description
BPROT[1:0]	Out	D4	Current bus master	Bus Protections These signals provide additional information about the transfer being performed. All write cycles are indicated as being Supervisor accesses. These signals have the same timing as the BA signals.
BSIZE[1:0]	Out	D4	Current bus master	Bus Size These signals indicate the size of the transfer, which may be byte, halfword or word. These signals have the same timing as the address bus.
BTRAN[1:0]	Out	D8	Bus master	Bus Transaction Type These signals indicate the type of the next transaction, which may be address-only, nonsequential or sequential. These signals are driven when AGNT is asserted, and are valid during the HIGH phase of BCLK before the transfer to which they refer.
BWAIT	InOut	D4	System decoder and current bus master	Bus Wait This signal is driven by the selected slave to indicate if the current transfer may complete. If BWAIT is HIGH, a further bus cycle is required. If BWAIT is LOW, the transfer may complete in the current bus cycle.
BWRITE	Out	D4	Current bus master	Bus Write When HIGH, this signal indicates a write cycle and when LOW, a read cycle. This signal has the same timing as the address bus.
DSEL	In	–	System decoder	Slave Select This signal puts the ARM core into a test mode so that vectors can be written in and out of the core.

Table 2-1: AMBA interface signal descriptions (Continued)

Signal Description

2.2 Coprocessor Interface Signals

Name	Type	Drive	Description
CPCLK	Out	D4	Coprocessor Clock This clock controls the operation of the coprocessor interface.
CPDATA[31:0]	InOut	D4	Coprocessor Data Bus Using this bus, data is transferred to and from the co-processor. Data is valid on the falling edge of CPCLK .
CPDBE	In	–	Coprocessor Data Bus Enable When HIGH, this signal indicates that the coprocessor intends to drive the coprocessor data bus CPDATA . If the coprocessor interface is not to be used then this signal should be tied LOW.
CPnWAIT	Out	D4	Coprocessor Not Wait The coprocessor clock CPCLK is qualified by CPnWAIT to allow the ARM740T to control the transfer of data on the coprocessor interface.
CPTESTREAD	In	–	Coprocessor Test Read This signal is used for test of a Piccolo coprocessor (if attached) and should only be used with the ARM740T held in reset. When HIGH, it enables DB to be driven on to CPDATA , and should normally be held LOW. It must never be asserted at the same time as CPTESTWRITE .
CPTESTWRITE	In	–	Coprocessor Test Write This signal is used for test of a Piccolo coprocessor (if attached) and should only be used with the ARM740T held in reset. When HIGH, it enables DB to be driven on to CPDATA , and should normally be held LOW. It must never be asserted at the same time as CPTESTREAD .
EXTCPA	In	–	External Coprocessor Absent A coprocessor that is capable of performing the operation that ARM740T is requesting (by asserting nCPI) should take EXTCPA LOW immediately. If EXTCPA is HIGH at the end of the LOW phase of the cycle in which nCPI went LOW, ARM740T aborts the coprocessor instruction and takes the undefined instruction trap. If EXTCPA is LOW and remains LOW, ARM740T busy-waits until EXTCPB is LOW and then completes the coprocessor instruction.

Table 2-2: Coprocessor interface signal descriptions

Signal Description

Name	Type	Drive	Description
EXTCPB	In	—	External Coprocessor Busy A coprocessor which is capable of performing the operation which ARM740T is requesting (by asserting nCPI), but cannot commit to starting it immediately, should indicate this by driving EXTCPB HIGH. When the coprocessor is ready to start it should take EXTCPB LOW. ARM740T samples ExtCPB at the end of the LOW phase of each cycle in which nCPI is LOW.
nCPI	Out	D4	Not Coprocessor Instruction When LOW, this signal indicates that the ARM740T is executing a coprocessor instruction.
nOPC	Out	D8	Not OPcode Fetch When LOW, this signal indicates that the processor is fetching an instruction from memory. When HIGH, data (if present) is being transferred. This signal is used by the coprocessor to track the ARM pipeline.

Table 2-2: Coprocessor interface signal descriptions (Continued)



Signal Description

2.3 JTAG Signals

Name	Type	Drive	Description
IR[3:0]	Out	D4	TAP Instruction Register These signals reflect the current instruction loaded into the TAP controller instruction register. These signals change on the falling edge of TCK when the TAP state machine is in the UPDATE-IR state. These signals may be used to add additional scan chains using the ARM740T TAP controller.
RSTCLKBS	Out	D4	Reset Boundary Scan Clock This signal denotes that either the TAP controller state machine is in the RESET state or that nTRST has been asserted. This may be used to reset boundary scan cells outside the ARM740T.
SCREG[3:0]	Out	D4	Scan Chain Register These signals reflect the ID number of the scan chain currently selected by the TAP controller. These signals change on the falling edge of TCK when the TAP state machine is in the UPDATE-DR state.
SDINBS	Out	D4	Boundary Scan Serial Data In This signal is the serial data to be applied to an external scan chain.
SDOUTBS	In	–	Boundary Scan Serial Data Out This signal is the serial data from an external scan chain. It allows a single TDO port to be used. If an external scan chain is not connected, this input should be tied LOW.
TAPSM[3:0]	Out	D4	TAP Controller State These signals represent the current state of the TAP controller state machine. These signals change on the rising edge of TCK and may be used to add additional scan chains using the ARM740T TAP controller.
TCK	In	–	Test Clock This is part of the IEEE 1149.1 JTAG standard.
TCK1	Out	D4	Test Clock 1 This clock represents the HIGH phase of TCK . TCK1 is HIGH when TCK is HIGH. This signal may be used to allow more scan chains to be added using the ARM740T TAP controller.

Table 2-3: JTAG signal descriptions

Signal Description

Name	Type	Drive	Description
TCK2	Out	D4	Test Clock 2 This clock represents the LOW phase of TCK . TCK2 is HIGH when TCK is LOW. This signal may be used to allow more scan chains to be added using the ARM740T TAP controller. TCK2 is the non-overlapping compliment of TCK1 .
TDI	In	–	Test Data In This is part of the IEEE 1149.1 JTAG standard.
TDO	Out	D3	Test Data Out This is part of the IEEE 1149.1 JTAG standard.
TMS	In	–	Test Mode Select This is part of the IEEE 1149.1 JTAG standard.
nTDOEN	Out	D4	Not Test Data Out Output Enable When LOW, this signal denotes that serial data is being driven out on the TDO .
nTRST	In	–	Not Test Reset When LOW, resets the JTAG interface.

Table 2-3: JTAG signal descriptions (Continued)

Signal Description

2.4 Debugger Signals

Name	Type	Drive	Description
BREAKPOINT	In	–	Breakpoint This signal allows external hardware to halt execution of the processor for debug purposes. When HIGH causes the current memory access to be breakpointed. If the memory access is an instruction fetch, the core enters debug state if the instruction reaches the execute stage of the core pipeline. If the memory access is for data, the core enters debug state after the current instruction completes execution. This allows extension of the internal breakpoints provided by the EmbeddedICE module.
COMMRX	Out	D4	Communication Receive Empty When HIGH, this signal denotes that the comms channel receive buffer is empty.
COMMTX	Out	D4	Communication Transmit Empty When HIGH, this signal denotes that the comms channel transmit buffer is empty.
DBGACK	Out	D4	Debug Acknowledge When HIGH, indicates that the ARM is in debug state.
DBGEN	In	–	Debug Enable This signal allows the debug features of ARM740T to be disabled. This signal should be LOW if debug is not required.
DBGREQ	In	–	Debug Requests This signal causes the core to enter debug state after executing the current instruction. This allows external hardware to force the core into debug state, in addition to the debugging features provided by the EmbeddedICE module.
EXTERN[1:0]	In	–	External Condition These signals allow breakpoints and/or watchpoints to be dependent on an external condition.
RANGEOUT[1:0]	Out	D4	Rangeout These signals indicate that the relevant EmbeddedICE watchpoint register has matched the conditions currently present on the address, data and control buses. These signals are independent of the state of the watchpoint enable control bits.

Table 2-4: Debugger signal descriptions

2.5 Miscellaneous Signals

Name	Type	Drive	Description
BIGEND	Out	D4	Big-endian Format When this signal is HIGH, the processor treats bytes in memory as being in big-endian format. When it is LOW, memory is treated as little-endian.
FASTBUS	In	–	Bus Clocking Mode Select Signal When LOW, the ARM740T operates from a single clock, BCLK , when HIGH selects fastbus mode operating from two clocks, BCLK and FCLK .
FCLK	In	–	Fast Clock This is used during the RAM and TAG tests, to enable efficient testing. In standard bus mode, is used to clock the core.
nUSER	Out	D8	Not User Mode When LOW, this signal indicates that the processor is in user mode. It is used by a coprocessor to qualify instructions.
nFIQ	In	–	ARM Fast Interrupt Request Typically there is only a single nFIQ signal in a system, although this may be disabled by the interrupt controller.
nIRQ	In	–	ARM Interrupt Request The interrupt controller mixes several interrupt sources and produces ARM nIRQ .
SnA	In	–	Synchronous / not Asynchronous In standard ARM bus mode this signal determines the bus interface mode and should be wired HIGH or LOW depending on the desired relationship between FCLK and BCLK in the application. See 10.3 Standard Mode on page 10-4. This pin is ignored when operating with the fastbus extension.
TBIT	Out	D4	THUMB Mode This signal when HIGH, indicates that the processor is executing the THUMB instruction set. When LOW, the processor is executing the ARM instruction set.

Table 2-5: Miscellaneous signal descriptions

Signal Description



3

Programmer's Model

This chapter describes the operating states of the ARM740T.

3.1	Processor Operating States	3-2
3.2	Data Types	3-2
3.3	Operating Modes	3-3
3.4	Memory Formats	3-4
3.5	Registers	3-5
3.6	Program Status Registers	3-9
3.7	Exceptions	3-11
3.8	Reset	3-15

Programmer's Model

3.1 Processor Operating States

From the programmer's point of view, the ARM740T can be in one of two states:

<i>ARM state</i>	which executes 32-bit, word-aligned ARM instructions.
<i>THUMB state</i>	which operates with 16-bit, halfword-aligned THUMB instructions. In this state, the PC uses bit 1 to select between alternate halfwords.

Note *Transition between these two states does not affect the processor mode or the contents of the registers.*

3.1.1 Switching state

Entering THUMB state

Entry into THUMB state happens:

- 1 On Execution of a BX instruction with the state bit (bit 0) set in the operand register.
- 2 On return from an exception (IRQ, FIQ, UNDEF, ABORT, SWI etc.), if the exception was entered with the processor in THUMB state.

Entering ARM state

Entry into ARM state happens:

- On execution of the BX instruction with the state bit clear in the operand register.
- On the processor taking an exception (IRQ, FIQ, RESET, UNDEF, ABORT, SWI etc.). In this case, the PC is placed in the exception mode's link register, and execution starts at the exception's vector address.

3.2 Data Types

ARM740T supports the following data types:

byte	(8-bit)
halfword	(16-bit). Halfwords must be aligned to 2-byte boundaries.
word	(32-bit). Words must be aligned to 4-byte boundaries.

3.3 Operating Modes

ARM740T supports seven modes of operation:

User (usr)	The normal ARM program execution state
FIQ (fiq)	Designed to support a data transfer or channel process
IRQ (irq)	Used for general-purpose interrupt handling
Supervisor (svc)	Protected mode for the operating system
Abort mode (abt)	Entered after a data or instruction prefetch abort
System (sys)	A privileged user mode for the operating system
Undefined (und)	Entered when an undefined instruction is executed

Mode changes may be made under software control, or may be brought about by external interrupts or exception processing.

Most application programs execute in User mode. The non-user modes—known as *privileged modes*—are entered in order to service interrupts or exceptions, or to access protected resources.

Programmer's Model

3.4 Memory Formats

The bigend bit in the Control Register sets whether the ARM740T treats words in memory as being stored in big-endian or little-endian format. See **Chapter 4, Configuration** for more information on the Control Register.

ARM740T views memory as a linear collection of bytes numbered upwards from zero. Bytes 0 to 3 hold the first stored word, bytes 4 to 7 the second and so on. ARM740T can treat words in memory as being stored either in big-endian or little-endian format.

3.4.1 Big-endian format

- In big-endian format:
- the most significant byte of a word is stored at the lowest numbered byte
 - the least significant byte at the highest numbered byte
- Byte 0 of the memory system is therefore connected to data lines 31 through 24.

Higher Address	31	24	23	16	15	8	7	0	Word Address
	8	9	10	11					8
	4	5	6	7					4
	0	1	2	3					0
Lower Address	- Most significant byte is at lowest address - Word is addressed by byte address of most significant byte								

Figure 3-1: Big-endian address of bytes within words

3.4.2 Little-endian format

In little-endian format the lowest numbered byte in a word is considered the word's least significant byte, and the highest numbered byte the most significant.

Byte 0 of the memory system is therefore connected to data lines 7 through 0.

Higher Address	31	24	23	16	15	8	7	0	Word Address
	11	10	9	8					8
	7	6	5	4					4
	3	2	1	0					0
Lower Address	- Least significant byte is at lowest address - Word is addressed by byte address of least significant byte								

Figure 3-2: Little-endian addresses of bytes with words



3.5 Registers

ARM740T has a total of 37 registers:

- 31 general-purpose 32-bit registers
- six status registers

These cannot all be seen at once. The processor state and operating mode dictate which registers are available to the programmer.

3.5.1 The ARM state register set

In ARM state, 16 general registers and one or two status registers are visible at any one time. In privileged (non-User) modes, mode-specific banked registers are switched in.

Figure 3-3: Register organization in ARM state shows which registers are available in each mode: the banked registers are marked with a shaded triangle.

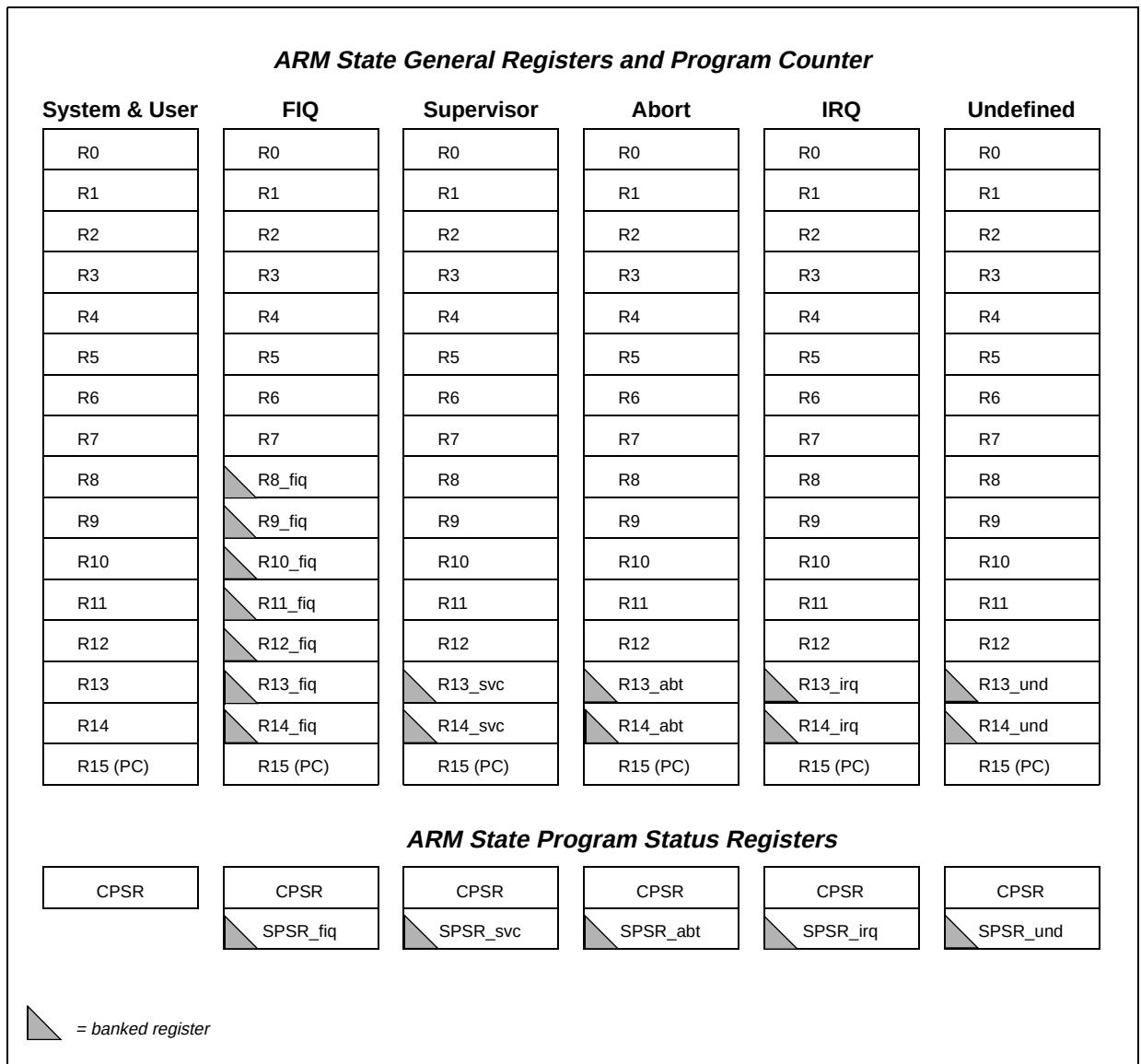


Figure 3-3: Register organization in ARM state

Programmer's Model

The ARM state register set contains 16 directly accessible registers: R0 to R15. All of these except R15 are general-purpose, and may be used to hold either data or address values. In addition to these, there is a 17th register used to store status information.

Pre-defined registers

Register 14	is used as the subroutine link register. This receives a copy of R15 when a Branch and Link (BL) instruction is executed. At all other times it may be treated as a general-purpose register. The corresponding banked registers R14_svc, R14_irq, R14_fiq, R14_abt and R14_und are similarly used to hold the return values of R15 when interrupts and exceptions arise, or when Branch and Link instructions are executed within interrupt or exception routines.
Register 15	holds the <i>Program Counter (PC)</i> . In ARM state, bits [1:0] of R15 are zero and bits [31:2] contain the PC. In THUMB state, bit [0] is zero and bits [31:1] contain the PC.
Register 16	is the <i>Current Program Status Register (CPSR)</i> . This contains condition code flags and the current mode bits.

FIQ mode

FIQ mode has seven banked registers mapped to R8 – 14 (R8_fiq – R14_fiq). In ARM state, many FIQ handlers do not need to save any registers. User, IRQ, Supervisor, Abort and Undefined mode each have two banked registers mapped to R13 and R14, allowing each of these modes to have a private stack pointer and link registers.

3.5.2 The THUMB state register set

The THUMB state register set is a subset of the ARM state set. The programmer has direct access to eight general registers, [R0:R7]:

- a Program Counter (PC)
- a stack pointer register (SP)
- a *link register* (LR), and the CPSR.

There are banked Stack Pointers, Link Registers and *Saved Process Status Registers* (SPSRs) for each privileged mode. This is shown in **Figure 3-4: Register organization in THUMB state**.

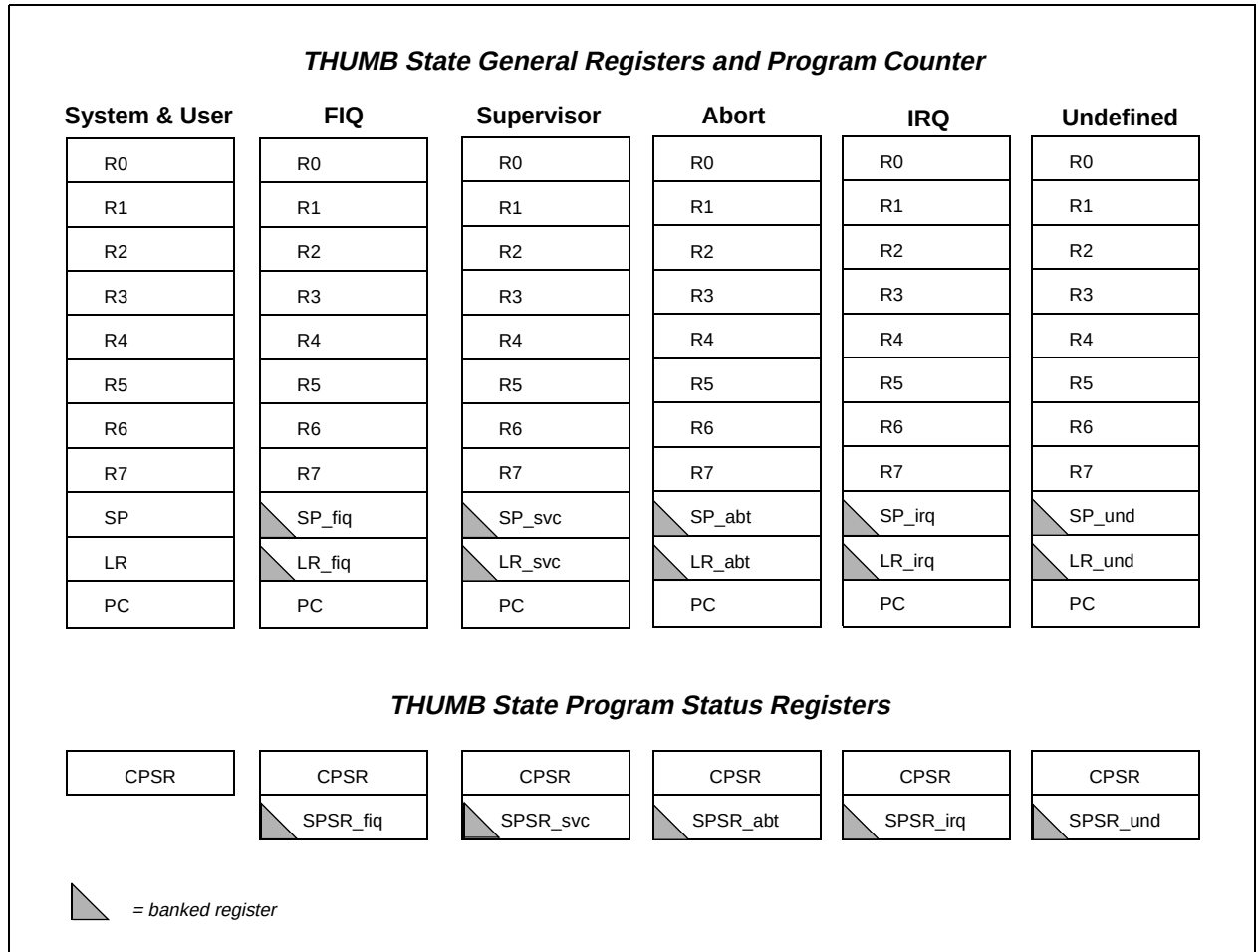


Figure 3-4: Register organization in THUMB state

3.5.3 The relationship between ARM and THUMB state registers

The THUMB state registers relate to the ARM state registers in the following way:

- THUMB state [R0:R7] and ARM state [R0:R7] are identical
- THUMB state CPSR and SPSRs and ARM state CPSR and SPSRs are identical
- THUMB state SP maps onto ARM state R13
- THUMB state LR maps onto ARM state R14

Programmer's Model

- The THUMB state Program Counter maps onto the ARM state Program Counter (R15)

This relationship is shown in **Figure 3-5: Mapping of THUMB state registers onto ARM state registers**.

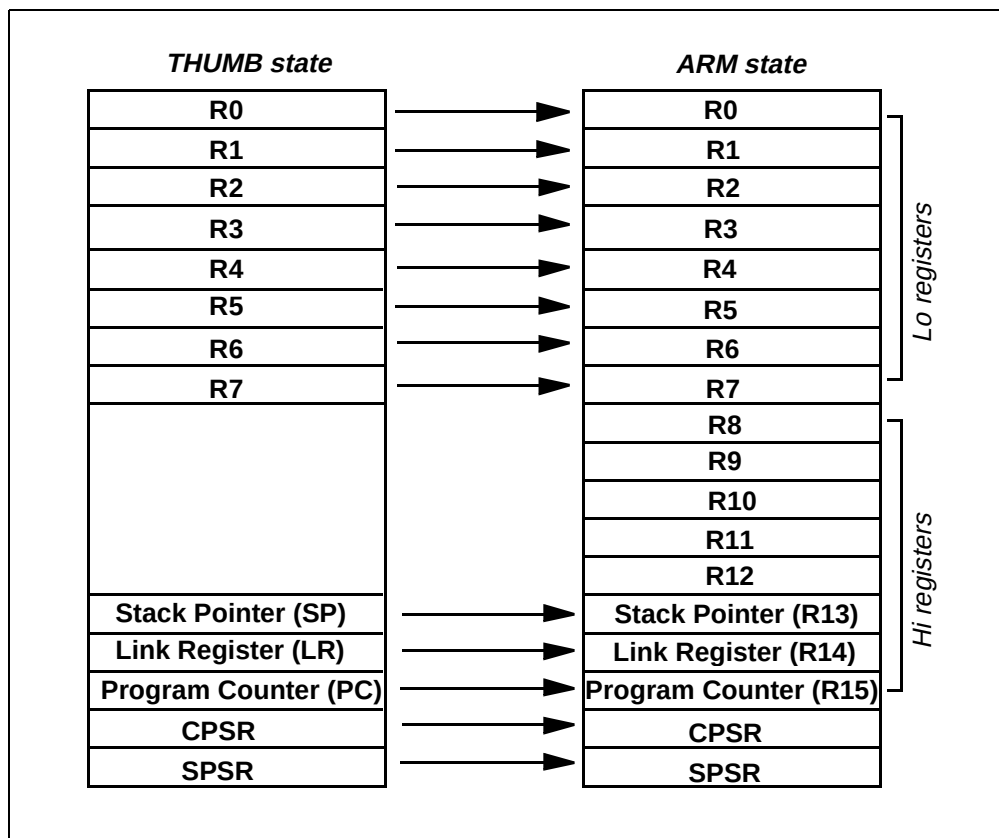


Figure 3-5: Mapping of THUMB state registers onto ARM state registers

3.5.4 Accessing Hi registers in THUMB state

In THUMB state, registers [R8:R15] (the *Hi registers*) are not part of the standard register set. However, the assembly language programmer has limited access to them, and can use them for fast temporary storage.

A value may be transferred from a register in the range [R0:R7] (a *Lo register*) to a Hi register, and from a Hi register to a Lo register, using special variants of the MOV instruction. Hi register values can also be compared against or added to Lo register values with the CMP and ADD instructions. See the information on high registers in the *ARM Architecture Reference Manual* (ARM DDI 0100) for details.

3.6 Program Status Registers

The ARM740T contains a *Current Program Status Register (CPSR)*, plus five *Saved Program Status Registers (SPSRs)* for use by exception handlers. These registers:

- hold information about the most recently performed ALU operation
- control the enabling and disabling of interrupts
- set the processor operating mode

The arrangement of bits is shown in **Figure 3-6: Program status register format**.

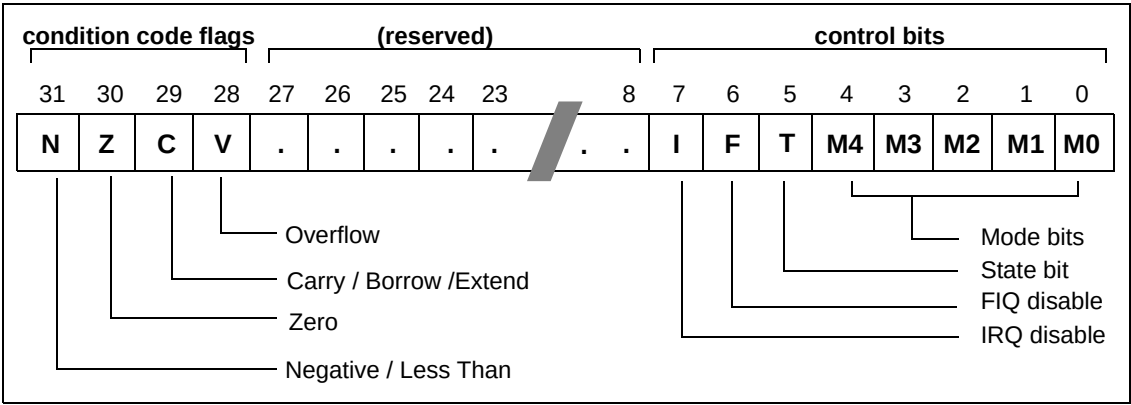


Figure 3-6: Program status register format

3.6.1 Condition code flags

The N, Z, C and V bits are the condition code flags. These may be changed as a result of arithmetic and logical operations, and may be tested to determine whether an instruction should be executed.

In ARM state, all instructions may be executed conditionally, in THUMB state, only the Branch instruction is capable of conditional execution: See the *ARM Architecture Reference Manual* (ARM DDI 0100) for details.

3.6.2 Control bits

The bottom 8 bits of a PSR (incorporating T, I, F and M[4:0]) are known collectively as the control bits. These will change when an exception arises. If the processor is operating in a privileged mode, they can also be manipulated by software.

- T bit
Reflects the operating state. When this bit is set, the processor is executing in THUMB state, otherwise it is executing in ARM state. This is reflected on the **TBIT** external signal. The software must never change the state of the **TBIT** in the CPSR. If this happens, the processor enters an unpredictable state.
- I and F bits
The interrupt disable bits. When set, these disable the IRQ and FIQ interrupts respectively.

Programmer's Model

M[4:0] bits

The mode bits. These determine the processor's operating mode, as shown in **Table 3-1: PSR mode bit values** on page 3-10. Not all combinations of the mode bits define a valid processor mode; you must use only those explicitly described.

Note: If any illegal value is programmed into the mode bits, M[4:0], the processor enters an unrecoverable state. If this occurs, reset should be applied.

M[4:0]	Mode	Visible THUMB state registers	Visible ARM state registers
10000	User	R7..R0, LR, SP PC, CPSR	R14..R0, PC, CPSR
10001	FIQ	R7..R0, LR_fiq, SP_fiq PC, CPSR, SPSR_fiq	R7..R0, R14_fiq..R8_fiq, PC, CPSR, SPSR_fiq
10010	IRQ	R7..R0, LR_irq, SP_irq PC, CPSR, SPSR_irq	R12..R0, R14_irq..R13_irq, PC, CPSR, SPSR_irq
10011	Supervisor	R7..R0, LR_svc, SP_svc, PC, CPSR, SPSR_svc	R12..R0, R14_svc..R13_svc, PC, CPSR, SPSR_svc
10111	Abort	R7..R0, LR_abt, SP_abt, PC, CPSR, SPSR_abt	R12..R0, R14_abt..R13_abt, PC, CPSR, SPSR_abt
11011	Undefined	R7..R0 LR_und, SP_und, PC, CPSR, SPSR_und	R12..R0, R14_und..R13_und, PC, CPSR
11111	System	R7..R0, LR, SP PC, CPSR	R14..R0, PC, CPSR

Table 3-1: PSR mode bit values

Reserved bits

The remaining bits in the PSRs are *reserved*. When changing a PSR's flag or control bits, you must ensure that these unused bits are not altered. Also, your program should not rely on their containing specific values, since in future processors they may read as one or zero.

3.7 Exceptions

Exceptions arise whenever the normal flow of a program has to be halted temporarily, for example to service an interrupt from a peripheral.

Before an exception can be handled, the current processor state must be preserved so that the original program can resume when the handler routine has finished.

Several exceptions may arise at the same time. If this happens, they are dealt with in a fixed order - see **3.7.10 Exception priorities** on page 3-14.

3.7.1 Action on entering an exception

When handling an exception, the ARM740T:

- 1 Preserves the address of the next instruction in the appropriate Link Register.
 - If the exception has been entered from ARM state, the address of the next instruction is copied into the Link Register (that is, current PC + 4 or PC + 8 depending on the exception. See **Table 3-2: Exception entry/exit** on page 3-12 for details).
 - If the exception has been entered from THUMB state, the value written into the Link Register is the current PC offset by a value such that the program resumes from the correct place on return from the exception. This means that the exception handler need not determine which state the exception was entered from.

For example, in the case of SWI, the instruction:

```
MOVS PC, R14_svc
```

always returns to the next instruction regardless of whether the SWI was executed in ARM or THUMB state.

- 2 Copies the CPSR into the appropriate SPSR.
- 3 Forces the CPSR mode bits to a value which depends on the exception.
- 4 Forces the PC to fetch the next instruction from the relevant exception vector.

ARM740T may also set the interrupt disable flags to prevent otherwise unmanageable nestings of exceptions.

Note *If the processor is in THUMB state when an exception occurs, it automatically switches into ARM state when the PC is loaded with the exception vector address.*

3.7.2 Action on leaving an exception

On completion, the exception handler:

- 1 Moves the Link Register, minus an offset where appropriate, to the PC. The offset varies depending on the type of exception.
- 2 Copies the SPSR back to the CPSR.
- 3 Clears the interrupt disable flags, if they were set on entry.

Note *An explicit switch back to THUMB state is never needed, since restoring the CPSR from the SPSR automatically sets the T bit to the value it held immediately prior to the exception.*

Programmer's Model

3.7.3 Exception entry/exit summary

Table 3-2: Exception entry/exit summarizes the PC value preserved in the relevant R14 on exception entry, and the recommended instruction for exiting the exception handler.

	Return Instruction	Previous State		Notes
		ARM R14_x	THUMB R14_x	
BL	MOV PC, R14	PC + 4	PC + 2	1
SWI	MOVS PC, R14_svc	PC + 4	PC + 2	1
UDEF	MOVS PC, R14_und	PC + 4	PC + 2	1
FIQ	SUBS PC, R14_fiq, #4	PC + 4	PC + 4	2
IRQ	SUBS PC, R14_irq, #4	PC + 4	PC + 4	2
PABT	SUBS PC, R14_abt, #4	PC + 4	PC + 4	1
DABT	SUBS PC, R14_abt, #8	PC + 8	PC + 8	3
RESET	NA	-	-	4

Table 3-2: Exception entry/exit

Notes

- 1 PC is the address of the BL/SWI/Undefined Instruction fetch which had the prefetch abort.
- 2 PC is the address of the instruction which did not get executed since the FIQ or IRQ took priority.
- 3 PC is the address of the Load or Store instruction which generated the data abort.
- 4 The value saved in R14_svc upon reset is unpredictable.

3.7.4 FIQ

The FIQ (Fast Interrupt Request) exception is designed to support a data transfer or channel process, and in ARM state has sufficient private registers to remove the need for register saving (thus minimising the overhead of context switching).

FIQ is externally generated by taking the **nFIQ** input LOW. **nFIQ** and **nIRQ** are considered asynchronous, and a cycle delay for synchronization is incurred before the interrupt can affect the processor flow.

Irrespective of whether the exception was entered from ARM or THUMB state, a FIQ handler should leave the interrupt by executing:

```
SUBS PC, R14_fiq, #4
```

FIQ may be disabled by setting the CPSR's F flag (but note that this is not possible from User mode). If the F flag is clear, ARM740T checks for a LOW level on the output of the FIQ synchronizer at the end of each instruction.

3.7.5 IRQ

The IRQ (Interrupt Request) exception is a normal interrupt caused by a LOW level on the **nIRQ** input. IRQ has a lower priority than FIQ and is masked out when a FIQ sequence is entered. It may be disabled at any time by setting the I bit in the CPSR, though this can only be done from a privileged (non-User) mode.

Irrespective of whether the exception was entered from ARM or THUMB state, an IRQ handler should return from the interrupt by executing:

```
SUBS PC, R14_irq, #4
```

3.7.6 Abort

An abort indicates that the current memory access cannot be completed. It can be signalled either by the Protection unit, or by the external **BERROR** input. ARM740T checks for the abort exception during memory access cycles.

There are two types of abort:

Prefetch abort	occurs during an instruction prefetch.
Data abort	occurs during a data access.

Prefetch abort

If a prefetch abort occurs, the prefetched instruction is marked as invalid, but the exception is not taken until the instruction reaches the head of the pipeline. If the instruction is not executed—for example because a branch occurs while it is in the pipeline—the abort does not take place.

Data abort

If a data abort occurs, the action taken depends on the instruction type:

- 1 Single data transfer instructions (LDR, STR) writeback-modified base registers: the Abort handler must be aware of this.
- 2 The swap instruction (SWP) is aborted as though it had not been executed.
- 3 Block data transfer instructions complete (LDM, STM). If writeback is set, the base is updated. If the instruction would have overwritten the base with data (ie. it has the base in the transfer list), the overwriting is prevented. All register overwriting is prevented after an abort is indicated, which means in particular that R15 (always the last register to be transferred) is preserved in an aborted LDM instruction.

Returning from an abort

After fixing the reason for the abort, the handler should execute the following irrespective of the state (ARM or THUMB), to restore both the PC and the CPSR, and retry the aborted instruction:

```
SUBS PC, R14_abt, #4 for a prefetch abort, or  
SUBS PC, R14_abt, #8 for a data abort
```

Note *Restrictions on the use of the external abort signal. are given in 7.5 External Aborts on page 7-11.*

3.7.7 Software interrupt

The software interrupt instruction (SWI) is used for entering Supervisor mode, usually to request a particular supervisor function. A SWI handler should return by executing the following irrespective of the state (ARM or THUMB):

```
MOV PC, R14_svc
```

This restores the PC and CPSR, and returns to the instruction following the SWI.

Programmer's Model

3.7.8 Undefined instruction

When ARM740T comes across an instruction which it cannot handle, it takes the undefined instruction trap. This mechanism may be used to extend either the THUMB or ARM instruction set by software emulation.

After emulating the failed instruction, the trap handler should execute the following irrespective of the state (ARM or THUMB), to restore the CPSR and return to the instruction following the undefined instruction:

```
MOVS PC, R14_und
```

3.7.9 Exception vectors

The following table shows the exception vector addresses.

Address	Exception	Mode on entry
0x00000000	Reset	Supervisor
0x00000004	Undefined instruction	Undefined
0x00000008	Software interrupt	Supervisor
0x0000000C	Abort (prefetch)	Abort
0x00000010	Abort (data)	Abort
0x00000014	Reserved	Reserved
0x00000018	IRQ	IRQ
0x0000001C	FIQ	FIQ

Table 3-3: Exception vector addresses

3.7.10 Exception priorities

When multiple exceptions arise at the same time, a fixed priority system determines the order in which they are handled:

- 1 Reset (Highest priority)
- 2 Data abort
- 3 FIQ
- 4 IRQ
- 5 Prefetch abort
- 6 Undefined Instruction, Software Interrupt. (Lowest priority)

Not all exceptions can occur at once

Undefined Instruction and Software Interrupt are mutually exclusive, as they each correspond to particular (non-overlapping) decodings of the current instruction.

If a data abort occurs at the same time as a FIQ, and FIQs are enabled (that is, the CPSR's F flag is clear), ARM740T enters the data abort handler and then immediately proceeds to the FIQ vector. A normal return from FIQ makes the data abort handler resume execution.

Placing data abort at a higher priority than FIQ is necessary to ensure that the transfer error does not escape detection. The time for this exception entry should be added to worst-case FIQ latency calculations.

3.8 Reset

When the **BnRES** signal goes LOW, ARM740T:

- 1 Abandons the executing instruction.
- 2 Flushes the Cache.
- 3 Disables the Write Buffer, Cache and Memory Management Unit.
- 4 Resets the Process Identifier.
- 5 Continues to fetch instructions from incrementing word addresses.

When **BnRES** goes HIGH again, ARM740T:

- 1 Overwrites R14_svc and SPSR_svc by copying the current values of the PC and CPSR into them. The value of the saved PC and SPSR is not defined.
- 2 Forces M[4:0] to 10011 (Supervisor mode), sets the I and F bits in the CPSR, and clears the CPSR's T bit.
- 3 Forces the PC to fetch the next instruction from address 0x00.
- 4 Resumes execution in ARM state.

4

Configuration

This chapter describes the ARM740T configuration.

4.1	Overview	4-2
4.2	Internal Coprocessor Instructions	4-3
4.3	Registers	4-4

Configuration

4.1 Overview

The operation and configuration of ARM740T is controlled via coprocessor 15 (CP15). Coprocessor instructions manipulate a number of on-chip registers which control the configuration of the following:

- Cache
- write buffer
- Protection Unit
- a number of other configuration options.

4.1.1 Compatibility

To ensure backwards compatibility of future CPUs, all reserved or unused bits in registers and coprocessor instructions should be programmed to '0'.

Invalid registers must not be read/written.

Note *The gray areas in the register diagrams are reserved and should be programmed 0 for future compatibility.*

4.2 Internal Coprocessor Instructions

The on-chip configuration registers may be read using MRC instructions and written using MCR instructions. These operations are only allowed in non-user modes and the undefined instruction trap is taken if accesses are attempted in user mode.

Note *The CP15 register map may change in later ARM processors. We strongly recommend you structure software such that any code accessing coprocessor 15 is contained in a single module. It can then be updated easily.*

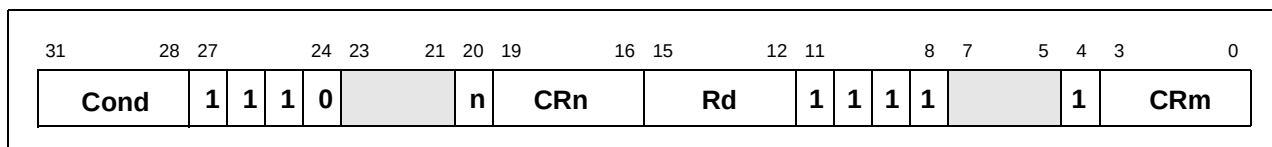


Figure 4-1: Format of internal coprocessor instructions MRC and MCR

where:

Cond	ARM condition codes
Crn	CP15 Source/Destination Register
Crm	CP15 Operand Register
Rd	ARM Register
n	1 = MRC register read 0 = MCR register write

The CRn field is normally used to determine which configuration register is being accessed.

Configuration

4.3 Registers

The configuration registers are accessed by CPRT instructions to CP15 with the processor in privileged mode.

Only some CRn registers are valid:

- an access to an invalid register causes neither the access nor an undefined instruction trap, and therefore should never be carried out
- an access to any of the registers [8:15] causes the undefined instruction trap to be taken

Register	Register Reads	Register Writes
0	ID Register	Reserved
1	Control	Control
2	Cacheable	Cacheable
3	Write Buffer Control	Write Buffer Control
4	Reserved	Reserved
5	Protection	Protection
6	Memory Area Definition	Memory Area Definition
7	Reserved	Flush unlocked Cache banks
8-15	Reserved	Reserved

Table 4-1: System control registers

4.3.1 Register 0: ID

Register 0 is a read-only identity register that returns the ARM code for this core. This code is 0x4180740x.

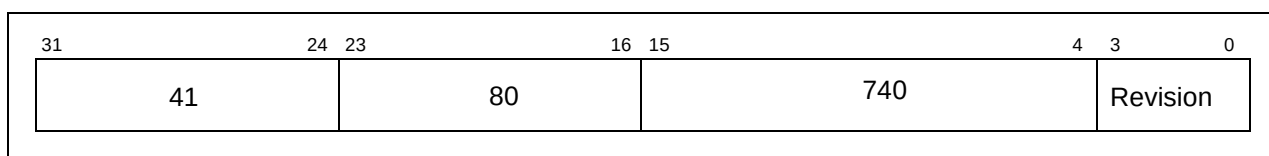


Figure 4-2: ID Code register

8KB cache variant

For the 8KB cache variant the id code is 0x41807400.

4KB cache variant

For the 4KB cache variant the ID code is 0x41817400.

4.3.2 Register 1: Control

Register 1 contains the control bits. All bits in this register are forced LOW by reset.

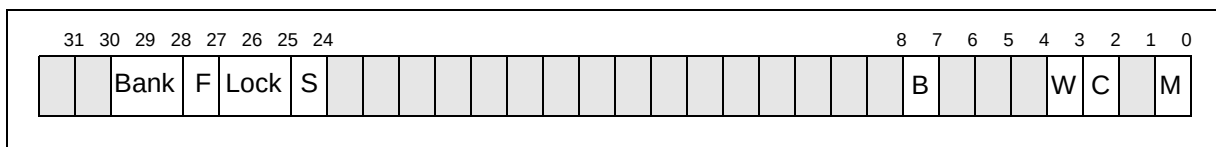


Figure 4-3: Control register

M Bit 0	Protection Unit Enable/disable
	0 on-chip Protection Unit turned off
	1 on-chip Protection Unit turned on.
C Bit 2	Cache enable/disable
	0 Cache turned off
	1 Cache turned on
W Bit 3	Write buffer enable/disable
	0 Write buffer turned off
	1 Write buffer turned on
B Bit 7	Big/little-endian
	0 little-endian operation
	1 big-endian operation
S Bit 24	Split Instruction Data Mode
	This bit controls the operating mode of the ARM740T Cache. Refer to 5.3.3 Split instruction data operation on page 5-5.
Lock Bits [26:25]	Lock Cache Lockdown control register
	This bit controls the ARM740T Cache. Refer to 5.3.2 Partially locked operation on page 5-5.
F Bit 27	Load Mode
	This bit controls the ARM740T Cache. Refer to 5.3.2 Partially locked operation on page 5-5.
Bank Bits [29:28]	Cache Bank select register
	These bits controls the ARM740T Cache. Refer to 5.3.2 Partially locked operation on page 5-5.

4.3.3 Register 2: Cacheable

Register 2 holds the current values of the Cacheable bit. See **7.2 Protection Unit Registers** on page 7-3 for a description of the operation of the Protection Unit.

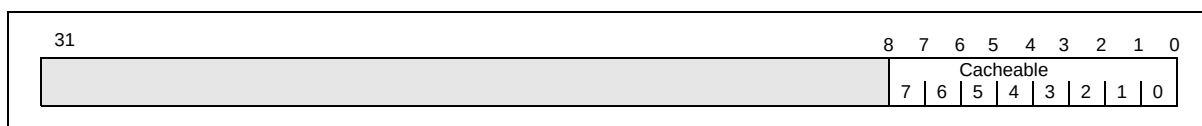


Figure 4-4: Cacheable register

5

Cache

This chapter describes the cache.

5.1	Overview	5-2
5.2	Control Registers	5-4
5.3	Operating Modes	5-5
5.4	Cache Operation	5-7

Cache

5.1 Overview

The ARM740T can incorporate either an 8KB or 4KB general purpose cache. Both variants are functionally equivalent.

The cache:

- is physically addressed
- is 4-way set associative
- is write through
- has four words and a valid flag per line
- uses a random replacement algorithm
- is filled by line

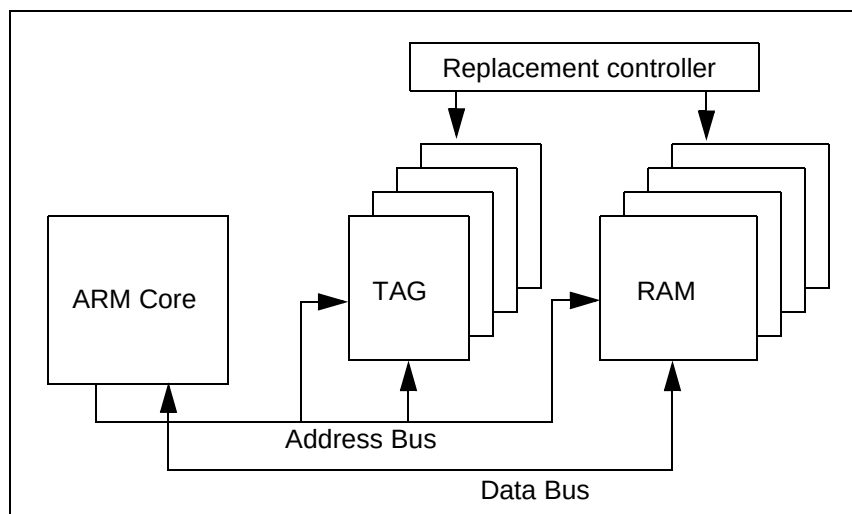


Figure 5-1: Cache architecture

Operating modes

Various operating modes are provided, to allow the cache to be tailored to the application:

- mixed instruction data
- partially locked
- split instruction data

Cache operation

The cache is enabled or disabled and configured via the ARM740T control register.

The operation of the cache is further controlled by the cacheable function of the protection unit. The protection unit must always be enabled if the cache is enabled or the behavior is undefined. The two functions may be enabled simultaneously, with a single write to the control register.

Replacement algorithm

The replacement algorithm of the cache is random. The various operating modes all use random allocation, though the replacement algorithm is constrained.

In all cases, the options only affect cache replacements. The complete cache is always searched for an address, and if the address is found, the data is used or updated. This ensures that the cache is guaranteed to be self-consistent, and coherent with external memory.

5.1.1 The 8KB variant

The 8KB variant has 128 lines per bank (set). The Tag field is 21 bits and the line index is 7 bits as shown below:

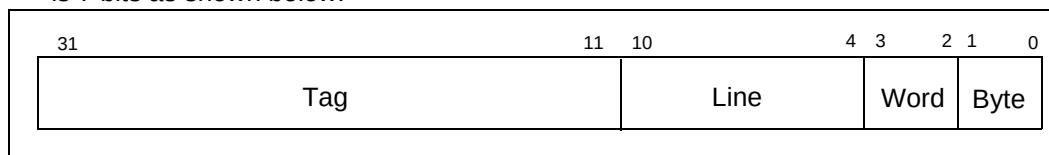


Figure 5-2: 8KB variant

5.1.2 The 4KB variant

The 4KB variant has 64 lines per bank (set). The Tag field is 22 bits and the line index is 6 bits as shown below:

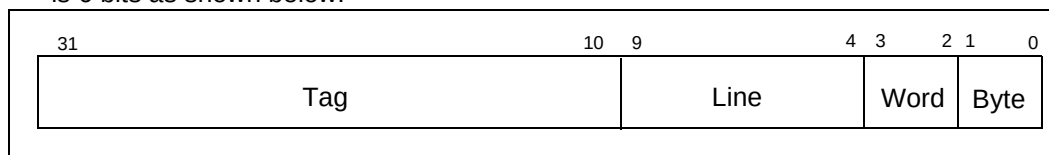


Figure 5-3: 4KB variant

5.1.3 Read-Lock-Write

The IDC treats the Read-Locked-Write instruction as a special case.

The read phase always forces a read of external memory, regardless of whether the data is contained in the cache.

The write phase is treated as a normal write operation (and if the data is already in the cache, the cache will be updated).

Externally the two phases are flagged as indivisible by asserting the **BLOK** signal.

5.1.4 Reset

The IDC is automatically disabled and flushed on **BnRES**. Once enabled, cacheable read accesses place lines in the cache.

Cache

5.2 Control Registers

The cache is controlled by the following bits in the control register.

bank[1:0]	These bits select the bank to be loaded when the F bit is set. Note: The cache banks are always locked starting from bank 0, so the order of loading should be 0, 1, 2. While bank 3 can be loaded, there is no mechanism for locking all four banks of cache.
C	Cache enable bit. The cache is filled when a cacheable (instruction or data) fetch is performed. The cache is loaded by a line fetch of four words.
F	This bit forces all linefetches to occur to the bank selected by bank[1:0]. When this bit is set, all instruction fetches are forced to be 'Uncacheable'—data fetches are still subject to the cacheable mapping in the protection unit.
lock[1:0]	These bits are used to set the number of banks locked, and when in Split Instruction Data mode, is it also used to program the split. The effect of lock[1:0] when used to lock banks of the cache is shown in Table 5-1: Cache banks locked by lock[1:0] .

lock[1:0]	Bank 3	Bank 2	Bank 1	Bank 0	Description
00	Cache	Cache	Cache	Cache	No Banks Locked
01	Cache	Cache	Cache	Locked	1 Bank Locked
10	Cache	Cache	Locked	Locked	2 Banks Locked
11	Cache	Locked	Locked	Locked	3 Banks Locked

Table 5-1: Cache banks locked by lock[1:0]

S	This is the split instruction/data bit. When this bit is set, the Cache is configured according to the value of the lock[1:0] bits. It is illegal to have F and S set simultaneously. The effects of the lock[1:0] bits when in split instruction/data mode is shown in Table 5-2: Bank allocation in Split Instruction / Data mode on page 5-5.
---	---

See **4.3.2 Register 1: Control** on page 4-5 for a full description of the configuration register.

5.3 Operating Modes

The following operating modes are provided, to allow the cache to be tailored to the application:

- mixed instruction data
- partially locked
- split instruction data

5.3.1 Mixed instruction data operation

This is the standard mode of operation of the cache. In this mode, the cache functions as a standard mixed instruction and data cache. Lines fetched into the cache are randomly placed into one of the cache banks.

5.3.2 Partially locked operation

The ARM740T cache allows critical code and data to be locked into the cache to ensure predictable high performance.

Locking code or data into the cache

To lock code or data into the cache:

- 1 Select the bank to be loaded using the bank[1:0] register, and set the F bit to 1. Cache banks are always locked starting from bank 0, hence should be loaded and locked in the order 0, 1, 2.
- 2 Perform a cache flush operation. This is necessary to ensure that the required instructions and data are loaded into the selected cache bank. If this is not performed, they may be elsewhere in the cache, and therefore are not loaded into the selected bank.
- 3 Load the instructions or data to be locked into the cache either using LDM or LDR instructions, one per line. While in load mode, all instruction fetches are uncacheable.
- 4 Set the F bit to zero.
- 5 Set the number of banks to be locked into the lock[1:0] register.

Once the lock register is set, the replacement algorithm is prevented from replacing in the locked banks. This has the effect of reducing the associativity of the cache to the number of banks remaining as cache.

5.3.3 Split instruction data operation

As a further option, the ARM740T cache can be operated in split instruction data mode. This forces instructions and data to be cached in separate banks of the cache. This can be used to improve performance where a small code set is processing a large data set. The split nature of the cache prevents the data from replacing the cached instructions. The banks of the cache are used as shown in **Table 5-2: Bank allocation in Split Instruction / Data mode** on page 5-5.

lock[1:0]	Bank 3	Bank 2	Bank 1	Bank 0	Description
00	-	-	-	-	Reserved
01	Data	Data	Data	Instr.	1 Bank Instruction, 2 Banks Data
10	Data	Data	Instr.	Instr.	2 Banks Instruction, 2 Banks Data
11	Data	Instr.	Instr.	Instr.	3 Banks Instruction, 1 Bank Data

Table 5-2: Bank allocation in Split Instruction / Data mode

Cache

It is not necessary to flush the cache before enabling split instruction / data mode. The complete cache is searched, regardless of the split selected.

- 1 Set the S bit.
- 2 Select the required split using the lock[1:0] register.

If required, this mechanism can be used to 'snapshot' contents of the instruction banks, and lock them into the cache. The required sequence of operations is as follows:

- 1 Set the S bit to 1, and select the required split using the lock[1:0] register.
- 2 Flush the cache to ensure that the code is loaded into the instruction banks.
- 3 Execute the required code fragment.
- 4 Set the S bit to 0, leaving the same value in the lock[1:0] register.

In all cases, when operating in split instruction / data mode, the associativity of each section of the cache is equal to the number of banks allocated to it.

Notes

It is illegal to simultaneously have the S bit and the F bit set.

It is illegal to have the S bit set, with a value of 00 in the lock[1:0] register.

5.4 Cache Operation

The cache is always searched regardless of whether it is enabled. If an address hits, then the data will be read or written. So when the cache is disabled it should also be flushed.

Cacheable reads	A linefetch of four words is performed when a 'cache-miss' occurs in a cacheable area of memory. This is placed in the cache according to the current mode of operation.
Uncacheable reads	An external memory access is performed and the cache is not written.
Writes	All writes updates the data in the cache if present, and are written through to the main memory.

5.4.1 Cacheable bit

The protection unit uses the appropriate cacheable bit in the cacheable register to determine whether data being read may be placed in the IDC and used for subsequent read operations.

Typically, main memory is marked as cacheable to improve system performance, and I/O space as non-cacheable to stop the data being stored in ARM740T's cache.

For example, if the processor is polling a hardware flag in I/O space, it is important that the processor is forced to read data from the external peripheral, and not a copy of the initial data held in the cache. See **Chapter 7, Protection Unit** for more details.

5.4.2 Software IDC flush

All unlocked banks of the Cache may be marked as invalid by writing to the ARM740T IDC Flush Register (Register 7), see **4.3.8 Register 7: IDC Flush** on page 4-6. The cache is flushed immediately the register is written, but note that the two instruction fetches following may come from the cache before the register is written.

6

Write Buffer

This chapter describes the Write Buffer.

- 6.1 Overview of the Write Buffer
- 6.2 Write Buffer Operation

6-2
6-3



Write Buffer

6.1 Overview of the Write Buffer

The ARM740T write buffer is provided to improve system performance. It can buffer up to eight words of data, and four independent addresses.

The write buffer may be enabled or disabled via the W bit (bit 3) in the ARM740T Control Register, and the buffer is disabled and flushed on reset.

For a write to use the write buffer, both the W bit in the Control Register, and the appropriate B bit in the Bufferable Register must be set.

It is not possible to abort buffered writes externally.

6.1.1 Bufferable bit

The operation of the write buffer is further controlled by the bufferable function of the protection unit. If the write buffer is enabled the protection unit must also be enabled. The two functions may however be enabled simultaneously, with a single write to the Control Register.

This bit controls whether a write operation may or may not use the write buffer. Typically, main memory is bufferable and I/O space unbufferable. The bufferable bit can be configured for each area of memory, see **7.2.3 Bufferable register** on page 7-4.

6.1.2 Bufferable write

If the write buffer is enabled and the processor performs a write to a bufferable area, the data is placed in the write buffer at **BCLK** speeds and the CPU continues execution. The write buffer then performs the external write in parallel.

If, however, the write buffer is full (either because there are already eight words of data in the buffer, or because there is no slot for the new address), the processor is stalled until there is sufficient space in the buffer.

- A single write requires one address slot and one data slot in the write buffer
- a sequential write of n words requires one address slot and n data slots.

The total of eight data slots in the buffer may be used as required. For example, there could be three non-sequential writes and one sequential write of five words in the buffer, and the processor could continue as normal: a fifth write or a sixth word in the forth write would stall the processor until the first write had completed.

6.1.3 Unbufferable writes

If the write buffer is disabled or the CPU performs a write to an unbufferable area, the processor is stalled until the write buffer empties and the write completes externally, which may require synchronisation and several external clock cycles.

6.1.4 Read-lock-write

The write phase of a read-lock-write sequence is treated as an unbuffered write, even if it is marked as buffered.

6.2 Write Buffer Operation

When the CPU performs a write operation, the *Bufferable* bit for that address is inspected and the state of the B bit determines the subsequent action. If the write buffer is disabled via the ARM740T Control Register, buffered writes are treated in the same way as unbuffered writes.

6.2.1 To enable the write buffer

To enable the write buffer:

- 1 Ensure the Protection Unit is enabled by setting bit 0 in the Control Register.
- 2 Enable the write buffer by setting bit 3 in the Control Register.

The Protection Unit and write buffer may be enabled simultaneously with a single write to the Control Register.

6.2.2 To disable the write buffer

To disable the write buffer, clear bit 3 in the Control Register.

Note *Any writes already in the write buffer complete normally.*

7

Protection Unit

This chapter describes the Protection Unit.

7.1	Overview	7-2
7.2	Protection Unit Registers	7-3
7.3	Protection Unit Operation	7-7
7.4	Support for Overlapping Regions	7-9
7.5	External Aborts	7-11
7.6	Interaction of the Protection Unit, Cache and Write Buffer	7-12

Protection Unit

7.1 Overview

The Protection Unit performs two primary functions by containing a description of the properties of areas of memory in the memory map:

- controlling the cache and write buffer
- controlling memory access permissions

7.1.1 Controlling individual memory areas

The protection unit provides individual control for eight areas of memory (numbered 0 to 7). For each area the following registers can be programmed:

- Cachable
- Bufferable
- Basic Protection
- Size
- Base Address

This allows the memory architecture of the system to be described in an easily programmable but flexible manner.

7.2 Protection Unit Registers

The ARM740T provides several registers which control the operation of the Protection Unit. The format of these registers is shown in **Table 7-1: System control registers**.

Register	Register Reads	Register Writes
0	ID Register	Reserved
1	Configuration	Configuration
2	Cacheable	Cacheable
3	Bufferable	Bufferable
4	Reserved	Reserved
5	Protection	Protection
6	Memory Area Definition	Memory Area Definition
7	Reserved	Flush unlocked Cache banks
8–15	Reserved	Reserved

Table 7-1: System control registers

For a complete description of the Control Coprocessor see **Chapter 4, Configuration**.

7.2.1 Control register

The Configuration register contains the protection enable bit M, which is shown in **Figure 7-1: Control register**. On reset, this bit is set to zero, disabling the protection mechanisms. This allows full access to all of memory, and all accesses are then uncacheable and unbufferable.

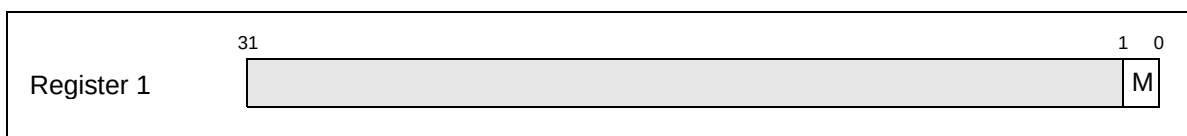


Figure 7-1: Control register

Note Other bits in the configuration register are also used for other functions. For a full description of the configuration register see **4.3.2 Register 1: Control** on page 4-5.

7.2.2 Cacheable register

The Cacheable register sets the cacheable bit for each of the eight areas of memory.

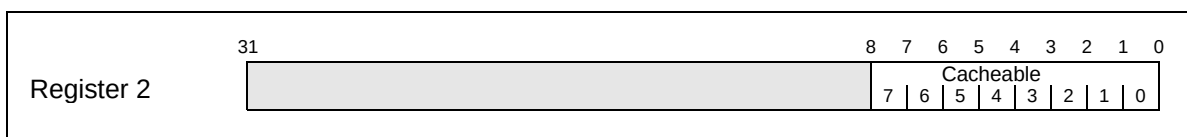


Figure 7-2: Cacheable register

The cacheable bit determines if a linefetch should be performed for an access to a given area of memory. The cache is always searched regardless of the state of this bit, and if the required address is found the copy of the data in the cache will be used.

Protection Unit

On reset all areas are marked as uncacheable.
Typically, main memory is marked as cacheable to provide maximum performance and peripherals are marked as uncacheable.

7.2.3 Bufferable register

The Bufferable register sets the bufferable bit for each of the eight areas of memory.

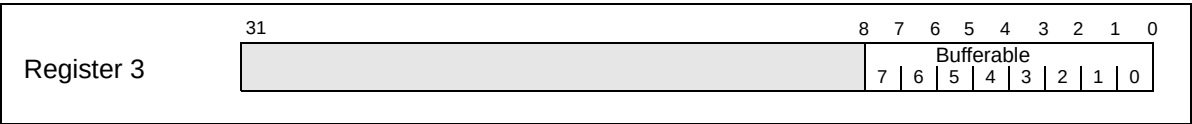


Figure 7-3: Bufferable register

The bufferable bit indicates that data at this address is written through the write buffer (if the write buffer is enabled). On reset, all areas are marked as unbufferable.

Note *The meaning of the cacheable and bufferable bits may change in later ARM processors. It is strongly recommended that you structure software so that code which manipulates the protection unit is contained in a single module. It can then be updated easily when you port it to a different ARM processor.*

7.2.4 Protection register

The Protection register controls the access permissions for the eight areas of memory.

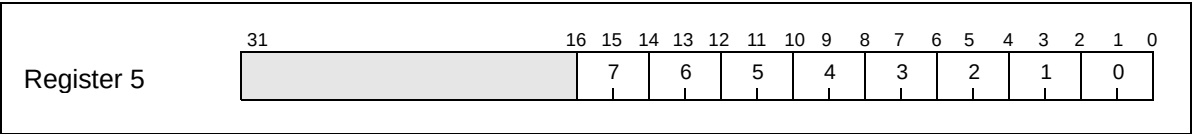


Figure 7-4: Protection register

For each area of memory, the access permissions are controlled by the value in the Protection register. These control access as shown in **Table 7-2: Access permissions**:

Value	Supervisor	User
00	No Access	No Access
01	Read / Write	No Access
10	Read / Write	Read Only
11	Read / Write	Read / Write

Table 7-2: Access permissions



7.2.5 Area registers

The Area registers control the parameters of the areas of memory controlled by the Protection Unit. These registers differ from the other CP15 registers in how the areas of memory are addressed. Rather than using separate bit-fields for each region of memory, one register is used for each area indexed by the coprocessor operand parameter in the instruction.

The number of the area of memory to be accessed should be placed in the CP15 operand field of the instruction. See **Figure 7-5: Format of internal coprocessor instructions MRC and MCR** on page 7-5.

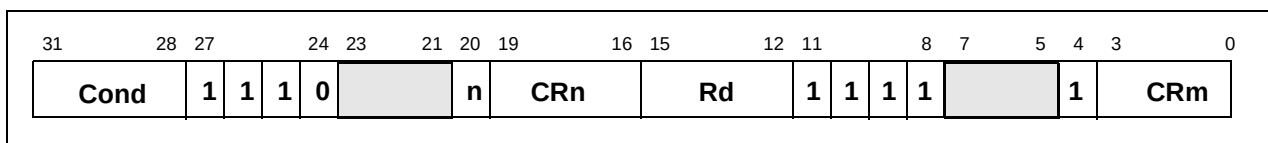


Figure 7-5: Format of internal coprocessor instructions MRC and MCR

where:

Cond	is an ARM condition code
CRn	is the CP15 Source/Destination Register (equal to 6 for the Area register)
CRm	is the CP15 Operand Register, set to the area to be accessed
Rd	is an ARM Register
n	1 = MRC register read 0 = MCR register write

Each Area register contains 3 fields to describe the location of the area of memory.

- The Base address of the Area
- The Size of the Area
- The enable bit, E

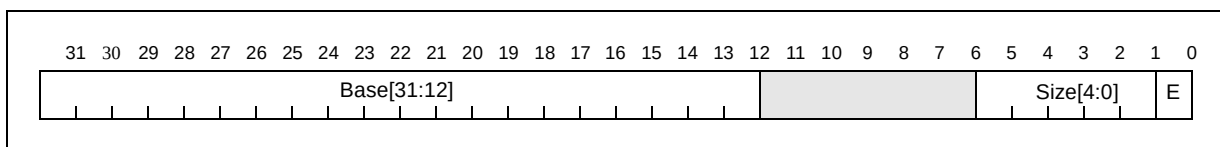


Figure 7-6: Size register

The enable bit E determines if a given area is active. If this bit is set to zero, the area is disabled.

The value in size[4:0] determines the size of a given area of memory, as shown in **Table 7-3: Area sizes**:

Size[4:0]	Area	Size[4:0]	Area
0b01011	4KB	0b10110	8MB
0b01100	8KB	0b10111	16MB

Table 7-3: Area sizes

Protection Unit

Size[4:0]	Area	Size[4:0]	Area
0b01101	16KB	0b11000	32MB
0b01110	32KB	0b11001	64MB
0b01111	64KB	0b11010	128MB
0b10000	128KB	0b11011	256MB
0b10001	256KB	0b11100	512MB
0b10010	512KB	0b11101	1GB
0b10011	1MB	0b11110	2GB
0b10100	2MB	0b11111	4GB
0b10101	4MB		

Table 7-3: Area sizes (Continued)

Base address

The base address of each area must be aligned with respect to the size of that area.

For example, if a region size is set to 16KB, then 0x8000 is a legal address for the region to start at, and 0x5000 is not legal.

The finest resolution for setting the location of a section is 4KB, as determined by the smallest region size setting. The behavior of the protection unit is undefined if this requirement is not met.

Accessing the area register

This register is accessed using MCR and MRC instructions as follows:

- To write the descriptor for an area of memory:

MCR p15, 0, Rd, c6, CRm, 0

where:

CRm is the area of memory to be defined

Rd is the ARM register containing the value to be written into the area register.

- To read back the descriptor:

MRC p15, 0, Rd, c6, CRm, 0

where:

CRm is the area of memory to be read

Rd is the ARM register where the descriptor is placed

7.3 Protection Unit Operation

The Protection Unit works by comparing the address generated by the ARM against the parameters of the eight areas of memory. This can cause one of three results:

No area hits	The access is aborted
One area hit	The properties of this area are applied to the access
Multiple areas hit	The properties of the highest priority area is applied to the access.

This is illustrated diagrammatically in **Figure 7-7: Protection Unit operation**.

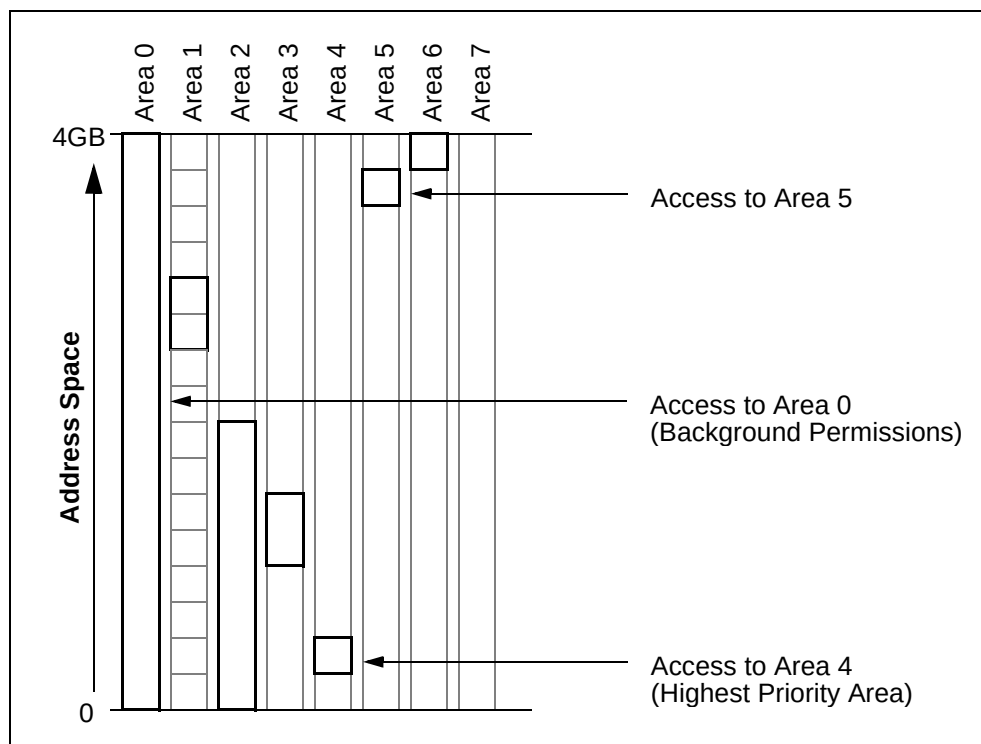


Figure 7-7: Protection Unit operation

7.3.1 Memory area properties

Each area of memory is defined in terms of the following properties:

- size
- base address
- access permissions
- bufferable bit
- cacheable bit

The Base address of the area of memory must be a multiple of the size of the area.

When an address matches multiple areas of memory, the properties of the highest priority area of memory are used. The priority ordering of the areas is fixed such that:

area 7	has the HIGHEST priority
area 0	has the LOWEST priority

Protection Unit

The bufferable and cacheable bits for the selected area of memory are used to determine if the cache and write buffer should be used (if enabled).

Property	Effect if Set
Bufferable	If the Access is a write, the write buffer will be used.
Cacheable	If the Access is a read, a cache linefill will be performed if the required word is not in the cache.

Table 7-4: Cacheable and bufferable properties

7.3.2 Access permissions

The access permission bits are checked against the access type. This decoding is detailed in **Table 7-2: Access permissions** on page 7-4.

- If the access is permitted, it continues.
- If the access is prohibited, the ARM is aborted and the access does not occur on the external bus.

7.3.3 Protection failures and external accesses

If the protection unit detects an access violation, it does so before the external memory access takes place, and it therefore inhibits the access. External aborts do not necessarily inhibit the external access, as described in **7.5 External Aborts** on page 7-11.

An internally aborting access may cause the address on the external address bus to change, even though the external bus cycle has been cancelled. No memory access is performed to this address.

7.3.4 Reset

The Protection Unit is disabled on **BnRES**. Before it is enabled, all the Protection Unit registers must be programmed. If this is not observed, unpredictable behaviour will result.

7.4 Support for Overlapping Regions

Overlapping regions can be used to allow greater flexibility over how logical memory regions are mapped into physical memory devices.

For example, consider the case where the system has 4KB of supervisor code and 28KB of user code, both of which must be mapped into a 32KB RAM.

If overlapping memory is not supported, four regions would have to be used to achieve this:

- one 4KB region for the supervisor code
- one 32KB region
- one 16KB region
- one 4KB region for the user code

This is as shown below in **Figure 7-8: Use of overlapping memory regions**.

Overlapping supervisor and code regions

If the supervisor and user code regions can be overlapped, this can be achieved using only two regions:

- one 4KB region for the supervisor code
- one 32KB region for the user code, as shown in **Figure 7-8: Use of overlapping memory regions**

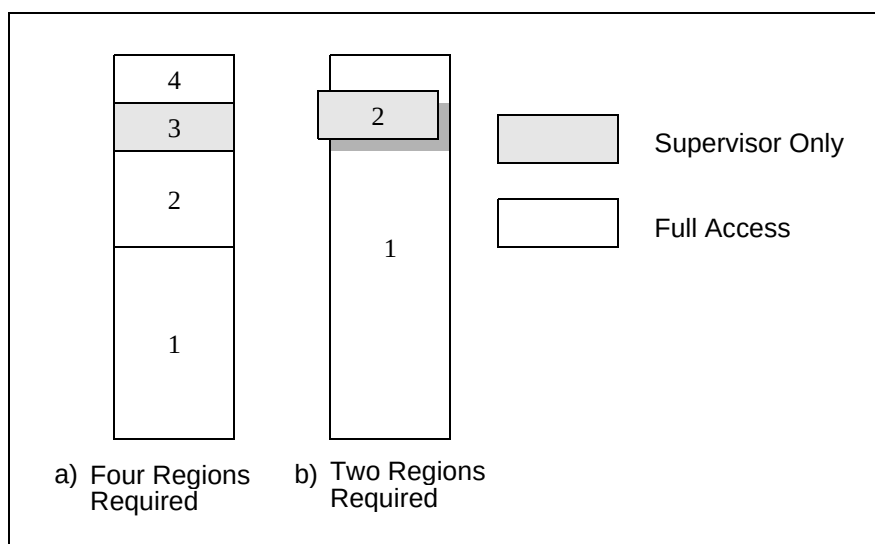


Figure 7-8: Use of overlapping memory regions

In this example, the supervisor code could be placed in Region 2, and the user code in Region 1. This ensures that the supervisor mapping takes precedence over the less strict user mapping.

Protection Unit

7.4.1 Undefined address space

The mechanism for overlapping segments can be used to allow the default protection for otherwise unmapped memory to be programmed. If the memory regions do not completely fill the 4GB address space of the ARM7TDMI, there are 'holes' in the address map. By configuring Region 0 (the lowest priority Region) to be 4GB in size, you can program what happens if an access is made to a hole.

For example, the attributes could be set to full access or no access. Alternatively, you may chose to ignore the holes, and any access to an area of memory not described by the protection unit results in an abort.

7.5 External Aborts

In addition to the aborts generated by the protection unit, ARM740T has an external abort input **BERROR** that may be used to flag an error on an external memory access. However, not all accesses can be aborted in this way, so this input must be used with great care.

7.5.1 Restrictions

The following accesses may be aborted and restarted safely.

- reads
- unbuffered writes
- read-lock-write sequence

If any of these are aborted, the external access ceases on the next cycle. In the case of a read-lock-write sequence in which the read aborts, the write does not happen.

7.5.2 Cacheable reads (linefetches)

A linefetch may be safely aborted on any word in the transfer.

- If an abort occurs during the linefetch, the cache is purged, so it does not contain invalid data.
- If the abort happens on a word that has been requested by the ARM740T, it is aborted, otherwise the cache line is purged but program flow is *not* interrupted.

The line is therefore purged under all circumstances.

7.5.3 Buffered writes

Buffered writes cannot be externally aborted. Therefore, the system should be configured so that it does not buffer writes to areas of memory that are capable of flagging an external abort.

Note *Areas of memory that can generate an external abort on a location that has previously been read successfully must not be marked as cacheable or unbufferable. If all writes to an area of memory abort, it is recommended that you mark it as read only in the Protection Unit, otherwise mark it as uncacheable and unbufferable.*

Protection Unit

7.6 Interaction of the Protection Unit, Cache and Write Buffer

The Protection Unit, cache and write buffer may be enabled and disabled independently. However, in order for the write buffer or the cache to be enabled the Protection Unit must also be enabled. There are no hardware interlocks on these restrictions, so invalid combinations cause undefined results.

Protection unit	Cache	Write buffer
off	off	off
on	off	off
on	on	off
on	off	on
on	on	on

Table 7-5: Valid protection unit, cache and write buffer combinations

The following procedures must be observed:

To enable the Protection Unit:

- 1 Program the Cacheable, Bufferable, Protection and Area registers as required.
- 2 Enable the Protection Unit by setting bit 0 in the Control register.

To disable the Protection Unit:

- 1 Disable the write buffer by clearing bit 3 in the Control register.
- 2 Disable the cache by clearing bit 2 in the Control register.
- 3 Disable the Protection Unit by clearing bit 0 in the Control register.

Disabling of all three functions may be done simultaneously with a single write to the control register.

8

Debug Interface

This chapter describes the ARM740T advanced debug interface.

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8.2	Debug Systems	8-3
8.3	Entering Debug State	8-4
8.4	Scan Chains and JTAG Interface	8-5
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Debug Interface

8.1 Overview

In this chapter ARM7TDM refers to the ARM7TDM core excluding the EmbeddedICE Macrocell. The ARM7TDM debug interface is based on IEEE Std. 1149.1 - 1990, *Standard Test Access Port and Boundary-Scan Architecture*. Please refer to this standard for an explanation of the terms used in this chapter and for a description of the TAP controller states.

8.1.1 Debug extensions

ARM7TDM contains hardware extensions for advanced debugging features. These are intended to ease the user's development of application software, operating systems, and the hardware itself.

The debug extensions allow the core to be stopped either on a given instruction fetch (breakpoint) or data access (watchpoint), or asynchronously by a debug-request. When this happens, ARM7TDM is said to be in *debug state*. At this point, the core's internal state and the system's external state may be examined. Once examination is complete, the core and system state may be restored and program execution resumed.

Debug state

ARM7TDM is forced into debug state either by a request on one of the external debug interface signals, or by an internal functional unit known as *EmbeddedICE*. Once in debug state, the core isolates itself from the memory system. The core can then be examined while all other system activity continues as normal.

Internal state

ARM7TDM's internal state is examined via a JTAG-style serial interface, which allows instructions to be serially inserted into the core's pipeline without using the external data bus. Thus, when in debug state, a store-multiple (STM) could be inserted into the instruction pipeline and this would dump the contents of ARM7TDM's registers. This data can be serially shifted out without affecting the rest of the system.

8.1.2 Pullup resistors

The IEEE 1149.1 standard effectively requires that **TDI** and **TMS** should have internal pullup resistors. In order to minimise static current draw, these resistors are *not* fitted to ARM7TDM. Accordingly, the four inputs to the test interface (the above four signals plus **TCK**) must all be driven to good logic levels to achieve normal circuit operation.

8.1.3 Instruction register

The instruction register is four bits in length.

There is no parity bit. The fixed value loaded into the instruction register during the CAPTURE-IR controller state is 0001.

8.2 Debug Systems

The ARM7TDM forms one component of a debug system that interfaces from the high-level debugging performed by the user to the low-level interface supported by ARM7TDM. Such a system typically has three parts:

The Debug Host	This is a computer, for example a PC, running a software debugger such as ARMSD. The debug host allows the user to issue high level commands such as “set breakpoint at location XX”, or “examine the contents of memory from 0x0 to 0x100”.
The Protocol Converter	The Debug Host is connected to the ARM7TDM development system via an interface (an RS232, for example). The messages broadcast over this connection must be converted to the interface signals of the ARM7TDM, and this function is performed by the protocol converter.
ARM7TDM	ARM7TDM, with hardware extensions to ease debugging, is the lowest level of the system. The debug extensions allow the user to stall the core from program execution, examine its internal state and the state of the memory system, and then resume program execution.

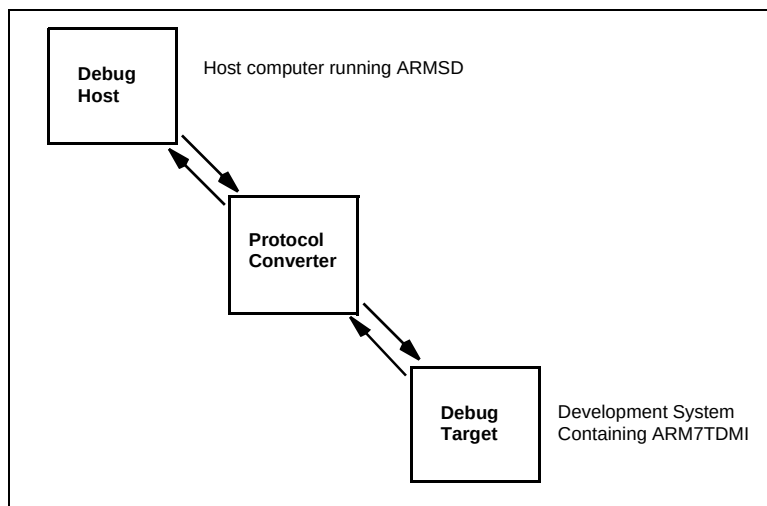


Figure 8-1: Typical debug system

The anatomy of ARM7TDM is shown in **Figure 8-2: ARM740T scan chain arrangement** on page 8-6. The major blocks are:

ARM7TDM	This is the CPU core, with hardware support for debug.
EmbeddedICE	This is a set of registers and comparators used to generate debug exceptions (eg. breakpoints). This unit is described in Chapter 9, EmbeddedICE Macrocell .
TAP controller	This controls the action of the scan chains via a JTAG serial interface.

The Debug Host and the Protocol Converter are system dependent. The rest of this chapter describes the ARM7TDM's hardware debug extensions.

Debug Interface

8.3 Entering Debug State

ARM7TDM is forced into debug state after a breakpoint, watchpoint or debug request has occurred. Conditions under which a breakpoint or watchpoint occur can be programmed using EmbeddedICE. Alternatively, external logic can monitor the address and data bus, and flag breakpoints and watchpoints via the BREAKPT pin.

8.3.1 Entering debug state on breakpoint

After an instruction has been breakpointed, the core does not enter debug state immediately. Instructions are marked as being breakpointed as they enter ARM7TDM's instruction pipeline. Thus ARM7TDM only enters debug state when (and if) the instruction reaches the pipeline's execute stage.

There are two reasons why a breakpointed instruction may not cause ARM7TDM to enter debug state:

- a branch precedes the breakpointed instruction. When the branch is executed, the instruction pipeline is flushed and the breakpoint is cancelled.
- an exception has occurred. Again, the instruction pipeline is flushed and the breakpoint is cancelled. However, the normal way to exit from an exception is to branch back to the instruction that would have executed next. This involves refilling the pipeline, and so the breakpoint can be re-flagged.

When a breakpointed conditional instruction reaches the execute stage of the pipeline, the breakpoint is *always* taken and ARM7TDM enters debug state, regardless of whether the condition was met.

Breakpointed instructions *are not* executed. Instead, ARM7TDM enters debug state. Thus, when the internal state is examined, the state *before* the breakpointed instruction is seen. Once examination is complete, the breakpoint should be removed and program execution restarted from the previously breakpointed instruction.

8.3.2 Entering debug state on watchpoint

Watchpoints occur on data accesses. A watchpoint is always taken, but the core may not enter debug state immediately. In all cases, the current instruction does complete. If this is a multi-word load or store (LDM or STM), many cycles may elapse before the watchpoint is taken.

Watchpoints can be thought of as being similar to data aborts. The difference is that if a data abort occurs, although the instruction completes, all subsequent changes to ARM7TDM's state are prevented. This allows the cause of the abort to be cured by the abort handler, and the instruction re-executed. In the case of a watchpoint, the instruction completes and all changes to the core's state occur (load data is written into the destination registers, and base writeback occurs). Thus, the instruction does not need to be restarted.

Watchpoints are *always* taken. If an exception is pending when a watchpoint occurs, the core enters debug state in the mode of that exception.

8.3.3 Entering debug state on debug-request

ARM7TDM may also be forced into debug state on debug request. This can be done either through EmbeddedICE programming (see **Chapter 9, EmbeddedICE Macrocell**) or by the assertion of the DBGREQ pin. This pin is an asynchronous input and is thus synchronised by logic inside ARM7TDM before it takes effect. Following synchronisation, the core normally enters debug state at the end of the current instruction. However, if the current instruction is a busy-waiting access to a coprocessor, the instruction terminates and ARM7TDM enters debug state immediately (this is similar to the action of **nIRQ** and **nFIQ**).

8.4 Scan Chains and JTAG Interface

There are several JTAG style scan chains inside ARM7TDM and chain inside the ARM740T. These allow testing, debugging, and EmbeddedICE programming.

In addition, support is provided for an optional scan chain. This is intended to be used for an external boundary scan chain around the pads of a packaged device. The control signals provided for this scan chain are described later.

The scan chains are controlled from a JTAG-style Test Access Port (TAP) controller. For further details of the JTAG specification, please refer to IEEE Standard 1149.1 - 1990 "Standard Test Access Port and Boundary-Scan Architecture".

Note *The scan cells are not fully JTAG-compliant. The following sections describe the limitations on their use.*

8.4.1 Scan limitations

The scan paths are shown in **Figure 8-2: ARM740T scan chain arrangement** on page 8-6.

Scan Chain 0	allows access to the entire periphery of the ARM7TDM core, including the data bus. The scan chain functions allow inter-device testing (EXTEST) and serial testing of the core (INTEST). The order of the scan chain (from SDIN to SDOUTMS) is: • data bus bits 0 through 3 • the control signals (see Table 8-3: Scan Chain 0 Bit Positions on page 8-28) • the address bus bits 31 through 0
Scan Chain 1	is a subset of the signals that are accessible through scan chain 0. Access to the core's data bus D[31:0] , and the BREAKPT signal is available serially. There are 33 bits in this scan chain; the order is (from serial data in to out): • data bus bits 0 through 31 • BREAKPT
Scan Chain 2	allows access to the EmbeddedICE registers. See Chapter 9, EmbeddedICE Macrocell for details.
Scan Chain 6	allows access to the TAG entries in the cache.
Scan Chain 15	allows access to the System Control Coprocessor registers.

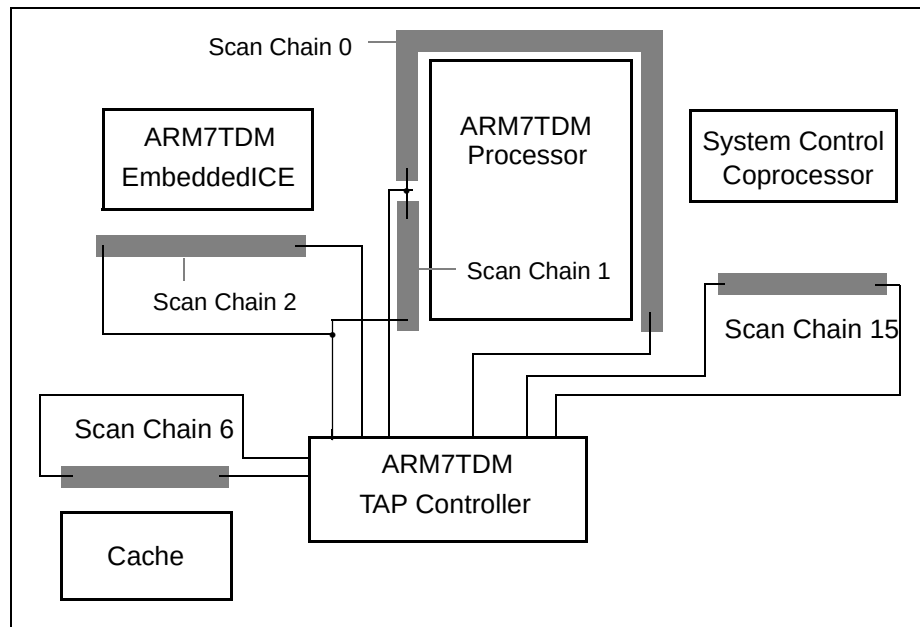


Figure 8-2: ARM740T *scan chain arrangement*

8.4.2 The JTAG state machine

The process of serial test and debug is best explained in conjunction with the JTAG state machine. **Figure 8-3: Test access port (TAP) controller state transitions** shows the state transitions that occur in the TAP controller. The state numbers are also shown on the diagram.

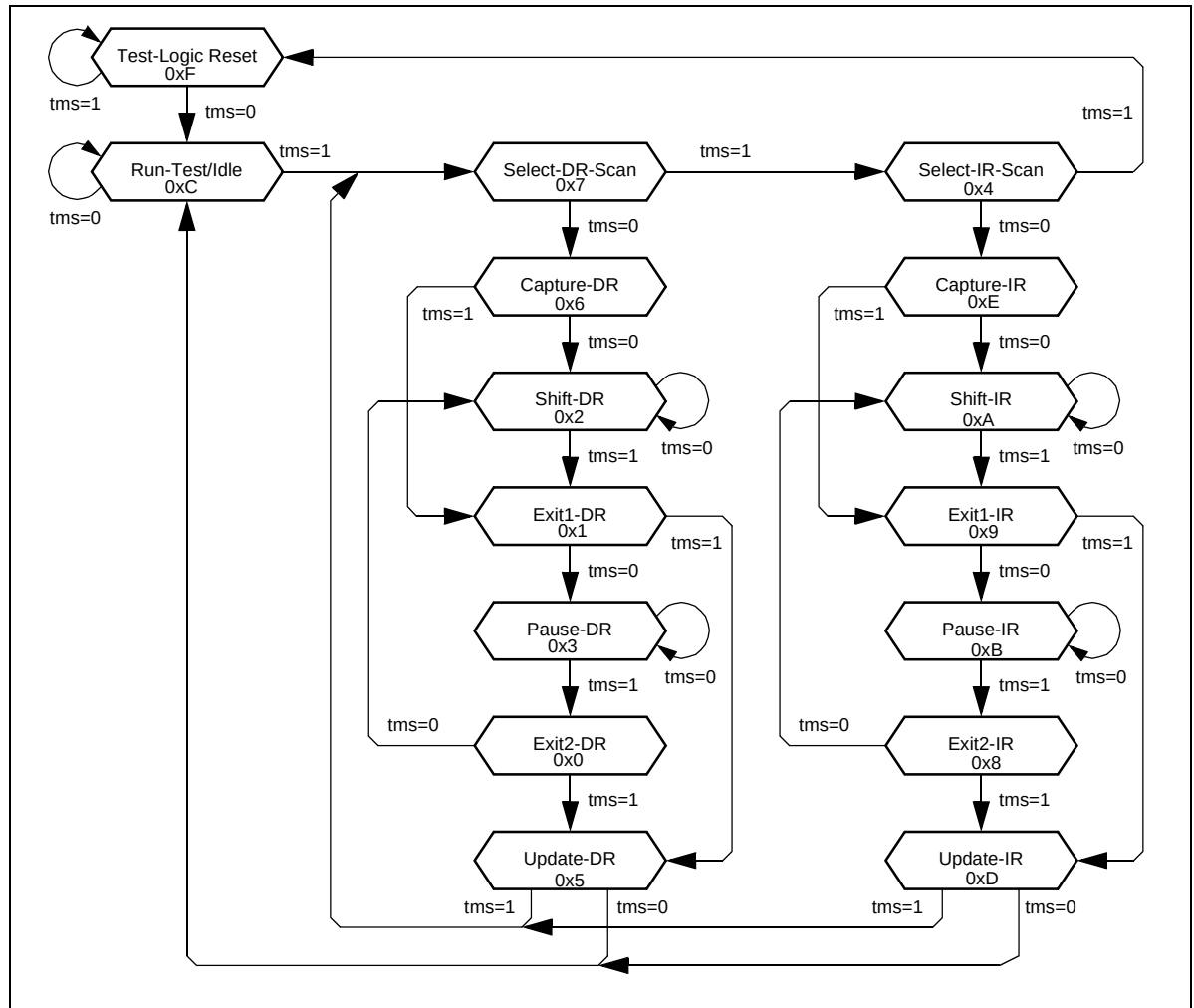


Figure 8-3: Test access port (TAP) controller state transitions

Debug Interface

8.5 Reset

The boundary-scan interface includes a state-machine controller (the TAP controller). In order to force the TAP controller into the correct state after power-up of the device, a reset pulse must be applied to the **nTRST** signal.

If the boundary scan interface is to be used, **nTRST** must be driven LOW, and then HIGH again. If the boundary scan interface is not to be used, the **nTRST** input may be tied permanently LOW.

Note *A clock on **TCK** is not necessary to reset the device.*

The action of reset is as follows:

- 1 System mode is selected (the boundary scan chain cells do *not* intercept any of the signals passing between the external system and the core).
- 2 The IDCODE instruction is selected. If the TAP controller is put into the Shift-DR state and **TCK** is pulsed, the contents of the ID register is clocked out of **TDO**.

8.6 Public Instructions

The public instructions are listed below. In the descriptions that follow, **TDI** and **TMS** are sampled on the rising edge of **TCK** and all output transitions on **TDO** occur as a result of the falling edge of **TCK**.

EXTEST	0000	places the selected scan chain in test mode. This instruction connects the selected scan chain between TDI and TDO . When the instruction register is loaded with EXTEST, all the scan cells are placed in their test mode of operation.
		CAPTURE-DR Inputs from the system logic and outputs from the output scan cells to the system are captured by the scan cells.
		SHIFT-DR The previously captured test data is shifted out of the scan chain via TDO , while new test data is shifted in via the TDI input. This data is applied immediately to the system logic and system pins.
SCAN_N	0010	connects the Scan Path Select Register between TDI and TDO . On reset, scan chain 3 is selected by default. The scan path select register is 4 bits long in this implementation, although no finite length is specified.
		CAPTURE-DR The fixed value 1000 is loaded into the register.
		SHIFT-DR The ID number of the desired scan path is shifted into the scan path select register
INTEST	1100	UPDATE-DR The scan register of the selected scan chain is connected between TDI and TDO , and remains connected until a subsequent SCAN_N instruction is issued.
		places the selected scan chain test mode. This instruction connects the selected scan chain between TDI and TDO . When the instruction register is loaded with this instruction, all the scan cells are placed in their test mode of operation. Single-step operation is possible using the INTEST instruction.
		CAPTURE-DR The value of the data applied from the core logic to the output scan cells, and the value of the data applied from the system logic to the input scan cells is captured.
		SHIFT-DR The previously captured test data is shifted out of the scan chain via the TDO pin, while new test data is shifted in via the TDI pin.

Debug Interface

IDCODE	1110	connects the device identification register (or ID register) between TDI and TDO. The ID register is a 32-bit register that allows the manufacturer, part number and version of a component to be determined through the TAP. See 8.7.2 ARM7TDM device identification (ID) code register on page 8-12 for the details of the ID register format. When the instruction register is loaded with this instruction, all the scan cells are placed in their normal (system) mode of operation. CAPTURE-DR The device identification code is captured by the ID register. SHIFT-DR The previously captured device identification code is shifted out of the ID register via the TDO pin, while data is shifted in via the TDI pin into the ID register. UPDATE-DR The ID register is unaffected.
BYPASS	1111	connects a 1 bit shift register (the bypass register) between TDI and TDO. When this instruction is loaded into the instruction register, all the scan cells are placed in their normal (system) mode of operation. This instruction has no effect on the system pins. Note: All unused instruction codes default to the <i>BYPASS instruction</i> CAPTURE-DR A logic 0 is captured by the bypass register. SHIFT-DR Test data is shifted into the bypass register via TDI and out via TDO after a delay of one TCK cycle. The first bit shifted out is a zero. UPDATE-DR The bypass register is not affected.
CLAMP	0101	connects a 1 bit shift register (the bypass register) between TDI and TDO. When this instruction is loaded into the instruction register, the state of all the output signals is defined by the values previously loaded into the currently loaded scan chain. Note: This instruction should only be used when scan chain 0 is the currently selected scan chain. CAPTURE-DR A logic 0 is captured by the bypass register. SHIFT-DR Test data is shifted into the bypass register via TDI and out via TDO after a delay of one TCK cycle. The first bit shifted out is a zero. UPDATE-DR The bypass register is not affected.

HIGHZ	0111	connects a 1 bit shift register (the bypass register) between TDI and TDO. When this instruction is loaded into the instruction register, the Address bus, A[31:0], the data bus, D[31:0], plus nRW, nOPC, LOCK, MAS[1:0] and nTRANS are all driven to the high impedance state and the external HIGHZ signal is driven HIGH. This is as if the signal TBE had been driven LOW. CAPTURE-DR A logic 0 is captured by the bypass register. SHIFT-DR Test data is shifted into the bypass register via TDI and out via TDO after a delay of one TCK cycle. Note that the first bit shifted out will be a zero. UPDATE-DR The bypass register is not affected.
CLAMPZ	1001	connects a 1 bit shift register (the bypass register) between TDI and TDO. When this instruction is loaded into the instruction register, all the 3-state outputs (as described above) are placed in their inactive state, but the data supplied to the outputs is derived from the scan cells. The purpose of this instruction is to ensure that, during production test, each output can be disabled when its data value is either a logic 0 or a logic 1. CAPTURE-DR A logic 0 is captured by the bypass register. SHIFT-DR Test data is shifted into the bypass register via TDI and out via TDO after a delay of one TCK cycle. Note that the first bit shifted out will be a zero. UPDATE-DR The bypass register is not affected.
RESTART	0100	restarts the processor on exit from debug state. It connects the bypass register between TDI and TDO and the TAP controller behaves as if the BYPASS instruction had been loaded. The processor resynchronizes back to the memory system once the RUN-TEST/IDLE state is entered.
SAMPLE/ PRELOAD	0011	Note: <i>This instruction is included for production test only, and should never be used.</i>

Debug Interface

8.7 Test Data Registers

The following registers may be connected between **TDI** and **TDO**:

- Bypass Register
- ID Code Register
- Instruction Register
- Scan Chain Select Register
- Scan chain 0, 1, 2, 3, 6, or 15

These are described in detail in the following sections.

8.7.1 Bypass register

This register bypasses the device during scan testing by providing a path between **TDI** and **TDO**. The bypass register is 1 bit in length.

Operating mode

When the **BYPASS** instruction is the current instruction in the instruction register, serial data is transferred from **TDI** to **TDO** in the **SHIFT-DR** state with a delay of one **TCK** cycle.

There is no parallel output from the bypass register.

A logic 0 is loaded from the parallel input of the bypass register in the **CAPTURE-DR** state.

8.7.2 ARM7TDM device identification (ID) code register

This register reads the 32-bit device identification code. No programmable supplementary identification code is provided. The register is 32 bits in length.

The format of the ID register is as follows:

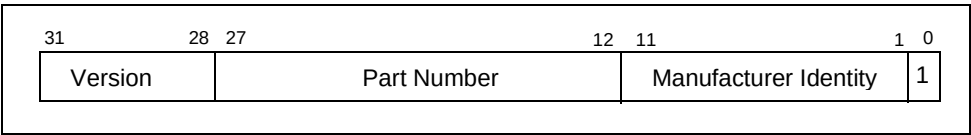


Figure 8-4: ID code register

Please contact your supplier for the correct Device Identification Code.

Operating mode

When the **IDCODE** instruction is current, the ID register is selected as the serial path between **TDI** and **TDO**.

There is no parallel output from the ID register.

The 32-bit device identification code is loaded into the ID register from its parallel inputs during the **CAPTURE-DR** state.



8.7.3 Instruction register

This register changes the current TAP instruction. The register is 4 bits in length

Operating mode

When in the SHIFT-IR state, the instruction register is selected as the serial path between **TDI** and **TDO**.

During the CAPTURE-IR state, the value 0001 binary is loaded into this register. This is shifted out during SHIFT-IR (lsb, least significant bit, first), while a new instruction is shifted in (lsb first).

During the UPDATE-IR state, the value in the instruction register becomes the current instruction.

On reset, IDCODE becomes the current instruction.

8.7.4 Scan chain select register

This register changes the current active scan chain. The register is 4 bits in length.

Operating mode

After SCAN_N has been selected as the current instruction, when in the SHIFT-DR state, the Scan Chain Select Register is selected as the serial path between **TDI** and **TDO**.

During the CAPTURE-DR state, the value 1000 binary is loaded into this register. This is shifted out during SHIFT-DR (lsb first), while a new value is shifted in (lsb first).

During the UPDATE-DR state, the value in the register selects a scan chain to become the currently active scan chain. All further instructions, such as INTEST, then apply to that scan chain.

The currently selected scan chain only changes when a SCAN_N instruction is executed, or a reset occurs. On reset, scan chain 3 is selected as the active scan chain.

The number of the currently selected scan chain is reflected on the **SCREG[3:0]** outputs. The TAP controller may be used to drive external scan chains in addition to those within the ARM7TDM macrocell. The external scan chain must be assigned a number and control signals for it can be derived from **SCREG[3:0]**, **IR[3:0]**, **TAPSM[3:0]**, **TCK1** and **TCK2**.

The list of scan chain numbers allocated by ARM is shown in **Table 8-1: Scan chain number allocation**. An external scan chain may take any other number. The serial data stream to be applied to the external scan chain is made present on **SDINBS** and the serial data back from the scan chain must be presented to the TAP controller on the **SDOUTBS** input. The scan chain present between **SDINBS** and **SDOUTBS** is connected between **TDI** and **TDO** whenever scan chain 3 is selected, or when any of the unassigned scan chain numbers is selected. If there is more than one external scan chain, a multiplexer must be built externally to apply the desired scan chain output to **SDOUTBS**. The multiplexer can be controlled by decoding **SCREG[3:0]**.

Scan Chain Number	Function	Length
0	Macrocell scan test	105
1	Debug	33
2	EmbeddedICE programming	38
3	External boundary scan	NA

Table 8-1: Scan chain number allocation

Debug Interface

Scan Chain Number	Function	Length
4	Reserved	NA
6	TAG	88 (8KB variant)
6	TAG	92 (4KB variant)
8	Reserved	NA
15	CP15	33

Table 8-1: Scan chain number allocation (Continued)

8.7.5 Overview of scan chains

These allow serial access to the core logic, and to EmbeddedICE for programming purposes. They are described in detail in the following sections.

Each scan chain cell is fairly simple, and consists of a serial register and a multiplexer. The scan cells perform two basic functions:

- capture
 - For input cells, the capture stage involves copying the value of the system input to the core into the serial register.
 - For output cells, capture involves placing the value of a core's output into the serial register.
- shift
 - For input cells, during shift, this value is output serially. The value applied to the core from an input cell is either the system input or the contents of the serial register, and this is controlled by the multiplexer.
 - For output cells, during shift, this value is serially output as before. The value applied to the system from an output cell is either the core output, or the contents of the serial register.

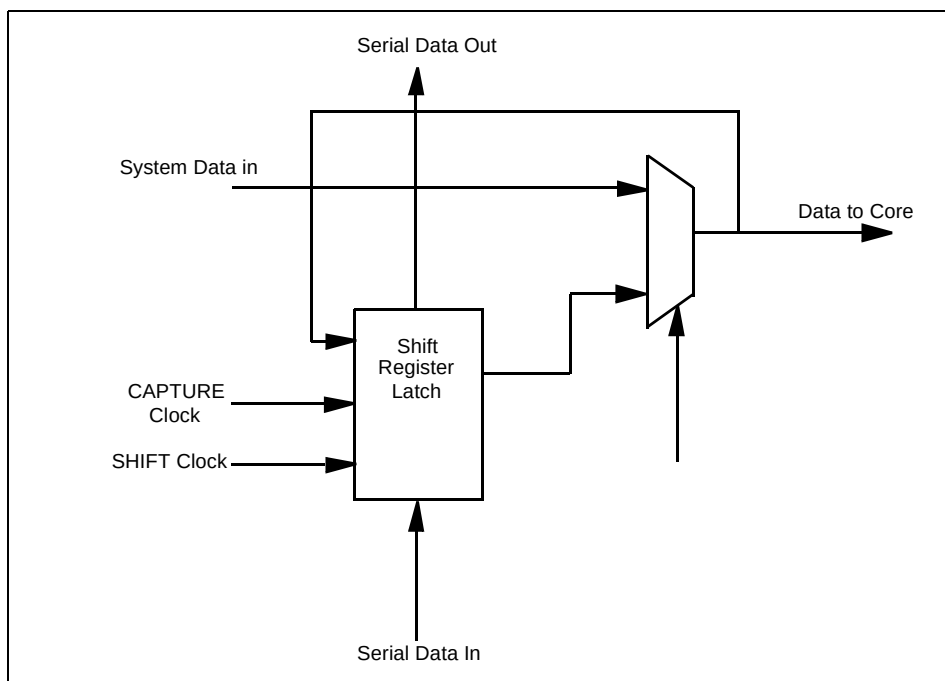


Figure 8-5: Input scan cell

All the control signals for the scan cells are generated internally by the TAP controller. The action of the TAP controller is determined by the current instruction, and the state of the TAP state machine. This is described in the following section.

Operating modes

The scan chains have three basic modes of operation. These are selected by the various TAP controller instructions:

SYSTEM mode	The scan cells are idle. System data is applied to inputs, and core outputs are applied to the system.
INTEST mode	The core is internally tested. The data serially scanned in is applied to the core, and the resulting outputs are captured in the output cells and scanned out.
EXTEST mode	Data is scanned onto the core's outputs and applied to the external system. System input data is captured in the input cells and then shifted out.

Note *The scan cells are not fully JTAG-compliant in that they do not have an Update stage. Therefore, while data is being moved around the scan chain, the contents of the scan cell is not isolated from the output. Thus the output from the scan cell to the core or to the external system could change on every scan clock.*

This does not affect ARM7TDM because its internal state does not change until it is clocked. However, the rest of the system needs to be aware that every output could change asynchronously as data is moved around the scan chain. External logic must ensure that this does not harm the rest of the system.

8.7.6 Scan chain 0

Scan chain 0 is intended primarily for inter-device testing (EXTEST), and testing the core (INTEST). Scan chain 0 is selected via the SCAN_N instruction, and is 105 bits long.

Serial testing the core

INTEST allows serial testing of the core. The TAP Controller must be placed in INTEST mode after scan chain 0 has been selected.

- During CAPTURE-DR, the current outputs from the core's logic are captured in the output cells.
- During SHIFT-DR, this captured data is shifted out while a new serial test pattern is scanned in, thus applying known stimuli to the inputs.
- During RUN-TEST/IDLE, the core is clocked. Normally, the TAP controller should only spend one cycle in RUN-TEST/IDLE.

The whole operation may then be repeated.

See **8.8 ARM7TDM Core Clocks** on page 8-19 for details of the core's clocks during test and debug.

Inter-device testing

EXTEST allows inter-device testing, which is useful for verifying the connections between devices on a circuit board. The TAP Controller must be placed in EXTEST mode after scan chain 0 has been selected.

- During CAPTURE-DR, the current inputs to the core's logic from the system are captured in the input cells.
- During SHIFT-DR, this captured data is shifted out while a new serial test pattern is scanned in, thus applying known values on the core's outputs.

Debug Interface

- During UPDATE-DR, the value shifted into the data bus **D[31:0]** scan cells appears on the outputs. For all other outputs, the value appears as the data is shifted round.

Note During RUN-TEST/IDLE, the core is not clocked.

The operation may then be repeated.

The ordering of signals on scan chain 0 is outlined in **Table 8-3: Scan Chain 0 Bit Positions** on page 8-28.

8.7.7 Scan chain 1

The primary use for scan chain 1 is for debugging, although it can be used for EXTEST on the data bus. Scan chain 1 is selected via the SCAN_N TAP Controller instruction. Debugging is similar to INTEST, and the procedure described above for scan chain 0 should be followed.

Scan chain length and purpose

This scan chain is 33 bits long—32 bits for the data value, plus the scan cell on the **BREAKPT** core input. This 33rd bit serves four purposes:

- 1 Under normal INTEST test conditions, it allows a known value to be scanned into the **BREAKPT** input.
- 2 During EXTEST test conditions, the value applied to the **BREAKPT** input from the system can be captured.
- 3 While debugging, the value placed in the 33rd bit determines whether ARM7TDM synchronises back to system speed before executing the instruction. See **8.10.5 System-speed access** on page 8-24 for further details.
- 4 After ARM7TDM has entered debug state, the first time this bit is captured and scanned out, its value tells the debugger whether the core entered debug state due to a breakpoint (bit 33 LOW), or a watchpoint (bit 33 HIGH).

8.7.8 Scan chain 2

This scan chain allows EmbeddedICE's registers to be accessed. The scan chain is 38 bits in length.

The order of the scan chain, from **TDI** to **TDO** is:

- read/write
- register address bits 4 to 0
- data value bits 31 to 0

See **Figure 9-2: EmbeddedICE block diagram** on page 9-5 for more information.

To access this serial register, scan chain 2 must first be selected via the SCAN_N TAP controller instruction. The TAP controller must then be placed in INTEST mode.

- No action is taken during CAPTURE-DR.
- During SHIFT-DR, a data value is shifted into the serial register. Bits 32 to 36 specify the address of the EmbeddedICE register to be accessed.
- During UPDATE-DR, this register is either read or written depending on the value of bit 37 (0 = read). Refer to **Chapter 9, EmbeddedICE Macrocell** for further details.

8.7.9 Scan chain 3

This scan chain allows ARM7TDM to control an external boundary scan chain. Scan chain 3 is provided so that an optional external boundary scan chain may be controlled via ARM7TDM. Typically, this would be used for a scan chain around the pad ring of a packaged device. Its length is user-defined.

The following control signals are provided. These are generated only when scan chain 3 has been selected. These outputs are inactive at all other times.

DRIVEBS	This would be used to switch the scan cells from system mode to test mode. This signal is asserted whenever either the INTEST, EXTEST, CLAMP or CLAMPZ instruction is selected.
PCLKBS	This is an update clock, generated in the UPDATE-DR state. Typically the value scanned into a chain would be transferred to the cell output on the rising edge of this signal.
ICAPCLKBS ECAPCLKBS	These are capture clocks used to sample data into the scan cells during INTEST and EXTEST respectively. These clocks are generated in the CAPTURE-DR state.
SHCLKBS SHCLK2BS	These are non-overlapping clocks generated in the SHIFTD state used to clock the master and slave element of the scan cells respectively. When the state machine is not in the SHIFT-DR state, both these clocks are LOW.
nHIGHZ	This signal may be used to drive the outputs of the scan cells to the high impedance state. This signal is driven LOW when the HIGHZ instruction is loaded into the instruction register, and HIGH at all other times.

External scan chains

In addition to the above control outputs, the following are provided for use when an external scan chain is in use:

SDINBS output	should be connected to the serial data input.
SDOUTBS input	should be connected to the serial data output.

8.7.10 Scan Chain 6

8KB variant

This scan chain is 88 bits long and across the Tag entries of the four banks. In operation, addresses may be scanned into the chain and the corresponding Tag entries scanned out. The cache is never written to by the scan chain, it is read only.

Although the scan chain is 88 bits long, only the seven bits of the line address are significant for addressing the cache. These bits are A[10:4] and are the last seven bits scanned into the chain. That is, A[4] is the very last bit scanned in and A[10] is the seventh last bit scanned in.

For the line address presented in this way, the corresponding 21 bits of the tag entries and their associated valid flags can be captured and scanned out. The first bit scanned out is Bank 3. Valid flags are followed by the Bank 3 Tag address (msb, most significant bit first) as shown in **Table 8-5: Tag scan chain** on page 8-31.

4KB variant

This scan chain is 92 bits long and across the Tag entries of the four banks. In operation, addresses may be scanned into the chain and the corresponding Tag entries scanned out. The cache is never written to by the scan chain, it is read only.

Although the scan chain is 92 bits long, only the six bits of the line address are significant for addressing the cache. These bits are A[9:4] and are the last six bits scanned into the chain. That is, A[4] is the very last bit scanned in and A[9] is the sixth last bit scanned in.

For the line addresses presented in this way, the corresponding 22 bits of the tag entries and their associated valid flags can be captured and scanned out. The first bit scanned out is the Bank 3 valid flag followed by the Bank 3 Tag address (msb first) as shown in **Table 8-5: Tag scan chain** on page 8-31.

8.7.11 Scan Chain 15

This scan chain is 33 bits long and sits on the CData bus at the interface of CP15.

The first bit scanned in is the Instruction flag followed by the most significant bit of the CData bus. If the Instruction flag is HIGH the scanned data value is considered to be a COP instruction, otherwise it is treated as data.

This scan chain can be used to present instructions and data to CP15.

8.8 ARM7TDM Core Clocks

ARM7TDM has two clocks:

- the memory clock, **MCLK**, generated by the ARM740T
- an internally **TCK**-generated clock, **DCLK**

During normal operation, the core is clocked by **MCLK**, and internal logic holds **DCLK** LOW.

There are two cases in which the clocks switch:

- during debugging
- during testing

8.8.1 Clock switch during debug

When ARM7TDM is in the debug state, the core is clocked by **DCLK** under the control of the TAP state machine, and **MCLK** may free run. The selected clock is output on the signal **ECLK** for use by the external system.

Note When the CPU core is being debugged and is running from **DCLK**, **nWAIT** has no effect.

When ARM7TDM enters debug state, it must switch from **MCLK** to **DCLK**. This is handled automatically by logic in the ARM7TDM. On entry to debug state, ARM7TDM asserts **DBGACK** in the HIGH phase of **MCLK**. The switch between the two clocks occurs on the next falling edge of **MCLK**. This is shown in **Figure 8-6: Clock Switching on entry to debug state**.

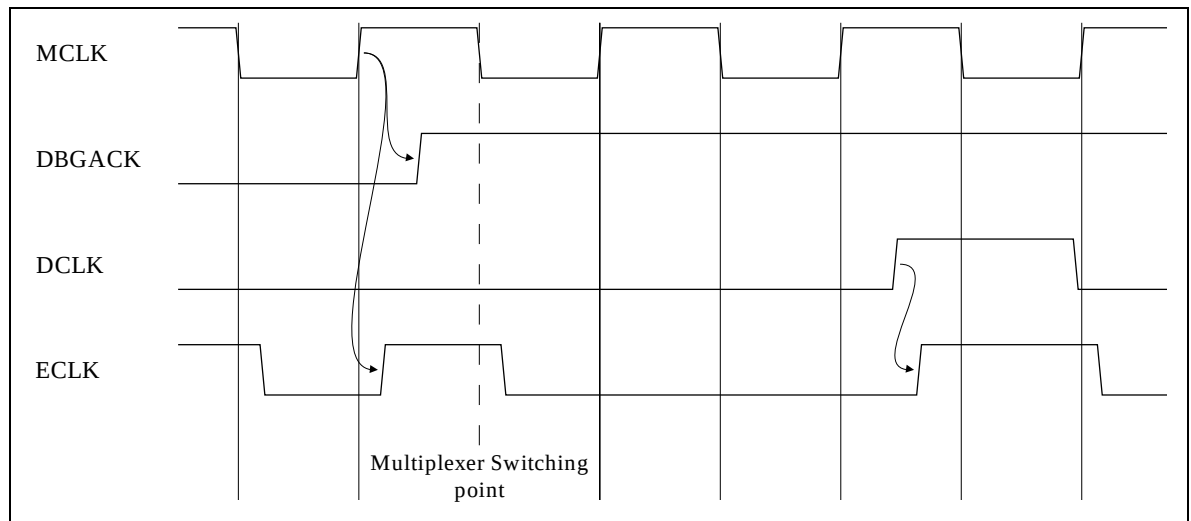


Figure 8-6: Clock Switching on entry to debug state

ARM7TDM is forced to use **DCLK** as the primary clock until debugging is complete. On exit from debug, the core must be allowed to synchronise back to **MCLK**. This must be done in the following sequence:

- 1 The final instruction of the debug sequence must be shifted into the data bus scan chain and clocked in by asserting **DCLK**.
- 2 At this point, **BYPASS** must be clocked into the TAP instruction register.
- 3 ARM7TDM now automatically resynchronizes back to **MCLK** and starts fetching instructions from memory at **MCLK** speed.

Please refer also to **8.9.4 Exit from debug state** on page 8-22.

Debug Interface

8.9 Determining the Core and System State

When ARM7TDM is in debug state, the core and system's state may be examined. This is done by forcing load and store multiples into the instruction pipeline.

ARM or THUMB state

Before the core and system state can be examined, the debugger must first determine whether the processor was in THUMB or ARM state when it entered debug. This is achieved by examining bit 4 of EmbeddedICE's Debug Status Register. If this is HIGH, the core was in THUMB state when it entered debug.

8.9.1 Determining the core's state

If the processor has entered debug state from THUMB state, the simplest course of action is for the debugger to force the core back into ARM state. Once this is done, the debugger can always execute the same sequence of instructions to determine the processor's state.

While in debug state, only the following instructions may legally be scanned into the instruction pipeline for execution:

- all data-processing instructions, except TEQP
- all load, store, load multiple and store multiple instructions
- MSR and MRS

Moving to ARM state

To force the processor into ARM state, the following sequence of THUMB instructions should be executed on the core:

```
STR R0, [R0]      ; Save R0 before use
MOV R0, PC        ; Copy PC into R0
STR R0, [R0]      ; Now save the PC in R0
BX PC             ; Jump into ARM state
MOV R8, R8        ; NOP
MOV R8, R8        ; NOP
```

As all THUMB instructions are only 16 bits long, the simplest course of action when shifting them into Scan Chain 1 is to repeat the instruction twice.

For example, the encoding for BX R0 is 0x4700. Therefore, if 0x47004700 is shifted into scan chain 1, the debugger does not have to keep track of which half of the bus the processor expects to read the data from.

From this point on, the processor's state can be determined by the sequences of ARM instructions described below.

In ARM state

Once the processor is in ARM state, the first instruction executed would typically be:

```
STM R0, {R0-R15}
```

This makes the contents of the registers visible on the data bus. These values can then be sampled and shifted out.

Note *The above use of R0 as the base register for STM is for illustration only: any register could be used.*

Accessing banked registers

After determining the values in the current bank of registers, it may be desirable to access the banked registers. This can only be done by changing mode. Normally, a mode change may only occur if the core is already in a privileged mode. However, while in debug state, a mode change from any mode into any other mode may occur.

Note *The debugger must restore the original mode before exiting debug state.*

For example, assume that the debugger had been asked to return the state of the USER and FIQ mode registers, and debug state was entered in supervisor mode.

The instruction sequence could be:

```
STM R0, {R0-R15}    ; Save current registers
MRS R0, CPSR
STR R0, R0           ; Save CPSR to determine current mode
BIC R0, 0x1F         ; Clear mode bits
ORR R0, 0x10         ; Select user mode
MSR CPSR, R0         ; Enter USER mode
STM R0, {R13,R14}    ; Save register not previously visible
ORR R0, 0x01         ; Select FIQ mode
MSR CPSR, R0         ; Enter FIQ mode
STM R0, {R8-R14}     ; Save banked FIQ registers
```

All these instructions are said to execute at *debug speed*. Debug speed is much slower than system speed because between each core clock, 33 scan clocks occur in order to shift in an instruction, or shift out data. Executing instructions more slowly than usual is fine for accessing the core's state because ARM7TDM is fully static. However, this same method cannot be used for determining the state of the rest of the system.

8.9.2 Determining system state

In order to meet the dynamic timing requirements of the memory system, any attempt to access system state must occur synchronously with it. Thus, ARM7TDM must be forced to synchronise back to system speed. This is controlled by the 33rd bit of scan chain 1.

Any instruction may be placed in scan chain 1 with bit 33 (the BREAKPT bit) LOW. This instruction is then executed at debug speed. To execute an instruction at system speed, the instruction prior to it must be scanned into scan chain 1 with bit 33 set HIGH.

After the system speed instruction has been scanned into the data bus and clocked into the pipeline, the BYPASS instruction must be loaded into the TAP controller. This makes the ARM7TDM automatically synchronize back to **MCLK** (the system clock), executes the instruction at system speed, and then re-enters debug state and switches itself back to the internally generated **DCLK**. When the instruction has completed, **DBGACK** is HIGH and the core will have switched back to **DCLK**. At this point, **INTEST** can be selected in the TAP controller, and debugging can resume.

In order to determine that a system speed instruction has completed the debugger must look at both **DBGACK** and **nMREQ**. In order to access memory, ARM7TDM drives **nMREQ** LOW after it has synchronised back to system speed. This transition is used by the memory controller to arbitrate whether ARM7TDM can have the bus in the next cycle. If the bus is not available, ARM7TDM may have its clock stalled indefinitely.

Therefore, the only way to tell that the memory access has completed, is to examine the state of both **nMREQ** and **DBGACK**. When both are HIGH, the access has completed. Usually, the debugger would be using EmbeddedICE to control debugging, and by reading EmbeddedICE's status register, the state of **nMREQ** and **DBGACK** can be determined. Refer to **Chapter 9, EmbeddedICE Macrocell** for more details.

By the use of system speed load multiples and debug speed store multiples, the state of the system's memory can be fed back to the debug host.

Debug Interface

Restrictions

There are restrictions on which instructions may have the 33rd bit set. The only valid instructions where this bit can be set are:

- loads
- stores
- load multiple
- store multiple

See also **8.9.4 Exit from debug state**.

When ARM7TDM returns to debug state after a system speed access, bit 33 of scan chain 1 is set HIGH. This gives the debugger information about why the core entered debug state the first time this scan chain is read.

8.9.3 Determining system control coprocessor state

In order to access the System Control Processor registers, debug state must be entered by a breakpoint, watchpoint or debug request. This ensures that the ARM7TDM core stops execution of code which may be dependent on the System Control Coprocessor.

Scan Chain 15 can then be selected via the SCAN_N instruction.

Instructions may then be scanned down the scan chain as if being executed from the ARM7TDM core. As the ARM7TDM is idle while Scan Chain 15 is being accessed, it is necessary to provide the register data via the scan chain. The instruction prior to the data must have the instruction/data flag cleared.

The data operation requires an additional clock from the TAP controller. This may be achieved by remaining in the RUN-TEST-IDLE state for an additional **TCK** cycle.

8.9.4 Exit from debug state

Leaving debug state involves:

- 1 restoring ARM7TDM's internal state.
- 2 branching to the next instruction to be executed.
- 3 synchronizing back to **MCLK**.

After restoring internal state, a branch instruction must be loaded into the pipeline. See **8.10 The PC During Debug** on page 8-23 for details on calculating the branch.

Bit 33 of scan chain 1 is used to force ARM7TDM to resynchronize back to **MCLK**. The penultimate instruction of the debug sequence is scanned in with bit 33 set HIGH. The final instruction of the debug sequence is the branch, and this is scanned in with bit 33 LOW.

The core is then clocked to load the branch into the pipeline. Now, the RESTART instruction is selected in the TAP controller. When the state machine enters the RUN-TEST-IDLE state, the scan chain reverts back to system mode and clock resynchronization to **MCLK** occurs within ARM7TDM. ARM7TDM then resumes normal operation, fetching instructions from memory. This delay, until the state machine is in the RUN-TEST-IDLE state, allows conditions to be set up in other devices in a multiprocessor system without taking immediate effect. Then, when the RUN-TEST-IDLE state is entered, all the processors resume operation simultaneously.

8.10 The PC During Debug

So that ARM7TDM may be forced to branch back to the place at which program flow was interrupted by debug, the debugger must keep track of what happens to the PC.

There are five cases:

- breakpoints
- watchpoints
- watchpoint when another exception occurs
- debug request
- system speed access

8.10.1 Breakpoints

Entry to the debug state from a breakpoint advances the PC by 4 addresses, or 16 bytes. Each instruction executed in debug state advances the PC by 1 address, or 4 bytes. The normal way to exit from debug state after a breakpoint is to remove the breakpoint, and branch back to the previously breakpointed address.

For example, if ARM7TDM entered debug state from a breakpoint set on a given address and two debug-speed instructions were executed, a branch of -7 addresses must occur (4 for debug entry, +2 for the instructions, +1 for the final branch).

The following sequence shows the data scanned into scan chain 1. This is msb first, and so the first digit is the value placed in the **BREAKPT** bit, followed by the instruction data:

```
0 E0802000; ADD R2, R0, R0
1 E1826001; ORR R6, R2, R1
0 EFFFFFF9; B -7 (2's complement)
```

Once in debug state, a minimum of two instructions must be executed before the branch, although these may both be NOPs, for example:

```
MOV R0, R0
```

For small branches, the final branch could be replaced by a subtract with the PC as the destination:

```
SUB PC, PC, #28
```

8.10.2 Watchpoints

Returning to program execution after entering debug state from a watchpoint is done in the same way as the procedure described above. Debug entry adds 4 addresses to the PC, and every instruction adds 1 address. The difference is that because the instruction that caused the watchpoint has executed, the program returns to the next instruction.

8.10.3 Watchpoint with another exception

If a watchpointed access simultaneously causes a data abort, ARM7TDM enters debug state in abort mode. Entry into debug is held off until the core has changed into abort mode, and fetched the instruction from the abort vector.

A similar sequence is followed when an interrupt, or any other exception, occurs during a watchpointed memory access. ARM7TDM enters debug state in the exception's mode, and so the debugger must check to see whether this happened. The debugger can deduce whether an exception occurred by looking at the current and previous mode (in the CPSR and SPSR), and the value of the PC. If an exception did take place, the user should be given the choice of whether to service the exception before debugging.

Debug Interface

Exiting from debug state

Exiting debug state if an exception occurred is slightly different from the other cases. Here, entry to debug state causes the PC to be incremented by 3 addresses rather than 4, and this must be taken into account in the return branch calculation.

For example, suppose that an abort occurred on a watchpointed access and 10 instructions had been executed to determine this. The following sequence could be used to return to program execution:

```
0 E1A00000; MOV R0, R0
1 E1A00000; MOV R0, R0
0 EAFFFFF0; B -16
```

This forces a branch back to the abort vector, causing the instruction at that location to be refetched and executed.

Note *After the abort service routine, the instruction which caused the abort and watchpoint is re-executed. This generates the watchpoint and ARM7TDM enters debug state again.*

8.10.4 Debug request

Entry into debug state via a debug request is similar to a breakpoint. However, unlike a breakpoint, the last instruction will have completed execution and so must not be refetched on exit from debug state. Therefore, entry to debug state adds 3 addresses to the PC, and every instruction executed in debug state adds 1.

For example, suppose that the user has invoked a debug request, and decides to return to program execution straight away. The following sequence could be used:

```
0 E1A00000; MOV R0, R0
1 E1A00000; MOV R0, R0
0 EAFFFFFA; B -6
```

This restores the PC, and restarts the program from the next instruction.

8.10.5 System-speed access

If a system-speed access is performed during debug state, the value of the PC is increased by 3 addresses. As system-speed instructions access the memory system, it is possible for aborts to take place. If an abort occurs during a system-speed memory access, ARM7TDM enters abort mode before returning to debug state.

This is similar to an aborted watchpoint except that the problem is much harder to fix, because the abort was not caused by an instruction in the main program, and the PC does not point to the instruction which caused the abort. An abort handler usually looks at the PC to determine the instruction which caused the abort, and hence the abort address. In this case, the value of the PC is invalid, but the debugger should know what location was being accessed. Thus the debugger can be written to help the abort handler fix the memory system.

8.10.6 Summary of return address calculations

The calculation of the branch return address can be summarized as follows:

- For normal breakpoint and watchpoint, the branch is:
 - $(4 + N + 3S)$
- For entry through debug request (**DBGREQ**), or watchpoint with exception, the branch is:
 - $(3 + N + 3S)$

where:

N is the number of debug speed instructions executed (including the final branch)

S is the number of system speed instructions executed.

Debug Interface

8.11 Priorities and Exceptions

Because the normal program flow is broken when a breakpoint or a debug request occurs, debug can be thought of as being another type of exception. Some of the interaction with other exceptions has been described in earlier sections. This section summarizes these priorities.

8.11.1 Breakpoint with prefetch abort

When a breakpointed instruction fetch causes a prefetch abort, the abort is taken and the breakpoint is disregarded. Normally, prefetch aborts occur when, for example, an access is made to a virtual address which does not physically exist, and the returned data is therefore invalid.

In such a case, the operating system's normal action is to swap in the page of memory and return to the previously invalid address. Here, when the instruction is fetched, and providing the breakpoint is activated (it may be data-dependent), ARM7TDM enters debug state.

In this case, the prefetch abort takes higher priority than the breakpoint.

8.11.2 Interrupts

When ARM7TDM enters debug state, interrupts are automatically disabled. If interrupts are disabled during debug, ARM7TDM is never forced into an interrupt mode. Interrupts only have this effect on watchpointed accesses. They are ignored at all times on breakpoints.

If an interrupt was pending during the instruction prior to entering debug state, ARM7TDM enters debug state in the mode of the interrupt. Thus, on entry to debug state, the debugger cannot assume that ARM7TDM is in the expected mode of the user's program. It must check the PC, the CPSR and the SPSR to fully determine the reason for the exception.

Thus, debug takes higher priority than the interrupt, although ARM7TDM *remembers* that an interrupt has occurred.

8.11.3 Data aborts

When a data abort occurs on a watchpointed access, ARM7TDM enters debug state in abort mode. Thus, the watchpoint has higher priority than the abort, although, as in the case of interrupt, ARM7TDM remembers that the abort happened.

8.12 Scan Interface Timing

This section describes the scan interface timing.

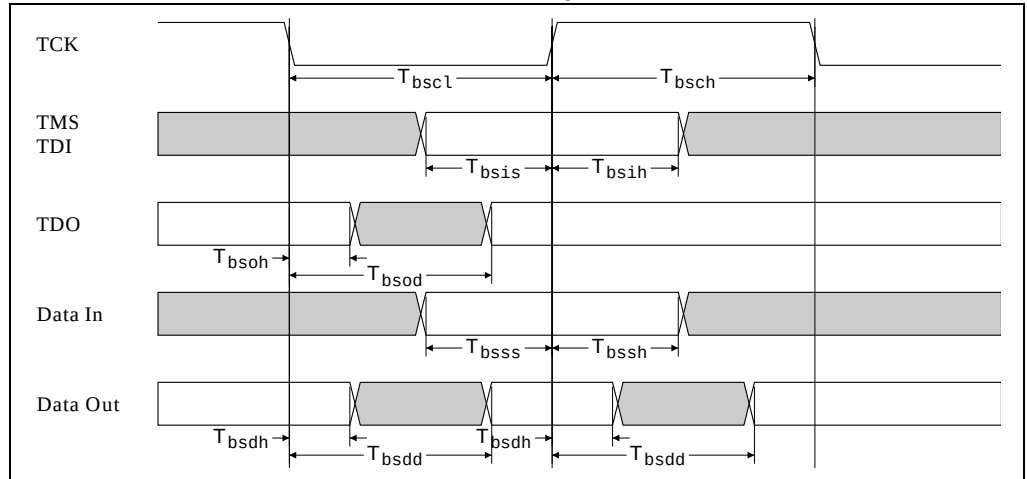


Figure 8-7: Scan general timing

In the following table, all units are ns. All delays are provisional and assume a process which achieves 33MHz **MCLK** maximum operating frequency.

Symbol	Parameter	Min	Type	Max	Notes
T_{bscl}	TCK low period				
T_{bsch}	TCK high period				
T_{bsis}	TDI,TMS setup to [TCr]				
T_{bsih}	TDI,TMS hold from [TCr]				
T_{bsoh}	TDO hold time				2
T_{bsod}	TCr to TDO valid				2
T_{bsss}	I/O signal setup to [TCr]				1
T_{bssh}	I/O signal hold from [TCr]				1
T_{bsdh}	data output hold time				2
T_{bsdd}	TCf to data output valid				2
T_{bsr}	Reset period				
T_{bse}	Output Enable time				2
T_{bsz}	Output Disable time				2

Table 8-2: JTAG Timing Parameters

Notes

- For correct data latching, the I/O signals (from the core and the pads) must be setup and held with respect to the rising edge of **TCK** in the CAPTURE-DR state of the INTEST and EXTEST instructions.
- Assumes that the data outputs are loaded with the AC test loads.

Debug Interface

Key

I	Input
O	Output
I/O	Input/Output

No	Signal	Type	No	Signal	Type
1	D[0]	I/O	28	D[27]	I/O
2	D[1]	I/O	29	D[28]	I/O
3	D[2]	I/O	30	D[29]	I/O
4	D[3]	I/O	31	D[30]	I/O
5	D[4]	I/O	32	D[31]	I/O
6	D[5]	I/O	33	BREAKPT	I
7	D[6]	I/O	34	NENIN	I
8	D[7]	I/O	35	NENOUT	O
9	D[8]	I/O	36	LOCK	O
10	D[9]	I/O	37	BIGEND	I
11	D[10]	I/O	38	DBE	I
12	D[11]	I/O	39	MAS[0]	O
13	D[12]	I/O	40	MAS[1]	O
14	D[13]	I/O	41	BL[0]	I
15	D[14]	I/O	42	BL[1]	I
16	D[15]	I/O	43	BL[2]	I
17	D[16]	I/O	44	BL[3]	I
18	D[17]	I/O	45	DCTL **	O
19	D[18]	I/O	46	nRW	O
20	D[19]	I/O	47	DBGACK	O
21	D[20]	I/O	48	CGENDBGACK	O
22	D[21]	I/O	49	nFIQ	I
23	D[22]	I/O	50	nIRQ	I
24	D[23]	I/O	51	nRESET	I
25	D[24]	I/O	52	ISYNC	I
26	D[25]	I/O	53	DBGRQ	I
27	D[26]	I/O	54	ABORT	I

Table 8-3: Scan Chain 0 Bit Positions

No	Signal	Type	No	Signal	Type
55	CPA	I	81	A[24]	O
56	nOPC	O	82	A[23]	O
57	IFEN	I	83	A[22]	O
58	nCPI	O	84	A[21]	O
59	nMREQ	O	85	A[20]	O
60	SEQ	O	86	A[19]	O
61	nTRANS	O	87	A[18]	O
62	CPB	I	88	A[17]	O
63	nM[4]	O	89	A[16]	O
64	nM[3]	O	90	A[15]	O
65	nM[2]	O	91	A[14]	O
66	nM[1]	O	92	A[13]	O
67	nM[0]	O	93	A[12]	O
68	nEXEC	O	94	A[11]	O
69	ALE	I	95	A[10]	O
70	ABE	I	96	A[9]	O
71	APE	I	97	A[8]	O
72	TBIT	O	98	A[7]	O
73	nWAIT	I	99	A[6]	O
74	A[31]	O	100	A[5]	O
75	A[30]	O	101	A[4]	O
76	A[29]	O	102	A[3]	O
77	A[28]	O	103	A[2]	O
78	A[27]	O	104	A[1]	O
79	A[26]	O	105	A[0]	O
80	A[25]	O			

Table 8-3: Scan Chain 0 Bit Positions (Continued)

Note **DCTL** is not described in this datasheet. **DCTL** is an output from the processor used to control the unidirectional data out latch, **DOUT[31:0]**. This signal is not visible from the periphery of ARM7TDM.

Debug Interface

8.13 Debug Timing

Symbol	Parameter	Min	Max
Ttdbgd	TCK falling to DBGACK , DBGRQI changing		
Ttpfd	TCKf to TAP outputs		
Ttpfh	TAP outputs hold time from TCKf		
Ttprd	TCKr to TAP outputs		
Ttprh	TAP outputs hold time from TCKr		
Ttckr	TCK to TCK1 , TCK2 rising		
Ttckf	TCK to TCK1 , TCK2 falling		
Tecapd	TCK to ECAPCLK changing		
Tdckf	DCLK induced: TCKf to various outputs valid		
Tdckfh	DCLK induced: Various outputs hold from TCKf		
Tdckr	DCLK induced: TCKr to various outputs valid		
Tdckrh	DCLK induced: Various outputs hold from TCKr		
Ttrstd	nTRSTf to TAP outputs valid		
Ttrsts	nTRSTr setup to TCKr		
Tsdttd	SDOUTBS to TDO valid		
Tclkbs	TCK to Boundary Scan Clocks		
Tshbsr	TCK to SHCLKBS, SHCLK2BS rising		
Tshbsf	TCK to SHCLKBS, SHCLK2BS falling		

Table 8-4: Debug Timing Parameters

Notes

- All delays are provisional and assume a process which achieves 33MHz **MCLK** maximum operating frequency.
- Assumes that the data outputs are loaded with the AC test loads.
All units are ns.

8.13.1 Tag scan chain

Bit	8KB Variant	Bit	4KB Variant
0	Bank3 Valid Flag	0	Bank3 Valid Flag
1	Bank3 Tag[20]	1	Bank3 Tag[21]
2	Bank3 Tag[19]	2	Bank3 Tag[20]
–	–	–	–
20	Bank3 Tag[1]	21	Bank3 Tag[1]
21	Bank3 Tag[0]	22	Bank3 Tag[0]
22	Bank2 Valid Flag	23	Bank2 Valid Flag
23	Bank2 Tag[20]	24	Bank2 Tag[21]
24	Bank2 Tag[19]	25	Bank2 Tag[20]
–	–	–	–
42	Bank2 Tag[1]	44	Bank2 Tag[1]
43	Bank2 Tag[0]	45	Bank2 Tag[0]
44	Bank1 Valid Flag	46	Bank1 Valid Flag
45	Bank1 Tag[20]	47	Bank1 Tag[21]
46	Bank1 Tag[19]	48	Bank1 Tag[20]
–	–	–	–
64	Bank1 Tag[1]	67	Bank1 Tag[1]
65	Bank1 Tag[0]	68	Bank1 Tag[0]
66	Bank0 Valid Flag	69	Bank0 Valid Flag
67	Bank0 Tag[20]	70	Bank0 Tag[21]
68	Bank0 Tag[19]	71	Bank0 Tag[20]
–	–	–	–
86	Bank0 Tag[1]	90	Bank0 Tag[1]
87	Bank0 Tag[0]	91	Bank0 Tag[0]

Table 8-5: Tag scan chain

9

EmbeddedICE Macrocell

This chapter describes the ARM740T EmbeddedICE module.

The ARM7TDM EmbeddedICE module, referred to simply as *EmbeddedICE*, provides integrated on-chip debug support for the ARM7TDM core.

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EmbeddedICE Macrocell

9.1 Overview

In this chapter ARM7TDM refers to the ARM7TDM core excluding the EmbeddedICE Macrocell.

EmbeddedICE is programmed in a serial fashion using the ARM7TDM TAP controller. It consists of two real-time watchpoint units, together with a control and status register. One or both watchpoint units can be programmed to halt the execution of instructions by the ARM7TDM core via its **BREAKPT** signal. Two independent registers, Debug Control and Debug Status, provide overall control of EmbeddedICE's operation.

Figure 9-1: ARM7TDM block diagram shows the relationship between the core, EmbeddedICE and the TAP controller.

Execution is halted when a match occurs between the values programmed into EmbeddedICE and the values currently appearing on the address bus, data bus and various control signals. Any bit can be masked so that its value does not affect the comparison.

Note Only those signals that are pertinent to EmbeddedICE are shown.

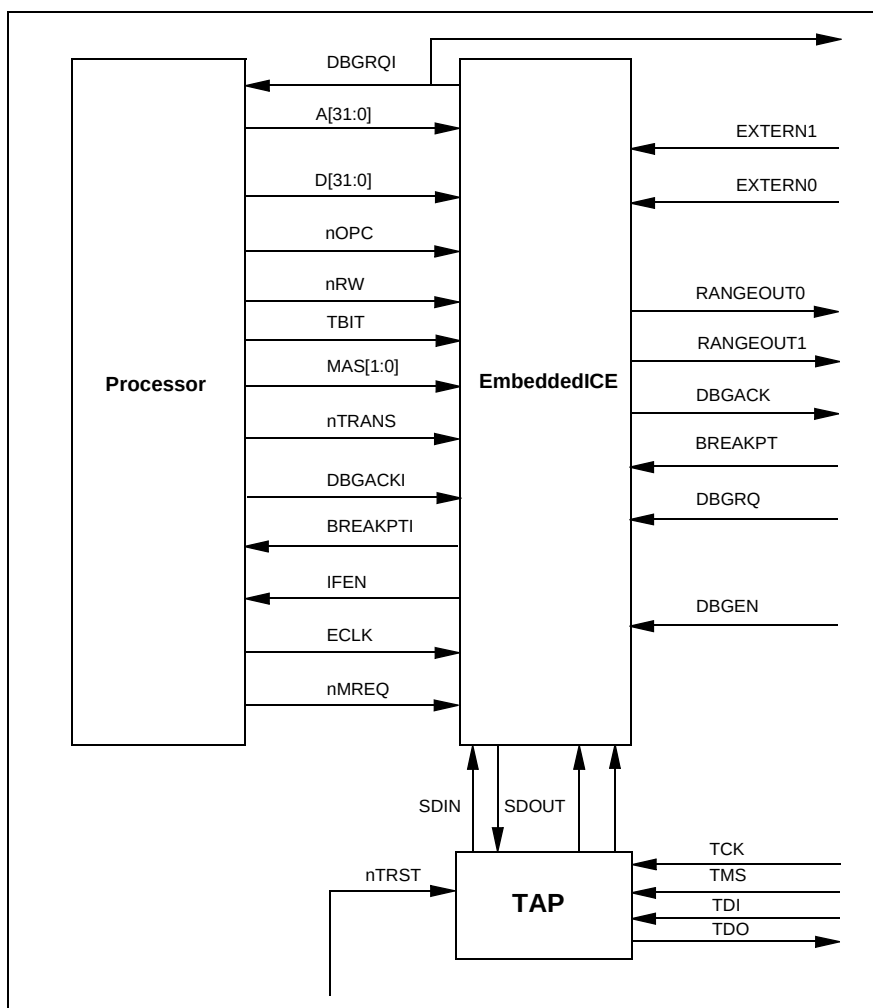


Figure 9-1: ARM7TDM block diagram

Either watchpoint unit can be configured to be a watchpoint (monitoring data accesses) or a breakpoint (monitoring instruction fetches). Watchpoints and breakpoints can be made to be data-dependent.

9.1.1 Disabling EmbeddedICE

EmbeddedICE may be disabled by wiring the **DBGEN** input LOW.

When **DBGEN** is LOW:

- **BREAKPT** and **DBGRQ** to the core are forced LOW
- **DBGACK** from the ARM7TDM is also forced LOW
- **IFEN** input to the core is forced HIGH, enabling interrupts to be detected by ARM7TDM

When **DBGEN** is LOW, EmbeddedICE is also put into a low-power mode.

9.1.2 EmbeddedICE timing

The **EXTERN1** and **EXTERN0** inputs are sampled by EmbeddedICE on the falling edge of **ECLK**. Sufficient set-up and hold time must therefore be allowed for these signals.

9.2 Watchpoint Registers

The two watchpoint units, known as *Watchpoint 0* and *Watchpoint 1*, each contain three pairs of registers:

- 1 Address Value and Address Mask
- 2 Data Value and Data Mask
- 3 Control Value and Control Mask

Each register is independently programmable and has its own address, as shown in **Table 9-1: Function and mapping of EmbeddedICE registers**:

Address	Width	Function
00000	3	Debug Control
00001	5	Debug Status
00100	6	Debug Comms Control Register
00101	32	Debug Comms Data Register
01000	32	Watchpoint 0 Address Value
01001	32	Watchpoint 0 Address Mask
01010	32	Watchpoint 0 Data Value
01011	32	Watchpoint 0 Data Mask
01100	9	Watchpoint 0 Control Value
01101	8	Watchpoint 0 Control Mask
10000	32	Watchpoint 1 Address Value
10001	32	Watchpoint 1 Address Mask
10010	32	Watchpoint 1 Data Value
10011	32	Watchpoint 1 Data Mask
10100	9	Watchpoint 1 Control Value
10101	8	Watchpoint 1 Control Mask

Table 9-1: Function and mapping of EmbeddedICE registers

9.2.1 Programming and reading watchpoint registers

A register is programmed by scanning data into the EmbeddedICE scan chain (scan chain 2). The scan chain consists of a 38-bit shift register comprising:

- a 32-bit data field
- a 5-bit address field
- a read/write bit

This is shown in **Figure 9-2: EmbeddedICE block diagram** on page 9-5.

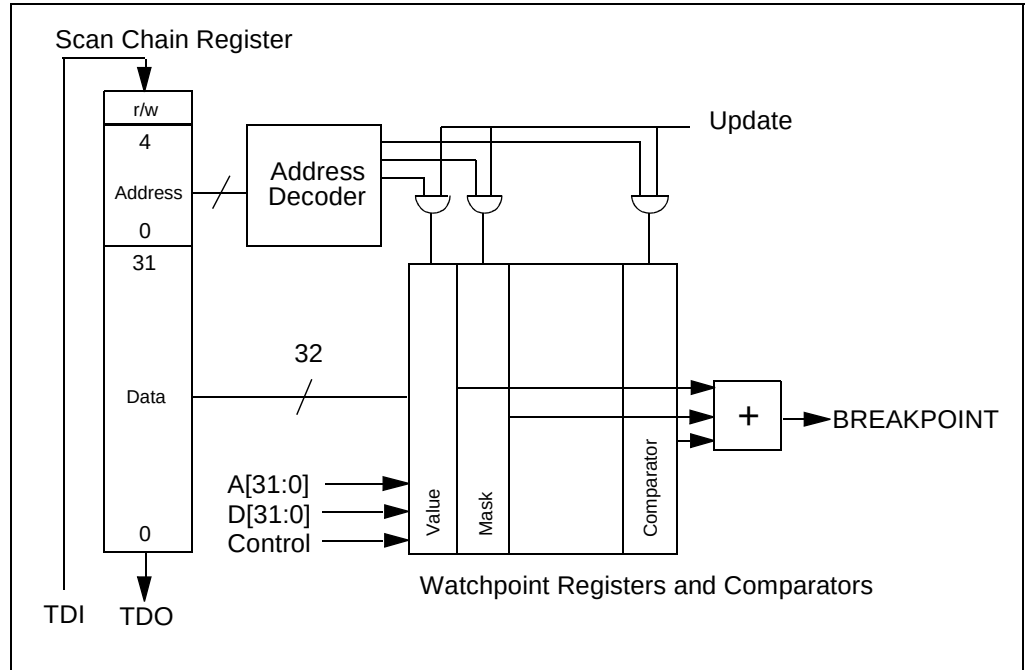


Figure 9-2: EmbeddedICE block diagram

The data to be written is scanned into the 32-bit data field, the address of the register into the 5-bit address field, and a 1 into the read/write bit.

A register is read by scanning its address into the address field and a 0 into the read/write bit. The 32-bit data field is ignored. The register addresses are shown in **Table 9-1: Function and mapping of EmbeddedICE registers** on page 9-4.

Note A read or write takes place when the TAP controller enters the UPDATE-DR state.

9.2.2 Using the mask registers

For each Value register in a register pair, there is a Mask register of the same format. Setting a bit to 1 in the Mask register has the effect of disregarding the corresponding bit in the Value register in the comparison. For example, if a watchpoint is required on a particular memory location but the data value is irrelevant, the Data Mask register can be programmed to 0xFFFFFFFF (all bits set to 1) to make the entire Data Bus field ignored.

Note The mask is an XNOR mask rather than a conventional AND mask. When a mask bit is set to 1, the comparator for that bit position always matches, irrespective of the value register or the input value.

Setting the mask bit to 0 means that the comparator only matches if the input value matches the value programmed into the value register.

9.2.3 The control registers

Control Value and Control Mask registers are mapped identically in the lower 8 bits. Bit 8 of the control value register is the **ENABLE** bit, which cannot be masked.

8	7	6	5	4	3	2	1	0
ENABLE	RANGE	CHAIN	EXTERN	nTRANS	nOPC	MAS[1]	MAS[0]	nRW

Figure 9-3: Watchpoint control value and mask format

The bits have the following functions:

- nRW

compares against the not-read/write signal from the core in order to detect the direction of bus activity. nRW is 0 for a read cycle and 1 for a write cycle.
- MAS[1:0]

compares against the **MAS[1:0]** signal from the core in order to detect the size of bus activity. The encoding is shown in the following table:

bit 1	bit 0	Data size
0	0	byte
0	1	halfword
1	0	word
1	1	(reserved)

Table 9-2: MAS[1:0] signal encoding

- nOPC

detects whether the current cycle is an instruction fetch (nOPC = 0) or a data access (nOPC = 1).
- nTRANS

compares against the not-translate signal from the core in order to distinguish between User mode (nTRANS = 0) and non-User mode (nTRANS = 1) accesses.
- EXTERN

is an external input to EmbeddedICE which allows the watchpoint to be dependent upon an external condition. The EXTERN input for Watchpoint 0 is labelled **EXTERN0** and the EXTERN input for Watchpoint 1 is labelled **EXTERN1**.
- CHAIN

can be connected to the chain output of another watchpoint in order to implement, for example, debugger requests of the form “breakpoint on address YYY only when in process XXX”.

In the ARM7TDM EmbeddedICE, the **CHAINOUT** output of Watchpoint 1 is connected to the CHAIN input of Watchpoint 0. The **CHAINOUT** output is derived from a latch; the address/control field comparator drives the write enable for the latch and the input to the latch is the value of the data field comparator. The **CHAINOUT** latch is cleared when the Control Value register is written or when nTRST is LOW.



RANGE	can be connected to the range output of another watchpoint register. In the ARM7TDM EmbeddedICE, the RANGEOUT output of Watchpoint 1 is connected to the RANGE input of Watchpoint 0. This allows the two watchpoints to be coupled for detecting conditions that occur simultaneously, for example, in range-checking.
ENABLE	If a watchpoint match occurs, the BREAKPT signal is asserted only when the ENABLE bit is set. This bit only exists in the value register: it cannot be masked.

For each of the bits [8:0] in the Control Value register, there is a corresponding bit in the Control Mask register. This removes the dependency on particular signals.

9.3 Programming Breakpoints

Breakpoints can be classified as hardware breakpoints or software breakpoints.

Hardware	These typically monitor the address value and can be set in any code, even in code that is in ROM or code that is self-modifying.
Software	These monitor a particular bit pattern being fetched from any address. One EmbeddedICE watchpoint can thus be used to support any number of software breakpoints. Software breakpoints can normally only be set in RAM because an instruction has to be replaced by the special bit pattern chosen to cause a software breakpoint.

9.3.1 Hardware breakpoints

To make a watchpoint unit cause hardware breakpoints (ie. on instruction fetches):

- 1 Program its Address Value register with the address of the instruction to be breakpointed.
- 2 Program the breakpoint bits as follows:

ARM state	program bits [1:0] of the Address Mask register to 1.
THUMB state	program bit 0 of the Address Mask to 1.

In both cases, the remaining bits are set to 0.
- 3 Program the Data Value register only if you require a data-dependent breakpoint, that is only if the actual instruction code fetched must be matched as well as the address. If the data value is not required, program the Data Mask register to 0xFFFFFFFF (all bits to 1), otherwise program it to 0x00000000.
- 4 Program the Control Value register with nOPC = 0.
- 5 Program the Control Mask register with nOPC = 0, all other bits to 1.
- 6 If you need to make the distinction between user and non-user mode instruction fetches, program the nTRANS Value and Mask bits as above.
- 7 If required, program the EXTERN, RANGE and CHAIN bits in the same way.

9.3.2 Software breakpoints

To make a watchpoint unit cause software breakpoints (that is, on instruction fetches of a particular bit pattern):

- 1 Program its Address Mask register to 0xFFFFFFFF (all bits set to 1) so that the address is disregarded.
- 2 Program the Data Value register with the particular bit pattern that has been chosen to represent a software breakpoint.

For a THUMB software breakpoint, the 16-bit pattern must be repeated in both halves of the Data Value register. For example, if the bit pattern is 0xDFFF, then 0xDFFFDFFF must be programmed. When a 16-bit instruction is fetched, EmbeddedICE only compares the valid half of the data bus against the contents of the Data Value register. In this way, a single Watchpoint register can be used to catch software breakpoints on both the upper and lower halves of the data bus.
- 3 Program the Data Mask register to 0x00000000.
- 4 Program the Control Value register with nOPC = 0.
- 5 Program the Control Mask register with nOPC = 0, all other bits to 1.

- 6 If you wish to make the distinction between user and non-user mode instruction fetches, program the nTRANS bit in the Control Value and Control Mask registers accordingly.

- 7 If required, program the EXTERN, RANGE and CHAIN bits in the same way.

Note *The address value register need not be programmed.*

Setting the breakpoint

To set the software breakpoint:

- 1 Read the instruction at the desired address and store it.
- 2 Write the special bit pattern representing a software breakpoint at the address.

Clearing the breakpoint

To clear the software breakpoint, restore the instruction to the address.

9.4 Programming Watchpoints

The above are just examples of how to program the watchpoint register to generate breakpoints and watchpoints; many other ways of programming the registers are possible. For instance, simple range breakpoints can be provided by setting one or more of the address mask bits.

To make a watchpoint unit cause watchpoints (ie. on data accesses):

- 1 Program its Address Value register with the address of the data access to be watchpointed.
- 2 Program the Address Mask register to 0x00000000.
- 3 Program the Data Value register only if you require a data-dependent watchpoint; ie. only if the actual data value read or written must be matched as well as the address. If the data value is irrelevant, program the Data Mask register to 0xFFFFFFFF (all bits set to 1) otherwise program it to 0x00000000.
- 4 Program the Control Value register with nOPC = 1, nRW = 0 for a read or nRW = 1 for a write, MAS[1:0] with the value corresponding to the appropriate data size.
- 5 Program the Control Mask register with nOPC = 0, nRW = 0, MAS[1:0] = 0, all other bits to 1. Note that nRW or MAS[1:0] may be set to 1 if both reads and writes or data size accesses are to be watchpointed respectively.
- 6 If you wish to make the distinction between user and non-user mode data accesses, program the nTRANS bit in the Control Value and Control Mask registers accordingly.
- 7 If required, program the EXTERN, RANGE and CHAIN bits in the same way.

9.4.1 Programming restriction

The EmbeddedICE watchpoint units should only be programmed when the clock to the core is stopped. This can be achieved by putting the core into the debug state.

The reason for this restriction is that if the core continues to run at **ECLK** rates when EmbeddedICE is being programmed at **TCK** rates, it is possible for the **BREAKPT** signal to be asserted asynchronously to the core.

This restriction does not apply if **MCLK** and **TCK** are driven from the same clock, or if it is known that the breakpoint or watchpoint condition can only occur some time after EmbeddedICE has been programmed.

Note *This restriction does not apply to the Debug Control or Status Registers.*

9.5 Debug Control Register

The Debug Control Register is 3 bits wide.

- If the register is accessed for a write (with the read/write bit HIGH), the control bits are written.
- If the register is accessed for a read (with the read/write bit LOW), the control bits are read.

The function of each bit in this register is as follows:

2	1	0
INTDIS	DBGRQ	DBGACK

Figure 9-4: Debug control register format

Bits 1 and 0 allow the values on **DBGRQ** and **DBGACK** to be forced.

DBGRQ

As shown in **Figure 9-6: Structure of TBIT, NMREQ, DBGACK, DBGRQ and INTDIS bits** on page 9-13, the value stored in bit 1 of the control register is synchronized and then ORed with the external **DBGRQ** before being applied to the processor. The output of this OR gate is the signal **DBGRQI** which is brought out externally from the macrocell.

The synchronization between control bit 1 and **DBGRQI** is to assist in multiprocessor environments. The synchronization latch only opens when the TAP controller state machine is in the RUN-TEST/IDLE state. This allows an *enter debug* condition to be set up in all the processors in the system while they are still running. Once the condition is set up in all the processors, it can then be applied to them simultaneously by entering the RUN-TEST/IDLE state.

DBGACK

In the case of **DBGACK**, the value of **DBGACK** from the core is ORed with the value held in bit 0 to generate the external value of **DBGACK** seen at the periphery of ARM7TDM. This allows the debug system to signal to the rest of the system that the core is still being debugged even when system-speed accesses are being performed (in which case the internal **DBGACK** signal from the core is LOW).

INTDIS

If bit 2 (INTDIS) is asserted, the interrupt enable signal (**IFEN**) of the core is forced LOW. Thus all interrupts (IRQ and FIQ) are disabled during debugging (**DBGACK** = 1) or if the INTDIS bit is asserted. The **IFEN** signal is driven according to the following table:

DBGACK	INTDIS	IFEN
0	0	1
1	x	0
x	1	0

Table 9-3: IFEN signal control

9.6 Debug Status Register

The Debug Status Register is 5 bits wide.

- If it is accessed for a write (with the read/write bit set HIGH), the status bits are written.
- If it is accessed for a read (with the read/write bit LOW), the status bits are read.

4	3	2	1	0
TBIT	nMREQ	IFEN	DBGQR	DBGACK

Figure 9-5: Debug status register format

The function of each bit in this register is as follows:

- | | |
|--------------|--|
| Bits 1 and 0 | allow the values on the synchronized versions of DBGQR and DBGACK to be read. |
| Bit 2 | allows the state of the core interrupt enable signal (IFEN) to be read. As the capture clock for the scan chain may be asynchronous to the processor clock, the DBGACK output from the core is synchronized before being used to generate the IFEN status bit. |
| Bit 3 | allows the state of the nMREQ signal from the core (synchronised to TCK) to be read. This allows the debugger to determine that a memory access from the debug state has completed. |
| Bit 4 | allows TBIT to be read. This enables the debugger to determine what state the processor is in, and which instructions to execute. |

The structure of the debug status register bits is shown in **Figure 9-6: Structure of TBIT, nMREQ, DBGACK, DBGQR and INTDIS bits** on page 9-13.

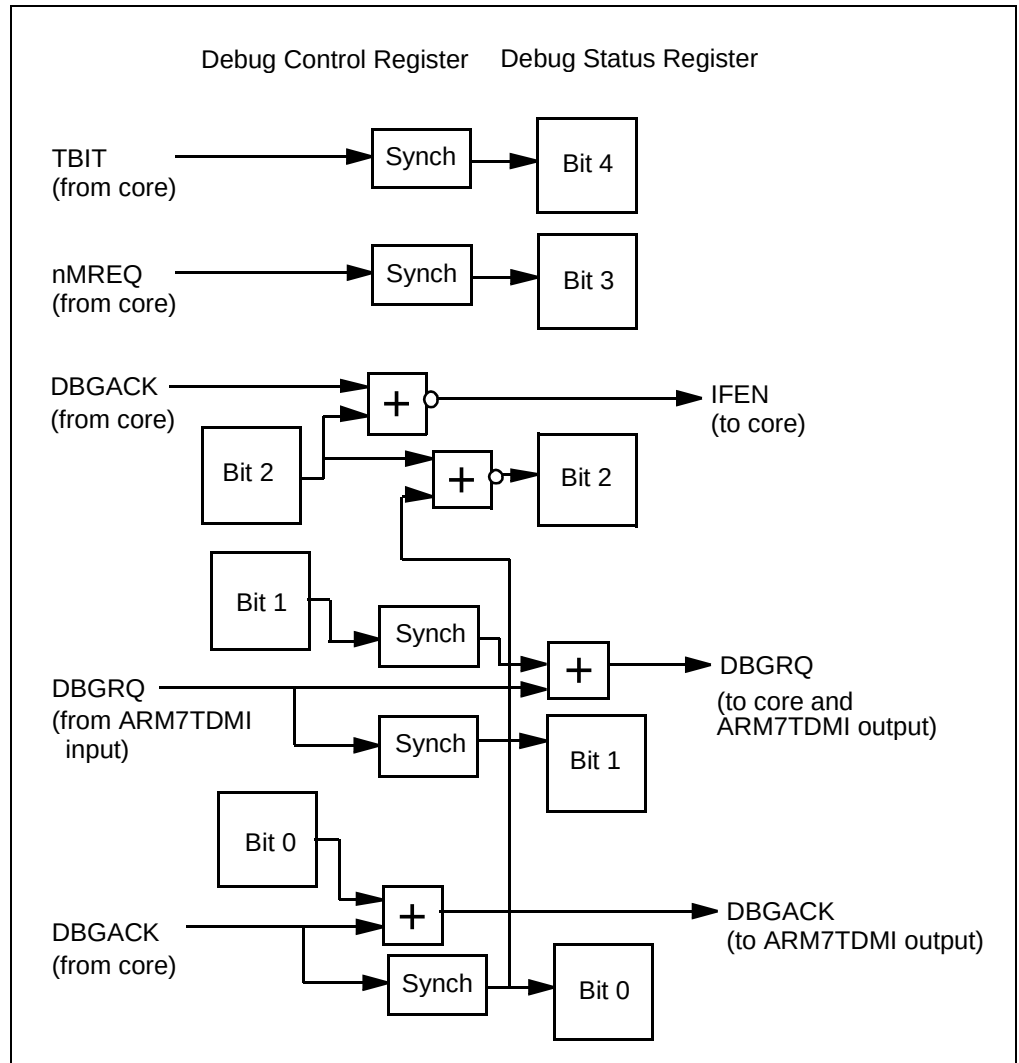


Figure 9-6: Structure of TBIT, NMREQ, DBGACK, DBGRQ and INTDIS bits

9.7 Coupling Breakpoints and Watchpoints

Watchpoint units 1 and 0 can be coupled together via the **CHAIN** and **RANGE** inputs.

CHAIN	enables watchpoint 0 to be triggered only if watchpoint 1 has previously matched.
RANGE	enables simple range checking to be performed by combining the outputs of both watchpoints.

9.7.1 Example

Let:

Av[31:0]	be the value in the Address Value Register
Am[31:0]	be the value in the Address Mask Register
A[31:0]	be the Address Bus from the ARM7TDM
Dv[31:0]	be the value in the Data Value Register
Dm[31:0]	be the value in the Data Mask Register
D[31:0]	be the Data Bus from the ARM7TDM
Cv[8:0]	be the value in the Control Value Register
C _m [7:0]	be the value in the Control Mask Register
C[9:0]	be the combined Control Bus from the ARM7TDM, other watchpoint registers and the EXTERN signal.

CHAINOUT signal

The **CHAINOUT** signal is then derived as follows:

WHEN (({Av[31:0], Cv[4:0]} XNOR {A[31:0], C[4:0]}) OR {Am[31:0], Cm[4:0]}) == 0xFFFFFFFF)

CHAINOUT = ((({Dv[31:0], Cv[6:4]} XNOR {D[31:0], C[7:5]}) OR {Dm[31:0], Cm[7:5]}) == 0x7FFFFFFF)

The **CHAINOUT** output of watchpoint register 1 provides the **CHAIN** input to Watchpoint 0. This allows for quite complicated configurations of breakpoints and watchpoints.

For example, the request by a debugger to breakpoint on the instruction at location YYY when running process XXX in a multiprocess system.

If the current process ID is stored in memory, the above function can be implemented with a watchpoint and breakpoint chained together. The watchpoint address is set to a known memory location containing the current process ID, the watchpoint data is set to the required process ID and the ENABLE bit is set to "off".

The address comparator output of the watchpoint is used to drive the write enable for the **CHAINOUT** latch, the input to the latch being the output of the data comparator from the same watchpoint. The output of the latch drives the **CHAIN** input of the breakpoint comparator. The address YYY is stored in the breakpoint register and when the **CHAIN** input is asserted and the breakpoint address matches, the breakpoint triggers correctly.

RANGEOUT signal

The **RANGEOUT** signal is then derived as follows:

$$\text{RANGEOUT} = (((\{Av[31:0], Cv[4:0]\} \text{ XOR } \{A[31:0], C[4:0]\}) \text{ OR } \{Am[31:0], Cm[4:0]\}) == 0xFFFFFFFF) \text{ AND } (((\{Dv[31:0], Cv[7:5]\} \text{ XOR } \{D[31:0], C[7:5]\}) \text{ OR } \{Dm[31:0], Cm[7:5]\}) == 0x7FFFFFFF)$$

The **RANGEOUT** output of watchpoint register 1 provides the **RANGE** input to watchpoint register 0. This allows two breakpoints to be coupled together to form range breakpoints.

Note *The selectable ranges are restricted to being powers of 2.*

Example

If a breakpoint is to occur when the address is in the first 256 bytes of memory, but not in the first 32 bytes, the watchpoint registers should be programmed as follows:

- 1 Watchpoint 1 is programmed with an address value of 0x00000000 and an address mask of 0x0000001F. The ENABLE bit is cleared. All other Watchpoint 1 registers are programmed as normal for a breakpoint. An address within the first 32 bytes causes the RANGE output to go HIGH but the breakpoint is not triggered.
- 2 Watchpoint 0 is programmed with an address value of 0x00000000 and an address mask of 0x000000FF. The ENABLE bit is set and the RANGE bit programmed to match a 0. All other Watchpoint 0 registers are programmed as normal for a breakpoint.

If Watchpoint 0 matches but Watchpoint 1 does not (for example, if the **RANGE** input to Watchpoint 0 is 0), the breakpoint is triggered.

9.8 Debug Communications Channel

ARM7TDM's EmbeddedICE contains a communication channel for passing information between the target and the host debugger. This is implemented as coprocessor 14.

The communications channel consists of:

- a 32-bit wide Comms Data Read register.
- a 32-bit wide Comms Data Write register.
- 6-bit wide Comms Control register for synchronized handshaking between the processor and the asynchronous debugger.

These registers live in fixed locations in EmbeddedICE's memory map (as shown in **Table 9-1: Function and mapping of EmbeddedICE registers** on page 9-4) and are accessed from the processor via MCR and MRC instructions to coprocessor 14.

9.8.1 Debug comms channel registers

The Debug Comms Control register is read-only and allows synchronized handshaking between the processor and the debugger.

31	30	29	28	...	1	0
0	0	0	1	...	W	R

Figure 9-7: Debug comms control register

The function of each register bit is described below:

- Bits [31:28]

contain a fixed pattern which denotes the EmbeddedICE version number, in this case 0001.
- Bit 1

denotes whether the Comms Data Write register is free (from the processor's point of view). From the processor's point of view:
 - if the Comms Data Write register is free (W=0), new data may be written.
 - if it is not free (W=1), the processor must poll until W=0.From the debugger's point of view, if W=1, new data has been written which may then be scanned out.
- Bit 0

denotes whether there is some new data in the Comms Data Read register. From the processor's point of view:
 - if R=1, there is some new data which may be read via an MRC instruction.From the debugger's point of view:
 - if R=0, the Comms Data Read register is free and new data may be placed there through the scan chain.
 - if R=1, this denotes that data previously placed there through the scan chain has not been collected by the processor and so the debugger must wait.
- From the debugger's point of view, the registers are accessed via the scan chain in the usual way. From the processor's point of view, these registers are accessed via coprocessor register transfer instructions.
- 9-16
- ARM740T Datasheet
- ARM DDI 0008E
-
- Open Access – Preliminary

Instructions

The following instructions should be used.

This instruction returns the Debug Comms Control register into Rd:

```
MRC CP14, 0, Rd, C0, C0
```

This instruction writes the value in Rn to the Comms Data Write register:

```
MCR CP14, 0, Rn, C1, C0
```

This instruction returns the Debug Data Read register into Rd:

```
MRC CP14, 0, Rd, C1, C0
```

Note *As the THUMB instruction set does not contain coprocessor instructions, it is recommended that these are accessed via SWI instructions when in THUMB state.*

9.8.2 Communications via the comms channel

Communication between the debugger and the processor occurs as follows:

- 1 When the processor wishes to send a message to EmbeddedICE, it first checks that the Comms Data Write register is free for use.
- 2 This is done by reading the Debug Comms Control register to check that the W bit is clear:
 - If it is clear, the Comms Data Write register is empty and a message is written by a register transfer to the coprocessor. The action of this data transfer automatically sets the W bit.
 - If it is set, this implies that previously-written data has not been picked up by the debugger and the processor must poll until the W bit is clear.
- 3 Because the data transfer occurs from the processor to the Comms Data Write register, the W bit is set in the Debug Comms Control register.
- 4 When the debugger polls this register it sees a synchronized version of both the R and W bit.
 - When the debugger sees that the W bit is set, it can read the Comms Data Write register and scan the data out.
 - The action of reading this data register clears the W bit of the Debug Comms Control register. At this point, the communications process may begin again.

9.8.3 Message transfer

Message transfer from the debugger to the processor is carried out in a similar fashion:

- 1 The debugger polls the R bit of the Debug Comms Control register:
 - If the R bit is LOW, the Data Read register is free and so data can be placed there for the processor to read.
 - If the R bit is set, previously deposited data has not yet been collected and so the debugger must wait.
- 2 When the Comms Data Read register is free, data is written there via the scan chain. The action of this write sets the R bit in the Debug Comms Control register.
- 3 When the processor polls this register, it sees an **MCLK** synchronized version.
 - If the R bit is set, this denotes that there is data waiting to be collected, and this can be read via a **CPRT** load. The action of this load clears the R bit in the Debug Comms Control register.
 - If the R bit is clear, this denotes that the data has been taken and the process may now be repeated.

10

Bus Clocking

This chapter describes the bus interface clocking.

10.1	Introduction	10-2
10.2	Fastbus Extension	10-3
10.3	Standard Mode	10-4

Bus Clocking

10.1 Introduction

The ARM740T bus interface can be operated using either:

- the standard mode of operation
- the new fastbus extension

As the ARM740T is a fully static design, the clock can be stopped indefinitely in either mode of operation. Care should be taken to ensure that the memory system does not dissipate power in the state in which it is stopped.

10.1.1 Standard mode

For designs using low-cost, low-speed memory, and wishing to operate the core at a faster speed, it is recommended that you use standard mode.

This mode consists of:

- two clocks, **FCLK** and **BCLK**
- synchronous or fully asynchronous operation

10.1.2 Fastbus extension

For new designs, you can operate the device using the fastbus extension. In fastbus mode, the device is clocked off a single clock, and the bus is operated at the same frequency as the core. This allows the bus interface to be clocked faster than if the device is operated in standard mode. It is recommended that you use this mode of operation in systems with high-speed memory and a single clock.

This mode consists of:

- single device clock
- increased maximum **BCLK** frequency

10.2 Fastbus Extension

Using the fastbus extension, the ARM740T has a single input clock, **BCLK**. This is used to clock the internals of the device, and qualified by **BWAIT**, controls the memory interface:

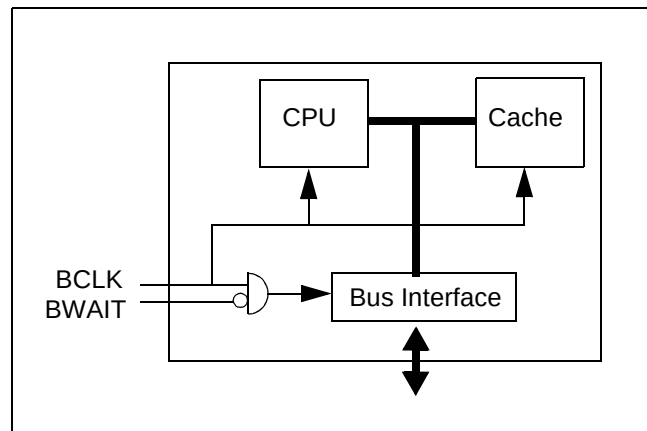


Figure 10-1: Conceptual device clocking using the fastbus extension

When operating the device with **FASTBUS** HIGH, the input **FCLK** and **SnA** are not used.

Note *To prevent unwanted power dissipation, ensure that they do not float to an undefined level. New designs should tie these signals LOW for compatibility with future products.*

10.2.1 Using **BWAIT**

The **BWAIT** signal is used to insert entire **BCLK** cycles into the bus cycle timing. **BWAIT** may only change when **BCLK** is LOW, and extends the memory access by inserting **BCLK** cycles into the access whilst **BWAIT** is asserted.

Figure 11-4: Use of the **BWAIT pin to stop ARM740T for 1 BCLK cycle** on page 11-8 shows the use of **BWAIT** in more detail.

Memory cycles

It is preferable to use **BWAIT** to extend memory cycles, rather than stretching **BCLK** externally to the device because it is possible for the core to be accessing the Cache while bus activity is occurring. This allows the maximum performance, as the Core can continue execution in parallel with the memory bus activity. All **BCLK** cycles are available to the CPU and Cache, regardless of the state of **BWAIT**.

In some circumstances, it may be desirable to stretch **BCLK** phases in order to match memory timing which is not an integer multiple of **BCLK**. There are certain cases where this results in a higher performance than using **BWAIT** to extend the access by an integer number of cycles.

CPU and Cache operation

CPU and Cache operation can only continue in parallel with buffered writes to the external bus. For all read accesses, the CPU is stalled until the bus activity has completed. So, if read accesses can be achieved faster by stretching **BCLK** rather than using **BWAIT**, this results in improved performance. An example of where this may be useful would be to interface to a ROM which has a cycle time of 2.5 times the **BCLK** period.

Asynchronous mode

In this mode, **FCLK** and **BCLK** may be completely asynchronous. You should select this mode by tying **SnA** LOW when the two clocks are of unrelated frequency.

There is a synchronisation penalty whenever the internal core clock switches between the two input clocks. This penalty is symmetrical and varies between nothing and a whole period of the clock to which the core is resynchronizing:

- when changing from **FCLK** to **BCLK**, the average resynchronization penalty is half an **BCLK** period
- when changing from **BCLK** to **FCLK**, the average resynchronization penalty is half an **FCLK** period.

Synchronous mode

You select this mode by tying **SnA** HIGH. In this mode, here is a tightly defined relationship between **FCLK** and **BCLK**, in that **BCLK** may only make transitions on the falling edge of **FCLK**. Some jitter between the two clocks is permitted, but **BCLK** must meet the setup and hold requirements relative to **FCLK**.

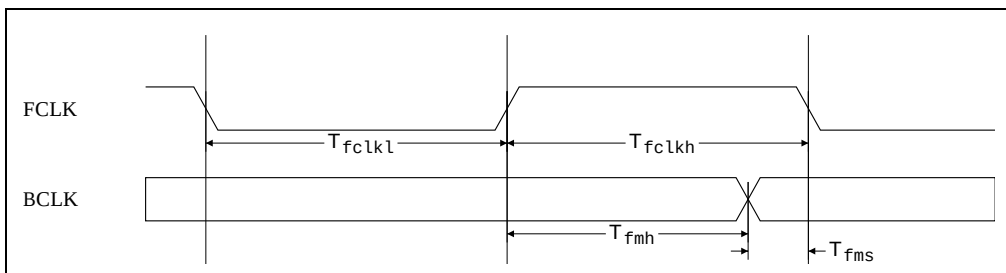


Figure 10-3: Relationship of FCLK and BCLK in synchronous mode

This chapter describes the operation of the AMBA bus interface.

In normal operation, the ARM740T is an *Advanced System Bus (ASB)* bus master. As a bus master it performs a subset of the possible ASB cycle types.

The ASB is further described in the *AMBA Specification*, ARM IHI 0001.

11.1	ASB Bus Interface Signals	11-2
11.2	Cycle Types	11-3
11.3	Addressing Signals	11-6
11.4	Memory Request Signals	11-6
11.5	Data Signal Timing	11-6
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11.7	Maximum Sequential Length	11-9
11.8	Read-Lock-Write	11-9
11.9	Big-Endian / Little-Endian Operation	11-10
11.10	Multi-Master Operation	11-13

AMBA Interface

11.1 ASB Bus Interface Signals

The signals in the ASB interface can be grouped into four categories:

Addressing signals	BA[31:0] BWRITE BSIZE BLOK
Memory request signals	BTRAN[1:0]
Data sampled signals	BD[31:0]
Slave response signals	BERROR BWAIT BLAST

System Arbiter

In addition to these signals, there are also three signals interfacing to the system arbiter and control logic:

AGNT	selects the ARM as a test bus master
AREQ	indicates that the ARM740T requires bus mastership
DSEL	selects the ARM as a test bus slave

11.2 Cycle Types

In normal operation, the ARM740T bus interface can perform two types of cycle:

- address cycles
- sequential cycles

These cycles are differentiated by the pipelined signal **BTRAN[1:0]**. Conventionally, cycles are considered to start from the falling edge of **BCLK**, and this is how they are shown in all diagrams.

These cycle types are a subset of the possible ASB cycle types. Other cycle types can be forced by the use of the Slave Response signals. See the (*AMBA Specification* ARM IHI 0001) for more details.

The Addressing and Memory Request signals are pipelined ahead of the Data Addressing by a phase (1/2 a cycle), and **BTRAN[1:0]** by a cycle. This advance information allows the implementation of efficient memory systems.

11.2.1 Single-word memory access

A simple single-word memory access is shown in **Figure 11-1: Simple single-cycle access**.

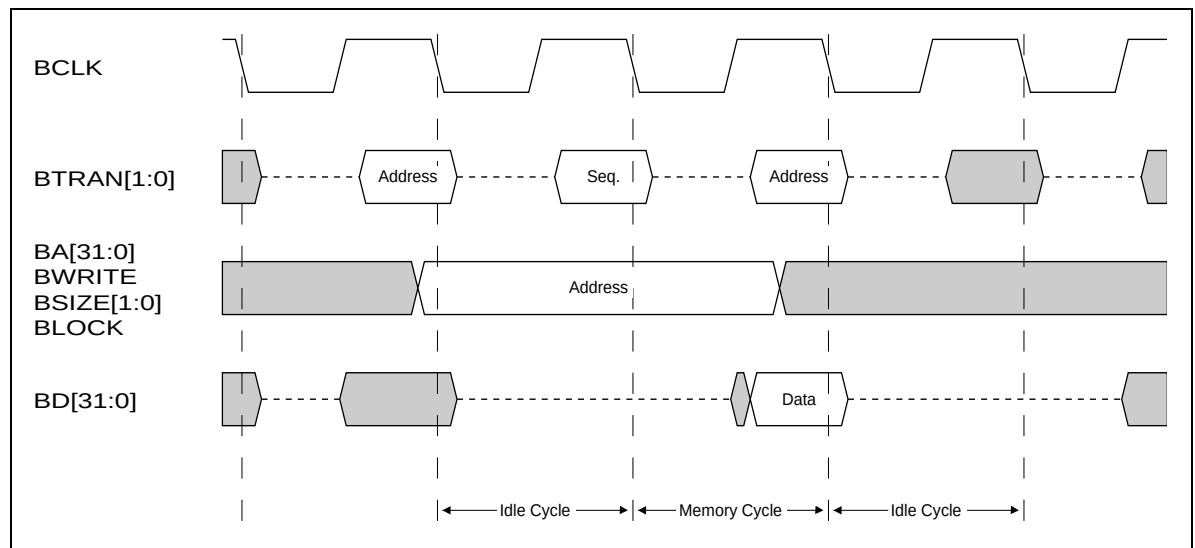


Figure 11-1: Simple single-cycle access

The access starts with the address being broadcast. This can be used for decoding, but the access is not committed until **BTRAN[1:0]** (Bus Transaction Type) signals a *sequential* cycle in the following HIGH phase of **BCLK**. This indicates that the next cycle is a memory access cycle.

In this example, **BTRAN[1:0]** returns to Address after a single cycle, indicating that there will be a single memory *access* cycle, followed by an address cycle. The data is transferred on the falling edge of **BCLK** at the end of the sequential cycle.

Therefore, a memory access consists of:

- an address cycle, with a valid address
- a memory cycle with the same address

The initial address cycle allows the memory controller more time to decode the address. See **Table 11-1: BTRAN[1:0] encoding** on page 11-6 for the encoding of **BTRAN[1:0]**.

AMBA Interface

11.2.2 Sequential accesses

ARM740T can perform sequential bursts of accesses. These consist of:

- an address cycle and a sequential cycle, as shown previously,
- further sequential cycles to:
 - incrementing word addresses (ie. a , $a+4$, $a+8$ etc.), or
 - halfword addresses (ie. a , $a+2$, $a+4$ etc.)

See **Figure 11-2: Simple sequential access** on page 11-4. After the initial address cycle, the address is pipelined by half a bus cycle from the data.

Note **BTRAN[1:0]** is pipelined by a bus cycle from the data. If **BWAIT** is being used to stretch cycles, **BTRAN[1:0]** no longer refers to the next **BCLK** cycle, but rather to the next bus cycle. See **11.6.2 BWAIT** on page 11-7.

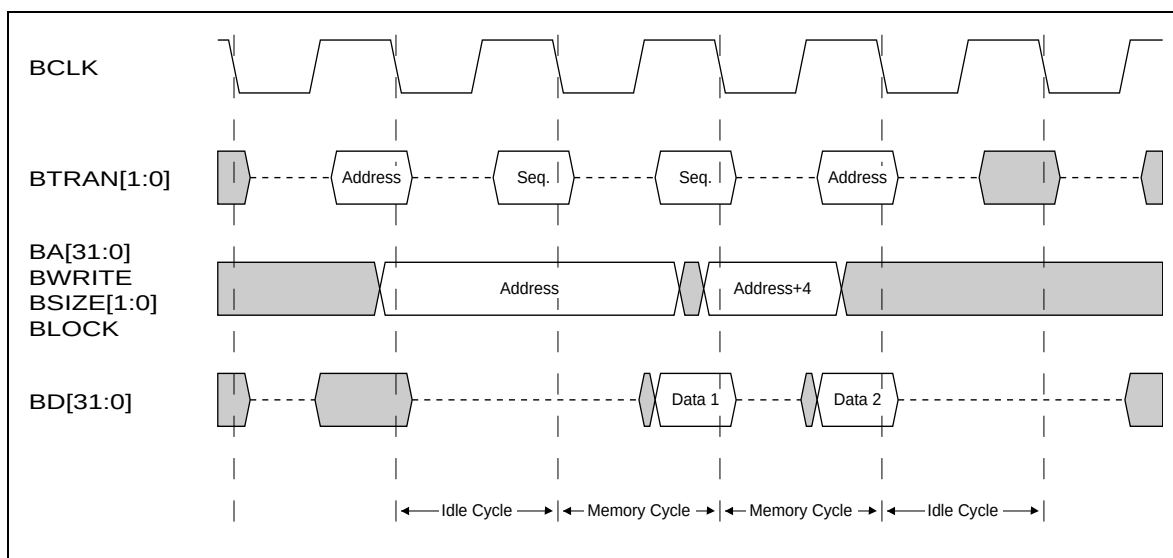


Figure 11-2: Simple sequential access

Sequential bursts can occur on word or halfword accesses, and are always in the same direction; that is Read (**BWRITE** LOW) or Write (**BWRITE** HIGH).

A memory controller should always qualify the use of the address with **BTRAN[1:0]**. There are certain circumstances in which a new address can be broadcast on the address bus, but **BTRAN[1:0]** does not signal a *sequential* access. This only happens when an internal (Protection Unit generated) abort occurs.

11.2.3 Bus accesses

The minimum interval between bus accesses can occur after a buffered write. In this case, there may only be a single address cycle between two memory cycles to nonsequential addresses. This means that the address for the second access is broadcast on **BA[31:0]** during the HIGH phase of the final memory cycle of the buffered write.

See **Figure 11-3: Minimum interval between bus accesses** for more information.

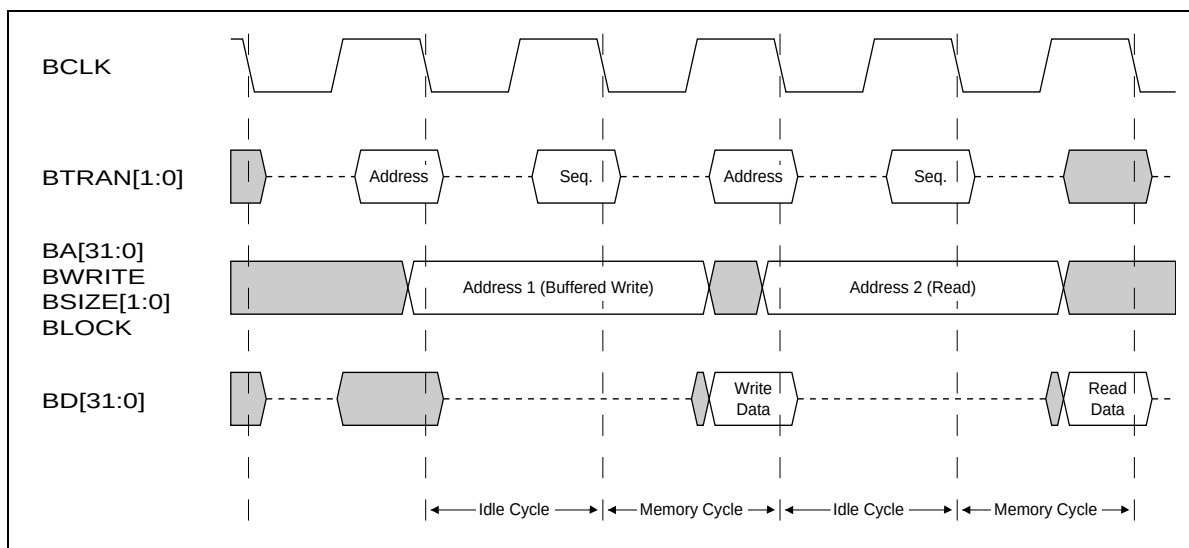


Figure 11-3: Minimum interval between bus accesses

This is the closest case of back-to-back cycles on the bus, and the memory controller should be designed to handle this case. In high-speed systems one solution is to use **BWAIT** to increase the decode and access time available for the second access.

Note Memory and peripheral strobes should not be direct decodes of the address bus. This could result in their changing during the last cycle of a write burst.

AMBA Interface

11.3 Addressing Signals

Memory accesses may be read or write, and are differentiated by the signal **BWRITE**.

BWRITE may not change during a sequential access, so if a read from address A is followed immediately by a write to address (A+4), the write to address (A+4) is performed on the bus as a non-sequential access.

In the same way, any memory access may be a word, a half-word or a byte. These are differentiated by the signal **BSIZE[1:0]**. Again, **BSIZE[1:0]** may not change during sequential accesses. It is not possible to perform sequential byte accesses.

In order to reduce system power consumption, the addressing signals are left with their current values at the end of an access, until the next access occurs.

After a buffered write, there may be only a single address cycle between the two memory cycles. In this case, the next non-sequential address is broadcast in the last cycle of the previous access. This is the worst case for address decoding, as shown in **Figure 11-3: Minimum interval between bus accesses** on page 11-5.

11.4 Memory Request Signals

The memory request signals, **BTRAN[1:0]** are pipelined by 1 bus cycle, and refer to the next bus cycle.

Care must be taken when de-pipelining these signals if **BWAIT** is being used, as they always refer to the following bus cycle, rather than the following **BCLK** cycle. **BWAIT** stretches the bus cycle by an integer number of **BCLK** cycles. See **11.6.2 BWAIT** on page 11-7.

BTRAN[1:0]	Cycle Type	Description	Note
00	Address	Address transfer or idle cycle	
01		Reserved	
10	Non-Sequential	Non-Sequential Data transfer cycle	1
11	Sequential	Sequential Data transfer cycle	

Table 11-1: BTRAN[1:0] encoding

Note 1 This cycle can only occur as a result of the slave response signals. In normal operation, ARM740T does not generate this cycle type.

11.5 Data Signal Timing

During a read access, the data is sampled on the falling edge of **BCLK** at the end of the sequential cycle. During a write access, the data on **BD[31:0]** is timed off the falling edge of **BCLK** at the start of the memory cycle. If **BWAIT** is being used to stretch this cycle, the data is valid from the falling edge of **BCLK** at the end of the previous cycle, when **BWAIT** was HIGH. See **11.6.2 BWAIT** on page 11-7.

Note *In a low-power system, you must ensure that the databus is not allowed to float to an undefined level. This causes power to be dissipated in the inputs of devices connected to the bus. This is particularly important when a system is put into a low-power sleep mode. It is recommended that one set of Databus drivers in the system is left enabled during sleep to hold the bus at a defined level.*

11.6 Slave Response Signals

11.6.1 BERROR

The **BERROR** signal is sampled on the rising edge of **BCLK** during a sequential cycle, on both read and write accesses. The effect of **BERROR** on the operation of the ARM740T is discussed in **3.7 Exceptions** on page 3-11.

BERROR can be flagged on any sequential cycle; however, it is ignored on buffered writes, which cannot be aborted.

Linefetches

The effect of **BERROR** during linefetches is slightly different to that during other access.

During a linefetch the ARM740T fetches four words of data, regardless of which words of data were requested by the ARM core, and the rest of the words are fetched speculatively.

- If **BERROR** is asserted on a word which was requested by the ARM core, the abort functions normally.
- If the abort is signalled on a word which was not requested by the ARM core, the access is not aborted, and program flow is not interrupted.

Regardless of which word was aborted, the line of data is not placed in the cache as it is assumed to contain invalid data.

11.6.2 BWAIT

The **BWAIT** pin can be used to extend memory accesses in whole cycle increments.

BWAIT is driven by the selected slave during the LOW phase of **BCLK**. When a slave cannot complete an access in the current cycle, it drives **BWAIT** HIGH to stall the ARM740T.

BWAIT does not prevent changes in **BTRAN[1:0]** and write data on **BD[31:0]** during the cycle in which it was asserted HIGH. Changes in these signals are then prevented until the **BCLK** HIGH phase after **BWAIT** was taken LOW. The addressing signals do not change from the rising **BCLK** edge when **BWAIT** goes HIGH, until the next **BCLK** HIGH phase after **BWAIT** returns LOW.

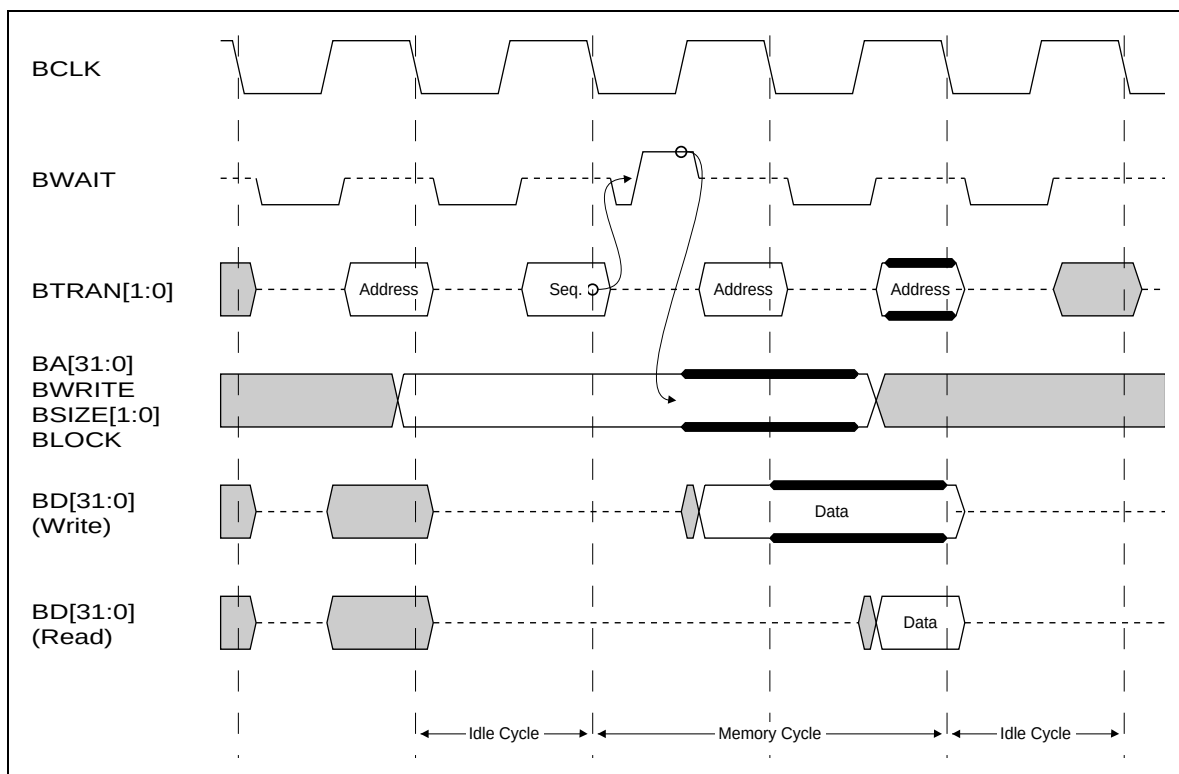


Figure 11-4: Use of the BWAIT pin to stop ARM740T for 1 BCLK cycle

In **Figure 11-4: Use of the BWAIT pin to stop ARM740T for 1 BCLK cycle**, the heavy bars indicate the cycle for which signals are stable as a result of asserting **BWAIT**.

The signal **BTRAN[1:0]** is pipelined by one bus cycle. This pipelining should be taken into account when these signals are being decoded. The value of **BTRAN[1:0]** indicates whether the next bus cycle is a data cycle or an address cycle.

As bus cycles are stretched by **BWAIT** the boundary between bus cycles is determined by the falling edge of **BCLK** when **BWAIT** was sampled as LOW on the rising edge of **BCLK**. A useful rule of thumb is to sample the value of **BTRAN[1:0]** on the *falling* edge of **BCLK** only when **BWAIT** was LOW on the previous *rising* edge of **BCLK**.

When **BWAIT** is used to stretch a sequential cycle, **BTRAN[1:0]** returns to signalling address during the first phase of the sequential cycle if a single word access is occurring. In this case, it is important that the memory controller does not interpret that an address cycle is signalled when it is a stretched memory cycle.

11.6.3 Other slave responses

Other slave response combinations including bus last, and bus retract are detailed in the *AMBA Specification* (ARM IHI 0001).

11.7 Maximum Sequential Length

The ARM740T may perform sequential memory accesses whenever the cycle is of the same type as the previous cycle (for example, read/write), and the addresses are consecutive. However, sequential accesses are interrupted on a 256-word boundary.

If a sequential access is performed over a 256-word boundary, the access to word 256 is turned into a non-sequential access, and further accesses continue sequentially as before.

This simplifies the design of the memory controller. Provided that peripherals and areas of memory are aligned to 256-word boundaries, sequential bursts are always local to one peripheral or memory device. This means that all accesses to a device always start with a non-sequential access.

A DRAM controller can take advantage of the fact that sequential cycles are always within a DRAM page, provided the page size is greater than 256.

11.8 Read-Lock-Write

The read-lock-write sequence is generated by a SWP instruction.

The **BLOK** signal indicates that the two accesses should be treated as an atomic unit. A memory controller should ensure that no other bus activity is allowed to happen between the accesses when **BLOK** is asserted. When the ARM has started a read-lock-write sequence, it cannot be interrupted until it has completed.

On the bus, the sequence consists of:

- a read access
- a write access to the same address

This sequence is differentiated by the **BLOK** signal. **BLOK**:

- goes HIGH in the HIGH phase of **BCLK** at the start of the read access
- always goes LOW at the end of the write access

The read cycle is always performed as a single, non-sequential, external read cycle, regardless of the contents of the cache.

The write is forced to be unbuffered, so that it can be aborted if necessary.

The cache is updated on the write.

AMBA Interface

11.9 Big-Endian / Little-Endian Operation

The ARM740T treats words in memory as being stored in big-endian or little-endian format depending on the value of the big-end bit in the control register, see **4.3.2 Register 1: Control** on page 4-5.

Load and store are the only instructions affected by the endianness. Refer to the *ARM Architecture Reference Manual* for details of the LDR and STR instructions.

Because the ARM740T duplicates the byte to be written across the databus and internally rotates bytes after reading them from the databus, a 32-bit memory system only needs to have control logic to enable the appropriate byte. There is no need to rotate or shift the data externally.

To ensure that all of the databus is driven during a byte read, it is valid to read a word back from the memory.

Little-endian format

In little-endian format:

- the lowest-numbered byte in a word is considered to be the least significant byte of the word.
- the highest-numbered byte is the most significant.

Byte 0 of the memory system should be connected to data lines 7 through 0 (**BD[7:0]**) in this format.

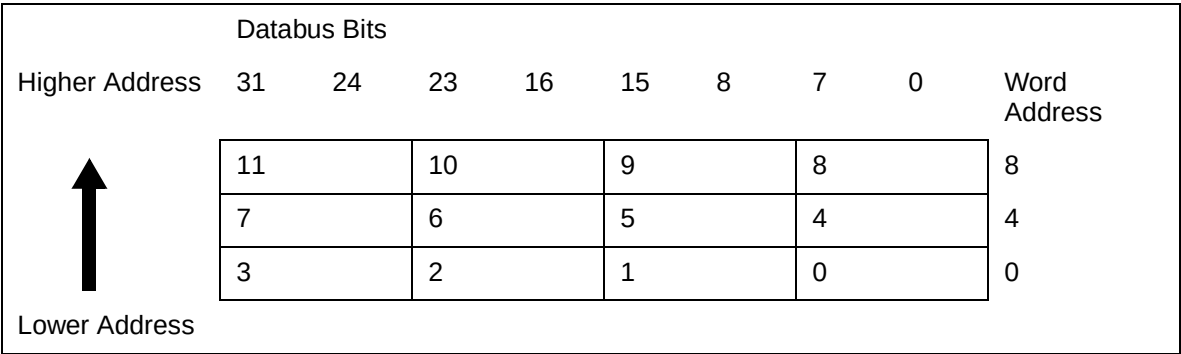


Figure 11-5: Little-endian addresses of bytes within word



Big-endian format

In big-endian format:

- the most significant byte of a word is stored at the lowest-numbered byte.
- the least significant byte is stored at the highest-numbered byte.

Byte 0 of the memory system should therefore be connected to data lines 31 through 24 (**BD[31:24]**).

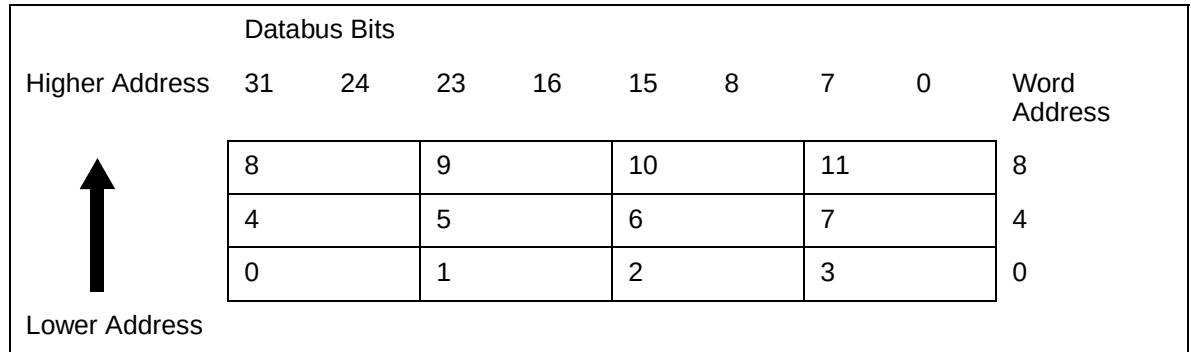


Figure 11-6: Big-endian addresses of bytes within words

11.9.1 Word operations

All word operations expect the data to be presented on data bus inputs 31 through 0. The external memory system should ignore the bottom two bits of the address if a word operation is indicated.

11.9.2 Halfword operations

A halfword store (STRH) repeats the bottom 16 bits of the source register twice across data bus outputs 31 through 0. The external memory system should activate the appropriate byte subsystems to store the data.

Little-endian operation

A halfword load (LDRH) expects the data on data bus inputs 15 through 0 if the supplied address is on a word boundary, or on data bus inputs 31 through 16 if it is a word address plus two bytes. The selected halfword is placed in the bottom 16 bits of the destination register. The other two bytes on the databus are ignored. See **Figure 11-5: Little-endian addresses of bytes within word** on page 11-10.

Big-endian operation

A halfword load (LDRH) expects the data on data bus inputs 31 through 16 if the supplied address is on a word boundary, or on data bus inputs 15 through 0 if it is a word address plus two bytes. The selected halfword is placed in the bottom 16 bits of the destination register. The other two bytes on the databus are ignored. See **Figure 11-6: Big-endian addresses of bytes within words** on page 11-11.

11.9.3 Byte operations

A byte store (STRB) repeats the bottom eight bits of the source register four times across data bus outputs 31 through 0. The external memory system should activate the appropriate byte subsystem to store the data.

Little-endian operation

A byte load (LDRB) expects the data on data bus inputs 7 through 0 if the supplied address is on a word boundary, on data bus inputs 15 through 8 if it is a word address plus one byte, and so on. The selected byte is placed in the bottom eight bits of the destination register. The other three bytes on the databus are ignored. See

Figure 11-5: Little-endian addresses of bytes within word on page 11-10.

Big-endian operation

A byte load (LDRB) expects the data on data bus inputs 31 through 24 if the supplied address is on a word boundary; on data bus inputs 23 through 16 if it is a word address plus one byte, and so on. The selected byte is placed in the bottom 8 bits of the destination register. The other three bytes on the databus are ignored. See

Figure 11-6: Big-endian addresses of bytes within words on page 11-11.

11.10 Multi-Master Operation

The AMBA bus specification supports multiple bus masters on the high performance *Advanced System Bus (ASB)*. A simple two wire request/grant mechanism is implemented between the arbiter and each bus master. The arbiter ensures that only one bus master is active on the bus and also ensures that when no masters are requesting the bus, a default master is granted.

The specification also supports a shared lock signal. This allows bus masters to indicate that the current transfer is indivisible from the following transfer and will prevent other bus masters from gaining access to the bus until the locked transfers have completed.

Arbitration

Efficient arbitration is important to reduce “dead-time” between successive masters being active on the bus. The bus protocol supports pipelined arbitration, such that arbitration for the next transfer is performed during the current transfer.

The arbitration protocol is defined, but the prioritization is flexible and left to the application. Typically, the Test Interface would be given the highest priority to ensure test access under all conditions. Every system must also include a default bus master, which is granted the bus when no bus masters are requesting it.

The request signal, **AREQ**, from each bus master to the arbiter indicates that the bus master requires the bus. The grant signal from the arbiter to the bus master, **AGNT**, indicates that the bus master is currently the highest priority master requesting the bus.

The bus master:

- Must drive the **BTRAN** signals during **BCLK** HIGH when **AGNT** is HIGH.
- Will become granted when **AGNT** is HIGH and **BWAIT** is LOW on a rising edge of **BCLK**.

The shared bus lock signal, **BLOK**, indicates to the arbiter that the following transfer is indivisible from the current transfer and no other bus master should be given access to the bus.

A bus master must always drive a valid level on the **BLOK** signal when granted the bus to ensure the arbitration process can continue, even if the bus master is not performing any transfers.

11.10.1 Arbiter

The arbiter functions as follows:

- 1 Bus masters assert **AREQ** during the HIGH phase of **BCLK**.
- 2 The arbiter samples all **AREQ** signals on the falling edge of **BCLK**.
- 3 During the LOW phase of **BCLK**, the arbiter also samples the **BLOK** signal and then asserts the appropriate **AGNT** signal.

If **BLOK** is LOW, the arbiter grants the highest priority bus master.

If **BLOK** is HIGH the arbiter keeps the same bus master granted.

The arbiter can update the grant signals every bus cycle; however, a new bus master can only become granted and start driving the bus when the current transfer completes, as indicated by **BWAIT** being LOW. Therefore, it is possible for the potential next bus master to change during waited transfers.

The **BLOK** signal is ignored by the arbiter during the single cycle of handover between two different bus masters. If no bus masters are requesting the bus then the arbiter must grant the default bus master.

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The arbitration protocol is defined, but the prioritization is flexible and left to the application. A simple fixed-priority scheme may be used; alternatively, a more complex scheme can be implemented if required by the application.

11.10.2 Bus Master Handover

Bus master handover occurs when a bus master, which is not currently granted the bus, becomes the new granted bus master.

A bus master becomes granted when **AGNT** is HIGH and **BWAIT** is LOW. **AGNT** HIGH indicates the bus master is currently the highest priority master requesting the bus and **BWAIT** LOW indicates the previous transfer has completed.

The handover process is as follows:

- 1 When **AGNT** is asserted, a bus master must drive the **BTRAN** signals during **BCLK** HIGH.
This may continue for many cycles if the previous transfer is waited.
Prior to handover, **BTRAN** must indicate an address-only cycle as the new bus master must commence with an address-only cycle to allow for bus turnaround.
- 2 When the previous transfer completes, the new bus master becomes granted.
- 3 In the last clock HIGH phase of the previous transfer, the address bus stops being driven by the previous bus master.
- 4 The new bus master starts to drive the address bus and control signals during the clock LOW phase.
- 5 The first transfer may then commence in the following bus cycle.

During a waited transfer, bus master handover may be delayed and it is possible that the **AGNTx** to a particular bus master may be asserted and then negated, if another higher priority bus master then requests the bus before the current transfer has completed.

11.10.3 Default Bus Master

If the ARM740T is to be the default bus master, as is the case in many systems. The **AREQ** signal from the ARM740T should not be used. In this case the arbiter should always allocate the bus to the ARM740T when not requested by higher priority bus masters.

This will result in a system with good bus performance, as the ARM740T will not have to wait for the bus to be granted when it wishes to perform a bus transfer.

12

AMBA Test

This chapter describes the test features of ARM740T.

12.1	Slave Operation (Test Mode)	12-2
12.2	ARM740T Test Mode	12-3
12.3	ARM7TDM Core Test Mode	12-3
12.4	RAM Test Mode	12-4
12.5	TAG Test Mode	12-5
12.6	Test Register Mapping	12-6

AMBA Test

12.1 Slave Operation (Test Mode)

When the ARM740T block is selected as a slave, it is possible to write and read test vectors to the core using the AMBA test methodology.

The ARM740T provides four test modes for this purpose:

- ARM740T test mode
- ARM7TDM Core test mode
- RAM test mode
- TAG test mode

To apply test vectors to the ARM740T, the ARM740T block must have been deselected as a master (**AGNT** goes LOW). The Test Interface Controller becomes the bus master, and the ARM740T is selected as a slave using the signal **DSELARM**. This places the ARM740T into test mode, and allows access to the test registers.

The tests are sequenced by the test state machine in the AMBA interface, which generates the appropriate control signals for the test modes.

A sample test sequence is shown in **Figure 12-1: Running a test vector on the processor core**.

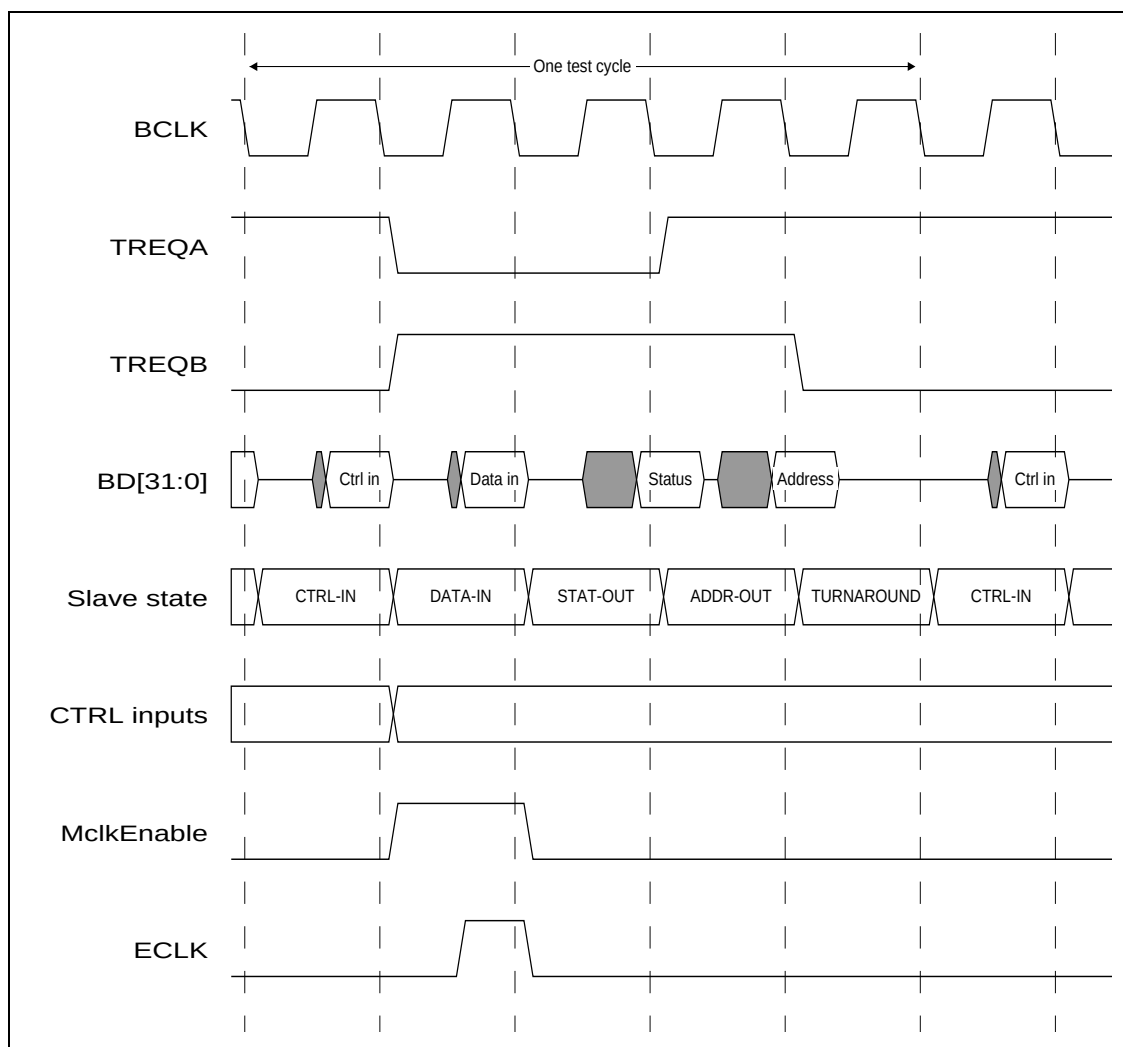


Figure 12-1: Running a test vector on the processor core

12.2 ARM740T Test Mode

The ARM740T test mode is used to test the functionality of:

- cache control logic
- write buffer
- protection unit
- cache

To perform this test control, stimuli are applied to the control register, see **Table 12-1: RAM test mode address packet bit positions** on page 12-4.

Data packets are read or written as appropriate and the address and status are read back (see **Table 12-1: RAM test mode address packet bit positions** on page 12-4).

The sequencing for this test mode is as shown in **Figure 12-2: State machine for ARM740T and ARM7TDMI test**. This is the default test mode, and is selected when the bits [31:29] of the control register are set LOW (see **Table 12-1: RAM test mode address packet bit positions** on page 12-4).

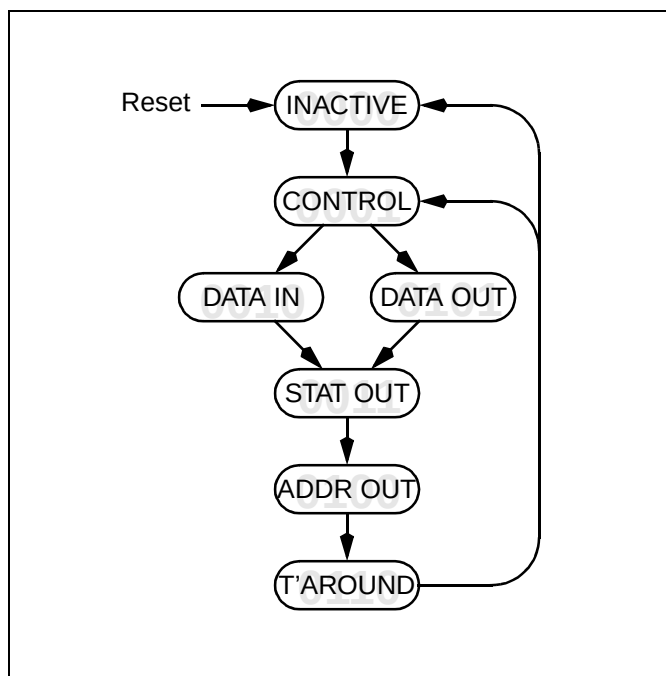


Figure 12-2: State machine for ARM740T and ARM7TDMI test

12.3 ARM7TDM Core Test Mode

The ARM7TDMI test places the ARM740T into a test mode so that the signals of the ARM7TDM are visible to the AMBA interface. In this mode, the rest of ARM740T is held in reset. The ARM740T is placed in the mode by setting bit 31 of the control register, see **Table 12-1: RAM test mode address packet bit positions** on page 12-4.

AMBA Test

12.4 RAM Test Mode

The RAM test mode is used to perform an intensive test of the RAM arrays, to provide full coverage of bit faults. In this test mode, the rest of the ARM740T is held in reset and direct access is provided to the data, address and control signals of the RAM.

To accommodate this, an alternative test sequence is used, see **Figure 12-3: State machine for RAM test mode**.

In this test mode, the RAM control signals are derived from unused address bits, as shown in **Table 12-1: RAM test mode address packet bit positions** on page 12-4.

To enter RAM test mode, bits 30 and 28 of the control packet should be set. This places the ARM740T into RAM test mode, and forces the RAM to be clocked from the **FCLK** input.

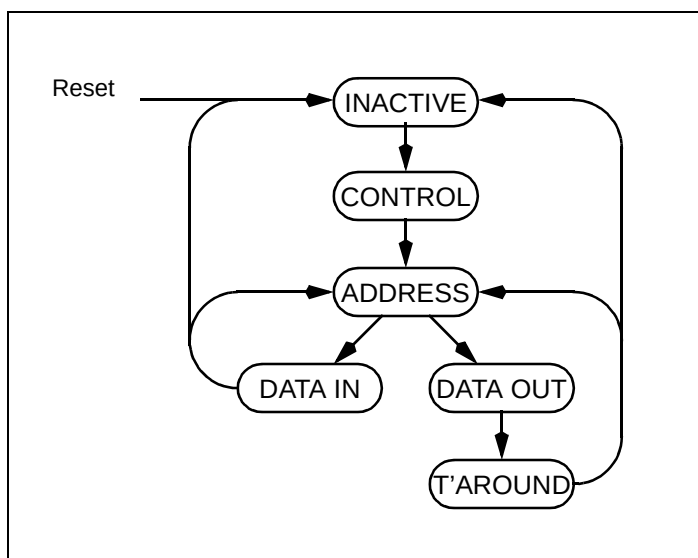


Figure 12-3: State machine for RAM test mode

Address packet bit	RAM signal	Description
[24:23]	MAS[1:0]	RAM access size
22	RSEQ	RAM sequential signal
21	IMMED	Immediate write signal, controls write pipeline, and selects between RAMSEL[3:0] and SETSEL[3:0].
20	WRITE	RAM write strobe
19	READ	RAM read strobe
[18:15]	RAMSEL[3:0]	RAM bank select signal, used when IMMED is LOW
[14:11]	SETSEL[3:0]	RAM bank select signal, used when IMMED is HIGH
[10:0]	ADDR[10:0]	RAM address

Table 12-1: RAM test mode address packet bit positions

12.5 TAG Test Mode

The TAG test mode is used to perform an intensive test of all of the cells of the TAG array, and to test the TAG comparators. In this test mode, the rest of the ARM740T is held in reset and direct access is provided to the data, address and control signals of the RAM. See **Figure 12-4: State machine for TAG test mode**.

In this test mode the TAG control signals are derived from the TAG CTL packet as shown in **Table 12-1: RAM test mode address packet bit positions** on page 12-4.

To enter TAG test mode, bits 29 and 28 of the control packet should be set. This places the ARM740T into TAG test mode, and forces the TAG to be clocked from the **FCLK** input.

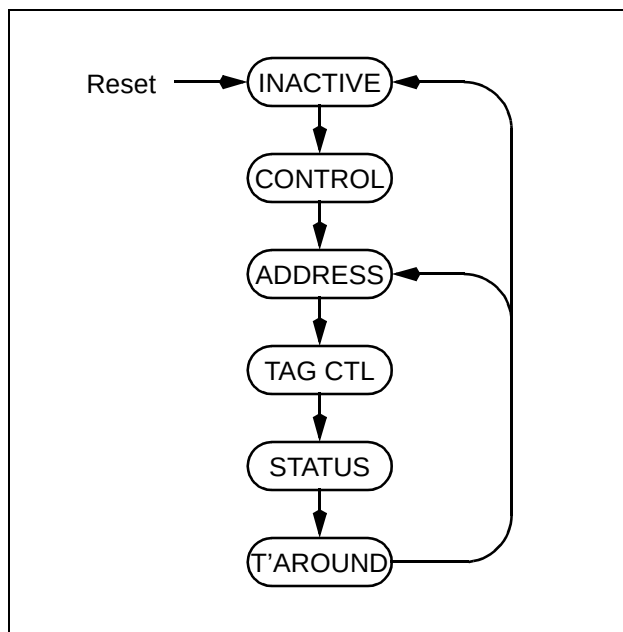


Figure 12-4: State machine for TAG test mode

TAG CTL packet bit	TAG signal	Description
[11:8]	FLUSH[3:0]	When asserted each bit flushes the appropriate TAG arrays
[7:4]	TAGSEL[3:0]	Tag select signal, each bit selects a TAG array
2	WRITE	TAG write strobe
1	READ	TAG read strobe
0	VALID	Valid input, the value on VALID is written into the valid cell in the array on a write.

Table 12-2: TAG test mode TAG CTL packet bit positions

AMBA Test

12.6 Test Register Mapping

The test registers are defined in the following tables:

- **Table 12-3: Status packet bit positions**
- **Table 12-4: Control Packet bit positions**

12.6.1 Status packet bit positions

Bit	ARM7TDMI Test	ARM740T Test	Notes
31	BUSDIS Bus Disable		
30	SCREG[2] Scan chain register	SCREG[2] Scan chain register	
29	SCREG[2] Scan chain register	SCREG[2] Scan chain register	
28	SCREG[1] Scan chain register	SCREG[1] Scan chain register	
27	SCREG[0] Scan chain register	SCREG[0] Scan chain register	
26	HIGHZ HIGHZ instruction in TAP controller	HIGHZ HIGHZ instruction in TAP controller	
25	nTDOEN not TDO enable	nTDOEN not TDO enable	
24	DBGREQI Internal debug request	DBGREQI Internal debug request	
23	RANGEOUT0 ICEbreaker Rangeout0	RANGEOUT0 ICEbreaker Rangeout0	
22	RANGEOUT1 ICEbreaker Rangeout1	RANGEOUT1 ICEbreaker Rangeout1	
21	COMMRX Communications channel receive	COMMRX Communications channel receive	
20	COMMTX Communications channel transmit	COMMTX Communications channel transmit	
19	DBGACK Debug acknowledge	DBGACK Debug acknowledge	
18	TDO Test data out	TDO Test data out	
17	nENOUT Not enable output.	nENOUT Not enable output	nENOUT is only valid during the data access cycle, so MclkEnable is used to clock a transparent latch that will capture the correct state.

Table 12-3: Status packet bit positions

Bit	ARM7TDMI Test	ARM740T Test	Notes
16	nENOUTI Not enable output	PROTWATCH[3] Protection Unit test output	nENOUTI as nENOUT
15	TBIT Thumb state	PROTWATCH[2] Protection Unit test output	
14	nCPI Not Coprocessor instruction.	PROTWATCH[1] Protection Unit test output	
13	nM[4] Not processor mode	PROTWATCH[0] Protection Unit test output	
12	nM[3] Not processor mode	CAMWATCH[1] Replacement test output	
11	nM[2] Not processor mode	CAMWATCH[0] Replacement test output	
10	nM[1] Not processor mode	IDCWATCH[3] Cache test output	
9	nM[0] Not processor mode	IDCWATCH[2] Cache test output	
8	nTRANS Not memory translate	IDCWATCH[1] Cache test output	
7	nEXEC Not executed	IDCWATCH[0] Cache test output	
6	LOCK Locked operation	LOCK Locked operation.	
5	MAS[1] Memory Access Size	MAS[1] Memory Access Size	
4	MAS[0] Memory Access Size	MAS[0] Memory Access Size	
3	nOPC Not op-code fetch	nENOUT Not enable output	
2	nRW Not read/write	nRW Not read/write	
1	nMREQ Not memory request	nMREQ Not memory request.	
0	SEQ Sequential address	SEQ Sequential address	

Table 12-3: Status packet bit positions (Continued)

AMBA Test

12.6.2 Control packet bit positions

Bit	ARM7TDMI Input	ARM740T Input	Notes
31	TESTCPU ARM7TDM test enable	TESTCPU ARM7TDMI test enable	
30		TAGTEST TAG test mode enable	
29		RAMTEST RAM test mode enable	
28	nENIN NOT enable input	FORCECLK Clock select override	nENIN is gated with MCLKENABLE , so it is only valid (LOW) during data access.
27	SDOUTBS Boundary scan serial output data		
26	TBE Test bus enable		
25	APE Address pipeline enable		
24	BL[3] Byte Latch Control		ANDed with MCLKENABLE , so will only be valid during data access cycle. Not Supported.
23	BL[2] Byte Latch Control		ANDed with MCLKENABLE , so is only valid during data access cycle. Not Supported.
22	BL[1] Byte Latch Control		ANDed with MCLKENABLE , so is only valid during data access cycle. Not Supported.
21	BL[0] Byte Latch Control		ANDed with MCLKENABLE , so is only valid during data access cycle. Not Supported.
20	TMS Test Mode Select	TMS Test Mode Select	
19	TDI Test Data in	TDI Test Data in	
18	TCK Test clock	TCK Test clock	ANDed with MCLKENABLE and BCLK .
17	nTRST Not Test Reset	nTRST Not Test Reset.	
16	EXTERN1 External input 1	EXTERN1 External input 1	

Table 12-4: Control Packet bit positions

Bit	ARM7TDMI Input	ARM740T Input	Notes
15	EXTERN0 External input 0	EXTERN0 External input 0.	
14	DBGREQ Debug request	DBGREQ Debug request	
13	BREAKPT Breakpoint	BREAKPT Breakpoint	
12	DBGEN Debug Enable	DBGEN Debug Enable	
11	ISYNC Synchronous interrupts	ISYNC Synchronous interrupts.	
10	BIGEND Big Endian configuration	BIGEND Big Endian configuration	
9	CPA Coprocessor absent	CPA Coprocessor absent	
8	CPB Coprocessor busy	CPB Coprocessor busy	
7	ABE Address bus enable	SnA Clock Configuration	This should normally be set HIGH, as if the address bus is tri-stated (ABE LOW), then it is not possible to read address values
6	ALE Address latch enable	ALE Address latch enable	
5	DBE Data Bus Enable	FASTBUS Clock configuration	DBE to the ARM7TDM is ANDed with the state machine generated DBE and BCLK to prevent bus conflict.
4	nFIQ Not fast interrupt request	nFIQ Not fast interrupt request	
3	nIRQ Not interrupt request	nIRQ Not interrupt request	
2	ABORT Memory Abort	ABORT Memory Abort	
1	nWAIT Not wait	nWAIT Not wait.	ANDed with MCLKENABLE , so that the core state can only change during the data access cycle.
0	nRESET Not reset	nRESET Not reset	

Table 12-4: Control Packet bit positions (Continued)